

Virtuoso® Layout Pro: T1 Environment and Basic Commands

Course Version IC 25.1

Lab Manual

Revision 1.0

© 1990-2025 Cadence Design Systems, Inc. All rights reserved.

Printed in the United States of America.

Cadence Design Systems, Inc. (Cadence), 2655 Seely Ave., San Jose, CA 95134, USA.

Trademarks: Trademarks and service marks of Cadence Design Systems, Inc. (Cadence) contained in this document are attributed to Cadence with the appropriate symbol. For queries regarding Cadence trademarks, contact the corporate legal department at the address shown above or call 1-800-862-4522.

All other trademarks are the property of their respective holders.

Restricted Print Permission: This publication is protected by copyright, and any unauthorized use of this publication may violate copyright, trademark, and other laws. Except as specified in this permission statement, this publication may not be copied, reproduced, modified, published, uploaded, posted, transmitted, or distributed in any way, without prior written permission from Cadence. This statement grants you permission to print one (1) hard copy of this publication subject to the following conditions:

- The publication may be used solely for personal, informational, and noncommercial purposes;

- The publication may not be modified in any way;

- Any copy of the publication or portion thereof must include all original copyright, trademark, and other proprietary notices and this permission statement; and

- Cadence reserves the right to revoke this authorization at any time, and any such use shall be discontinued immediately upon written notice from Cadence.

Disclaimer: Information in this publication is subject to change without notice and does not represent a commitment on the part of Cadence. The information contained herein is the proprietary and confidential information of Cadence or its licensors, and is supplied subject to, and may be used only by Cadence customers in accordance with, a written agreement between Cadence and the customer.

Except as may be explicitly set forth in such agreement, Cadence does not make and expressly disclaims, any representations or warranties as to the completeness, accuracy or usefulness of the information contained in this document. Cadence does not warrant that use of such information will not infringe any third party rights, nor does Cadence assume any liability for damages or costs of any kind that may result from the use of such information.

Restricted Rights: Use, duplication, or disclosure by the Government is subject to restrictions as set forth in FAR52.227-14 and DFAR252.227-7013 et seq. or its successor.

Table of Contents

Virtuoso Layout Pro: T1 Environment and Basic Commands

Module 1:	About This Course	6
	There are no labs for this module.....	6
Module 2:	The Design Environment.....	7
Lab 2-1	Opening Your Designs	8
	Lab Setup	8
	Invoking the <i>virtuoso</i> Executable.....	8
	Viewing the Fields in the Open File Form	9
	Using an Alternative Way to Open a Cellview	11
	Generating Thumbnails.....	12
	Opening the Layout.....	13
	Summary.....	17
Lab 2-2	Using the Library Path Editor	18
	Opening the Library Path Editor	18
	Viewing the Library Path Information	19
	Viewing the Color Definitions Information	20
	Summary.....	21
Lab 2-3	Using the Library Manager.....	22
	Invoking the Library Manager	22
	Using the Filter and Search Options	23
	Using the Filter Combo Box to View Information.....	24
	Using the Search Option	27
	Viewing the Contents of the <i>ether</i> Library with the Show Categories Option.....	27
	Creating a New Category	28
	Creating a New Library.....	29
	Copying a Cell	30
	Deleting a Layout/Cell View	32
	Hiding a Cell	33
	Showing the Hidden Cell	36
	Viewing the Current Cellview Status.....	38
	Using the Cdsenv Editor (Optional Exercise)	39
	Summary.....	40
Lab 2-4	Setting User Preferences.....	41
	Setting the Window Options	41
	Changing the Window Controls	42
	Summary.....	44
Lab 2-5	Using History and Bookmarks.....	45
	Using the History	45
	Adding the Bookmarks	45
	Summary.....	47
Lab 2-6	Using World View	48
	Using the World View Assistant.....	48
	Summary.....	49

Lab 2-7	Floating the Palettes Away from Cellviews and Saving Workspaces	50
	Saving the Workspace	50
	Summary	51
Module 3:	User Interface.....	52
Lab 3-1	Using the Layers, Objects, and Grids Assistants	53
	Lab Setup	53
	Starting the session.....	53
	Setting the Current Entry Layer in the Layers Assistant	54
	Making the Layers Visible/Invisible	56
	Customizing the Valid Layers and Saving the Setting for Later Use	56
	Using the Objects Assistant.....	59
	Using the Grids Assistant.....	61
	Summary	62
Lab 3-2	Loading and Verifying the Bindkeys	63
	Viewing the Loaded Bindkeys in the Layout Canvas	63
	Loading the Bindkey Files from the .cdsinit Startup File.....	63
	Verifying the Bindkey Actions	64
	Viewing the Bindkey Definitions in Bindkey Editor	65
	Summary	66
Lab 3-3	Saving the Defaults.....	67
	Saving the Defaults Settings for the Tools	67
	Summary	68
Lab 3-4	Using the Toolbar Manager	69
	Hiding an Icon in a Toolbar	70
	Creating a New Custom Toolbar.....	71
	Summary	74
Module 4:	Basic Layout Commands.....	75
Lab 4-1	Viewing Your Designs.....	76
	Lab Setup	76
	Starting the session.....	76
	Using the Pan Command to Move Around a Design	77
	Saving and Restoring a Zoomed Window.....	77
	Moving Around with the Zoom Commands	78
	Summary	79
Lab 4-2	Selecting and Deselecting the Objects	80
	Using the Mouse to Select Objects in Full Selection Mode	80
	Choosing Between Objects	81
	Adding or Subtracting Objects to the Selected List	82
	Using the Drag Box to Select or Deselect Objects.....	82
	Using the Mouse to Select Objects in Partial Selection Mode	82
	Summary	84
Lab 4-3	Using the Alignment Commands	85
	Using Quick Align to Abut Transistors.....	85
	Stretching Wires Using <i>Partial Select</i> and <i>Quick Align</i>	87
	Using the Align Toolbar.....	89
	Summary	91

Module 5:	iPegasus DRC and Fill for Virtuoso Studio	92
Lab 5-1	Setting Up and Running Virtuoso DRD and iPegasus DRC.	93
	Lab Setup	93
	Setting Up and Running Virtuoso DRD.....	94
	Running iPegasus DRC – XL/EXL/MXL Mode (Limited Mode DRC)	96
	Running iPegasus DRC – Full Mode	101
	Running iPegasus DRC – With Snapshot	102
	Running iPegasus DRC Within the Specified Area	108
	Running iPegasus DRC Only Over the Changed Area	109
	Running iPegasus DRC with Rule Filter.....	112
	Running iPegasus DRC with Palette Filter	115
Lab 5-2	Setting Up and Running iPegasus Fill	117
	Run iPegasus Fill – With Snapshot.....	117
	Running iPegasus Fill – With Exclude Cells	124
	Running iPegasus Fill – Create Symmetrical Fill	127
	Using Net Aware Feature.....	132

Module 1: About This Course

There are no labs for this module.

Module 2: The Design Environment

Lab 2-1 Opening Your Designs

Objective: To use the Open File form, Library Manager, and Thumbnails feature to open a design window.

Before you can begin creating or editing your design in Virtuoso Studio Design Environment, you need to set preferences as to how you want to see the icons, graphics, assistants, colors, and bindkeys that you will be working with. This module gives you the basics on how to do that. However, your preferences might change as you gain more expertise with the Virtuoso® platform.

You can open a design window using the Open File form or the Library Manager. The Library Manager lets you edit the data inside of the library, such as changing properties or renaming cells. Once you have access to the Library Manager, you can navigate to the desired library, cell, and view by clicking with your mouse. You will run the command to install the thumbnails of the cellviews in the forms.

Lab Setup

Follow the steps below to set up the environment and perform the lab:

1. Untar the lab database file, using the following command.

```
tar -xzf <filename.tar.gz>
```

2. Update the following line in the /VLPT1_IC_25_1/.cshrc_VLPT1 file with the appropriate valid local tool paths for Virtuoso.

```
setenv CDSHOME /ccstools/cdsind1/Software/IC251Base_lnx86
```

3. Save and close the file.

4. Source the updated class setup file.

```
source .cshrc_VLPT1
```

Invoking the *virtuoso* Executable

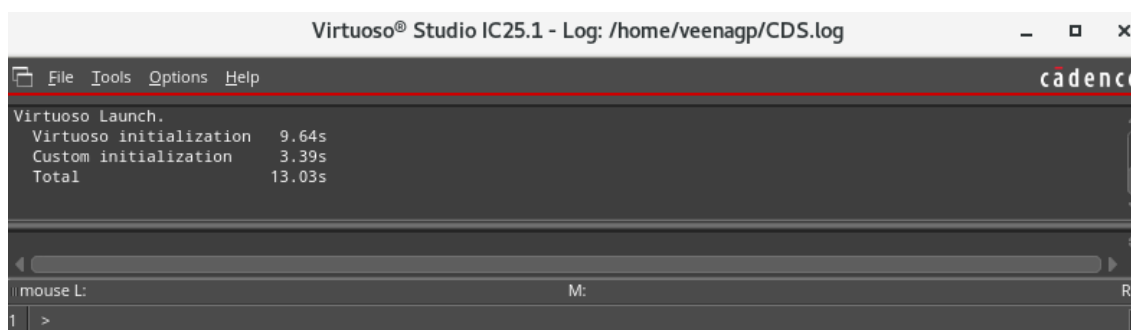
1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

```
cd ./M02_VLPT1_DesignEnv
```

2. To start the Cadence® Virtuoso software, enter this command in the same terminal window:

```
virtuoso &
```

The Command Interpreter Window (CIW) appears:



When you start the Virtuoso executable, the first tool window displayed is the Command Interpreter Window (CIW). From the CIW, you can open designs, set global options, view the history of the current session, and input expressions written in the SKILL[®] programming language. It is a good practice to have the CIW visible on top to view any messages that are displayed from commands that you have invoked.

Viewing the Fields in the Open File Form

1. Move your cursor into the CIW and choose **File – Open**.

The Open File form appears.

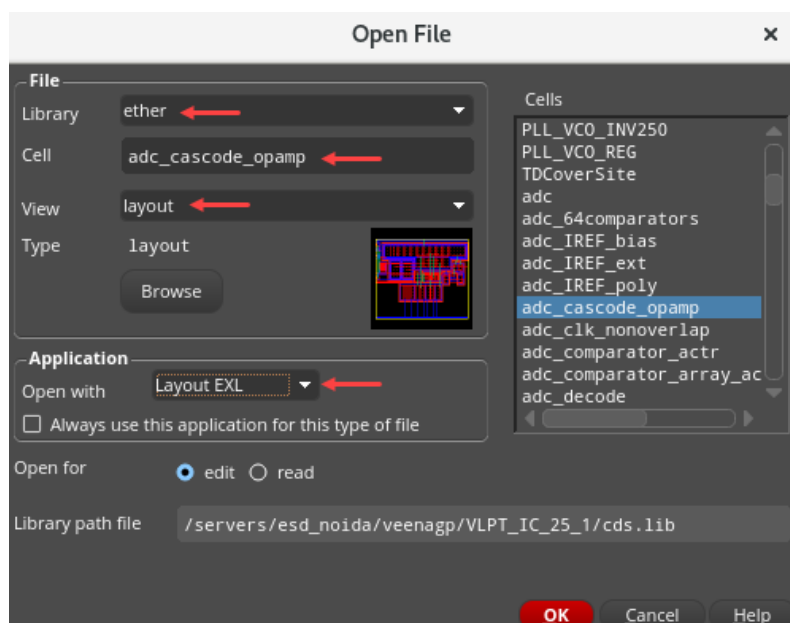
2. In the Open File form, set the *Library* name to **ether**. Click on the **Library** name cyclic field to view the listing of libraries available for editing.
3. In the *Cell* field, choose **adc_cascode_opamp**.

The listing of cell names is alphabetical, so you may need to use the scroll bar.

4. In the *View* field, click the listing of available views. Select the **layout** view.

Make sure that *Application – Open with* is set to **Layout EXL**. If it is not, use the cyclic field selector to set the application to **Layout EXL**.

The form looks like this:



5. Click **OK** to open the cell.

The *ether/adc_cascode_opamp/schematic* and *ether/adc_cascode_opamp/layout* cellviews open in EXL mode. Minimize the schematic view and maximize the layout view.

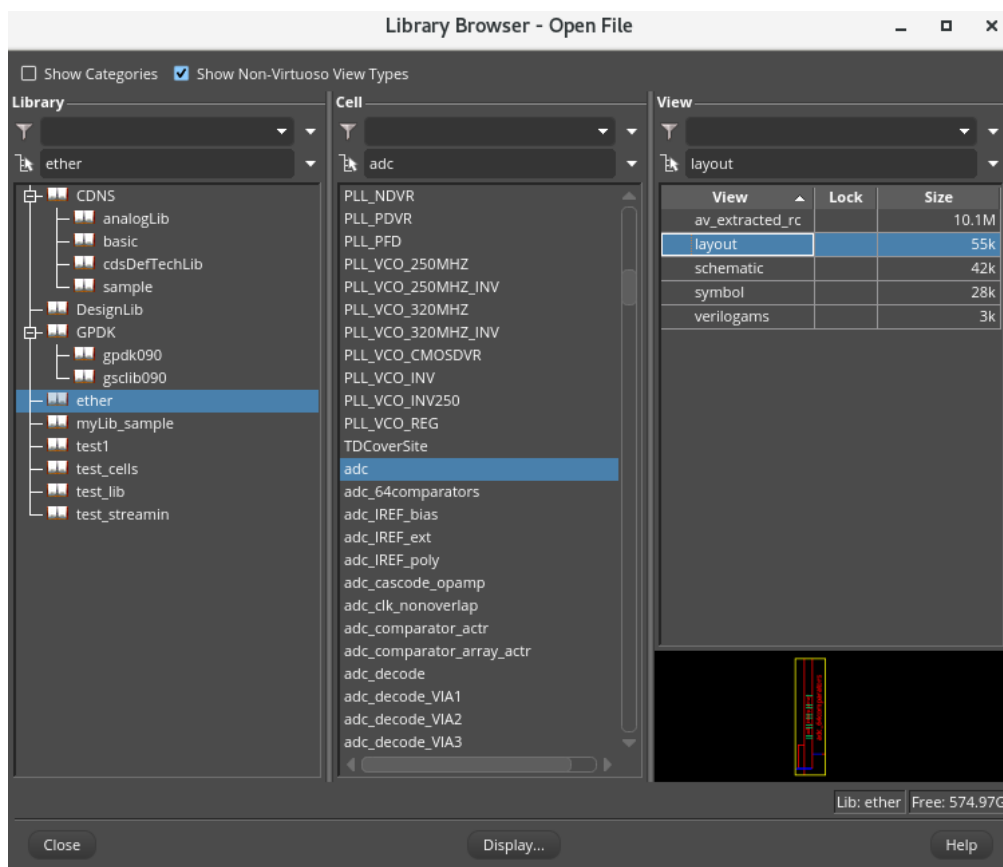
You can use bindkeys to change the level of detail in the display, zoom in to a portion of the cellview, and fit the screen to view the entire layout cellview.

6. Click in the window banner or cellview canvas to make the window active.
7. Press the bindkey **Shift+F** to view all the levels of the hierarchy in the layout canvas.
8. **Right**-click and drag the mouse to zoom in to a portion of the cellview.
A rubber band zoom window appears when dragging while **right**-clicking.
9. Press the **F** bindkey to fit the cellview extents.
10. Practice the **right**-click-drag-zoom function a few times to see how this embedded command works without a pull-down menu or bindkey.
11. Choose **File – Close** to close the schematic and layout views. Do **NOT** save.

Using an Alternative Way to Open a Cellview

1. In the CIW, choose **File – Open**.
2. Click **Browse** in the Open File form.

The Library Browser – Open File form appears.



3. Use the Library Browser – Open File form to populate the Open File form with the *ether* library, *adc* cell, and *layout* view, as shown here.
4. Click **Close** on the Library Browser – Open File form.
5. Click **OK** in the Open File form.

The *ether/adc/schematic* and *ether/adc/layout* cellviews open in EXL mode.

- Choose **File – Close** to close the schematic and layout views. Do **NOT** save.

Note: If you have any unsaved edits, the system prompts you to save them. Do **NOT** save.

Generating Thumbnails

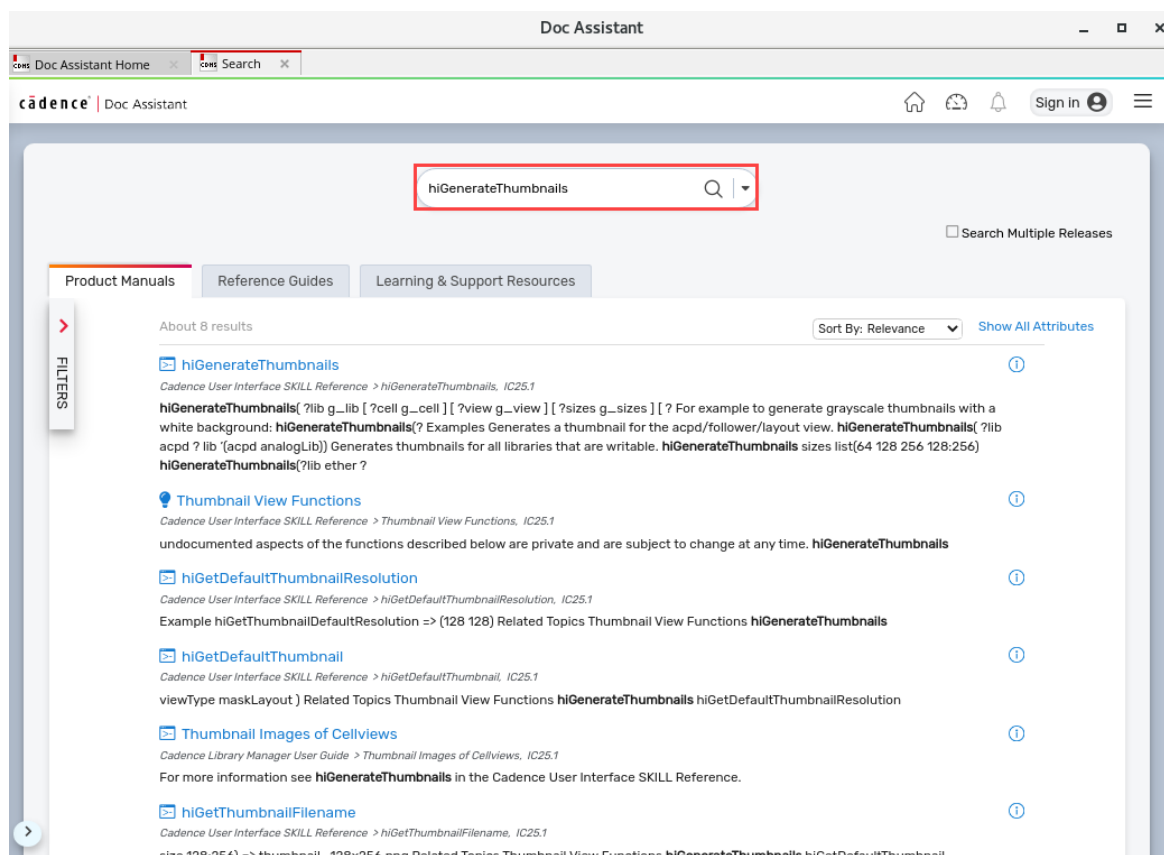
You create the thumbnail images for use in the Library Manager. You use the *Doc Assistant* system to find and use the **hiGenerateThumbnails** SKILL command.

- In the terminal window, enter:

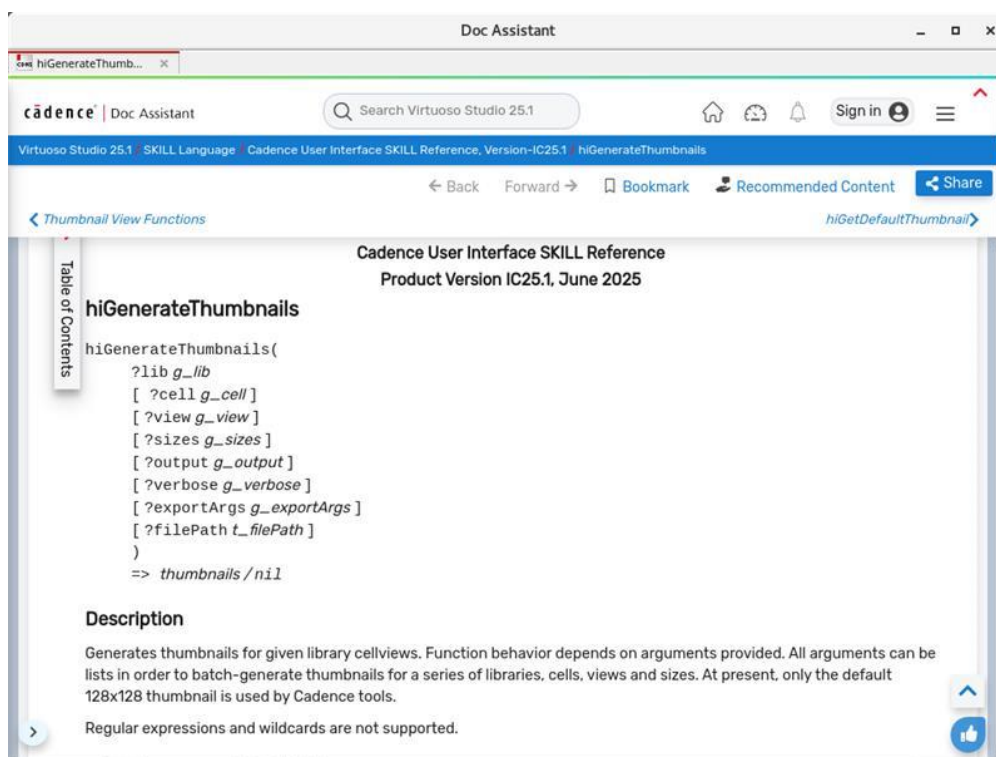
```
cda &
```

The Doc Assistant window opens.

- In the Documentation Browser, search for **hiGenerateThumbnails**, as shown here:



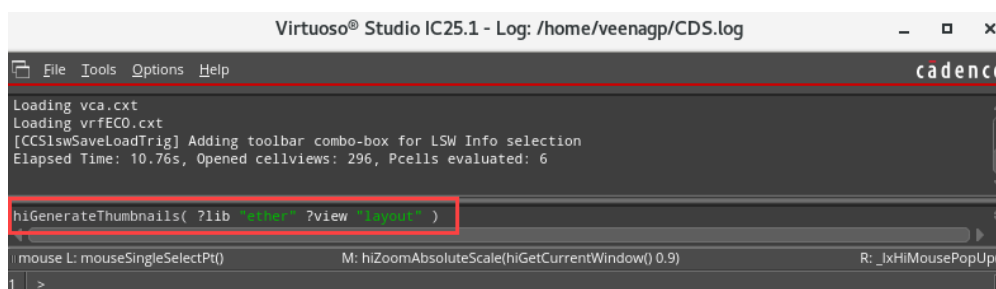
- Click on *hiGenerateThumbnails* and observe the syntax of the SKILL function:



- In the CIW input pane, enter:

```
hiGenerateThumbnails( ?lib "ether" ?view "layout" )
```

Note: This operation has already been done for this library. Thumbnails will appear for each appropriate layout view in this design. However, you can use this command to create and include thumbnails on other designs. The command takes a couple of minutes to generate all the layout thumbnails for the *ether* library.



After the command finishes, the Library Manager and Open File forms display thumbnails for the layout views of the library that you want to work in.

Opening the Layout

- Choose **File – Open (CIW)** or **Tools – Library Manager** GUIs to open the *ether/adc/schematic* and *ether/adc/layout* cellviews again in EXL mode.

2. Press the **Tab** key (starts the **View – Pan** command) and click in the **upper-right** corner of the design.

The cellview is panned to bring the **upper-right** area to the center of the canvas.

3. Press the **Tab** key and click to pan around the canvas.

The click point becomes the new center of the screen.

Note: You can also press the arrow keys on your keyboard to pan around your design.

4. Choose **File – Close** to close the schematic and layout views. Do **NOT** save.

5. If the Library Manager is not already open, in the CIW, choose the **Tools – Library Manager** menu command to open the Library Manager.

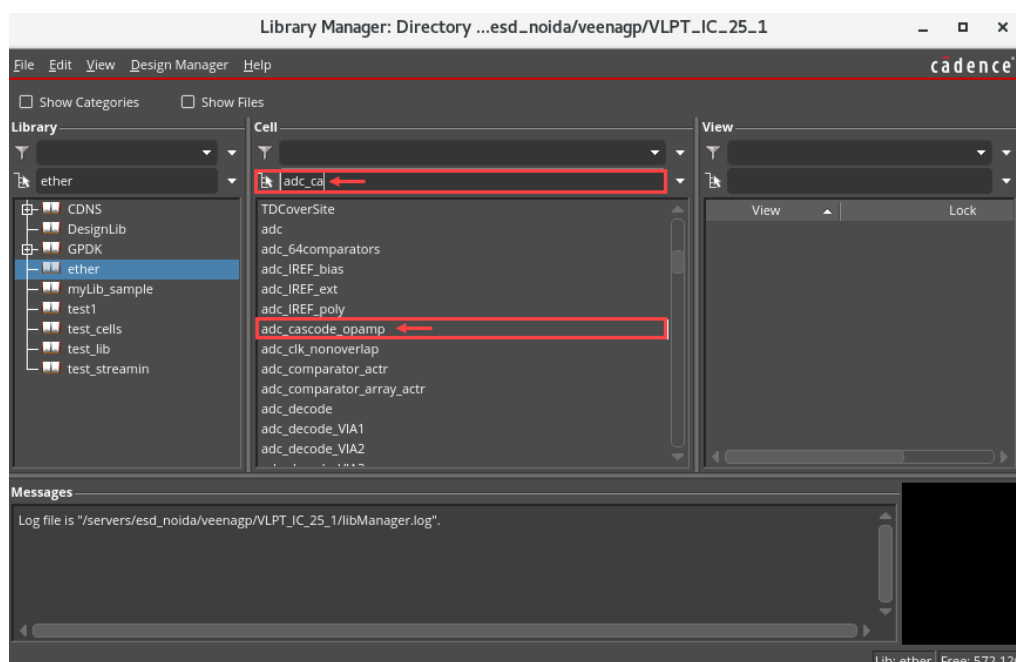
Notice the Library, Cell, and View columns in the Library Manager. Also, notice the thumbnail feature, which gives you a preview of the cellview as you click in the **View** column of the Library Manager.

6. Select **ether** from the Library column.

The list of cells in the *ether* library populates the Cell field.

7. In the Cell field, enter **adc_ca**.

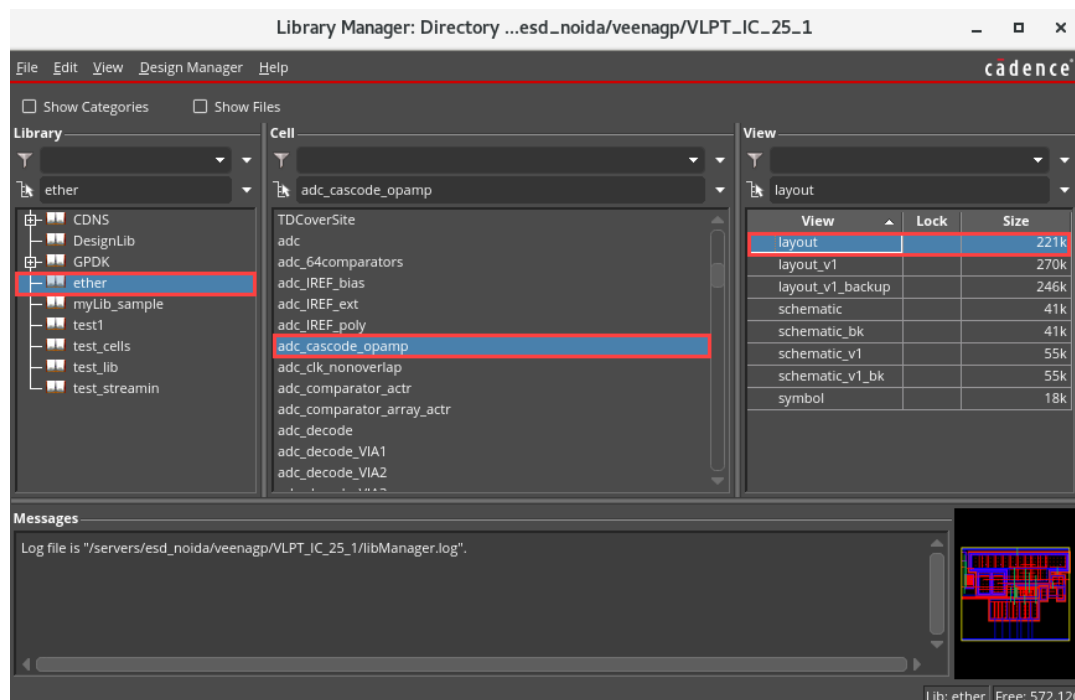
The Cell listing highlights the *adc_cascode_opamp* cell.



- Click on the **adc_cascode_opamp** cell in the Cell column.

The View field populates with the views available for the *ether/adc_cascode_opamp* cell.

- Click the **layout** view.



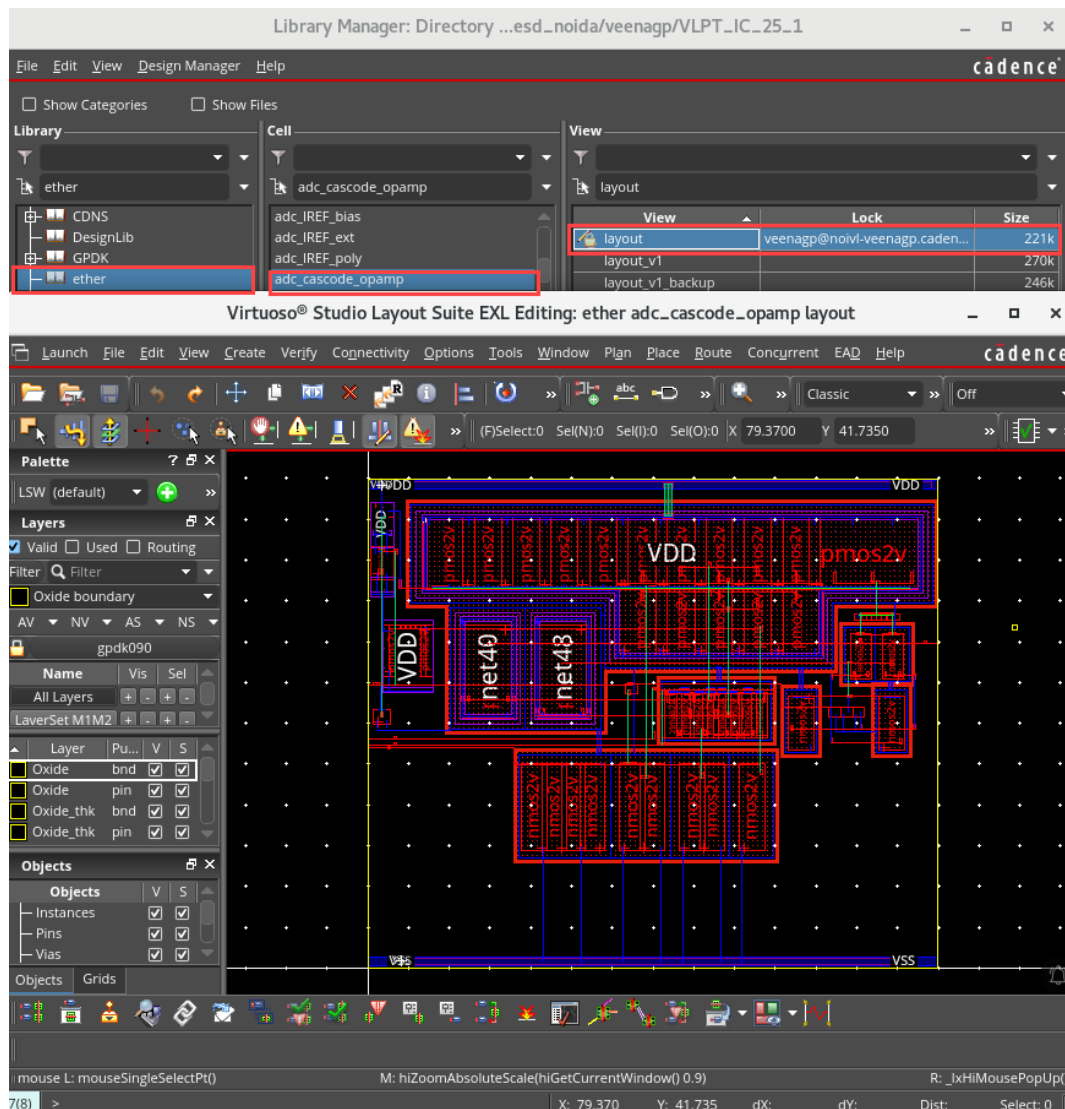
Notice that the View field includes the Lock column. At the moment, the cell is available for edit. No login appears in the Lock column.

- Right-click** the layout view name and select **Open**.

Alternatively, you can **double-click** the layout view in the Cell column or **double-click** the thumbnail to open the layout cellview in edit mode.

- Use the arrow keys to pan the canvas up, down, left, and right.

12. Look in the Library Manager again. The cellview is now locked with the user login posted.



13. Close the schematic and layout cellviews by choosing **File – Close**.

14. Choose **File – Exit** in the Library Manager.

15. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you learned how to

- ◆ Start the Virtuoso software
- ◆ View the fields in the Open File form
- ◆ Create thumbnails
- ◆ Open the layout
- ◆ Know who has a cellview checked out for editing
- ◆ Open and close a design window



Lab 2-2 Using the Library Path Editor

Objective: To use the Library Path Editor.

The Library Path Editor uses a graphical user interface (GUI) for editing the *cds.lib* file.

In this lab, you explore how to use the Library Path Editor.

Opening the Library Path Editor

1. From the CIW, choose **Tools – Library Path Editor**.

The Library Path Editor form appears.



The Library Path Editor reads the information from the *cds.lib* file. This file must reside in the directory where you start the software (the working directory). When you start the Virtuoso software, the *cds.lib* file is read to provide path information for the libraries available.

2. In the Library Path Editor form, choose **Edit – Exclusive Lock**.

By setting the exclusive lock from the Library Path Editor, you are the only one editing the *cds.lib* file.

3. In the same terminal window where you started the Virtuoso software, open the **cds.lib** file for viewing by entering:

```
more cds.lib
```

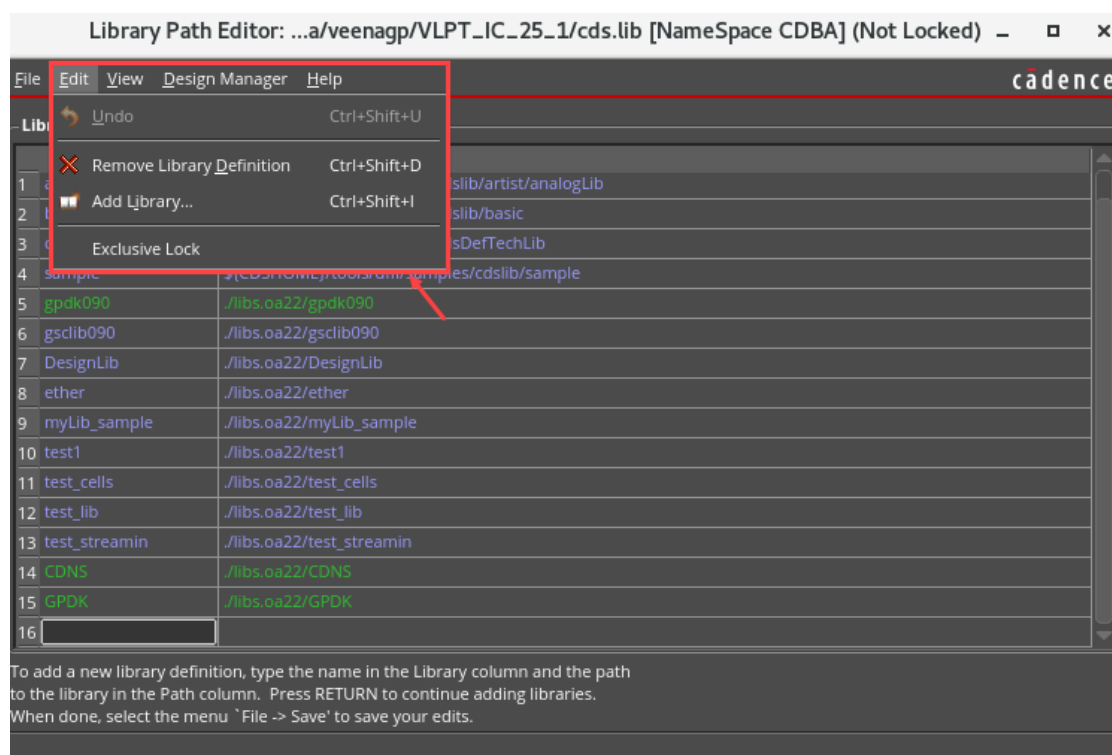
or

```
vi cds.lib
```

Do the cds.lib file matches the library paths in the Library Path Editor form?

Answer: You can observe that they are matching.

4. In the Library Path Editor form, choose **Edit**.



With this form, you can add or remove library paths to the *cds.lib* file.

By using the exclusive lock, you can be sure that you are the only person making changes to the *cds.lib* file.

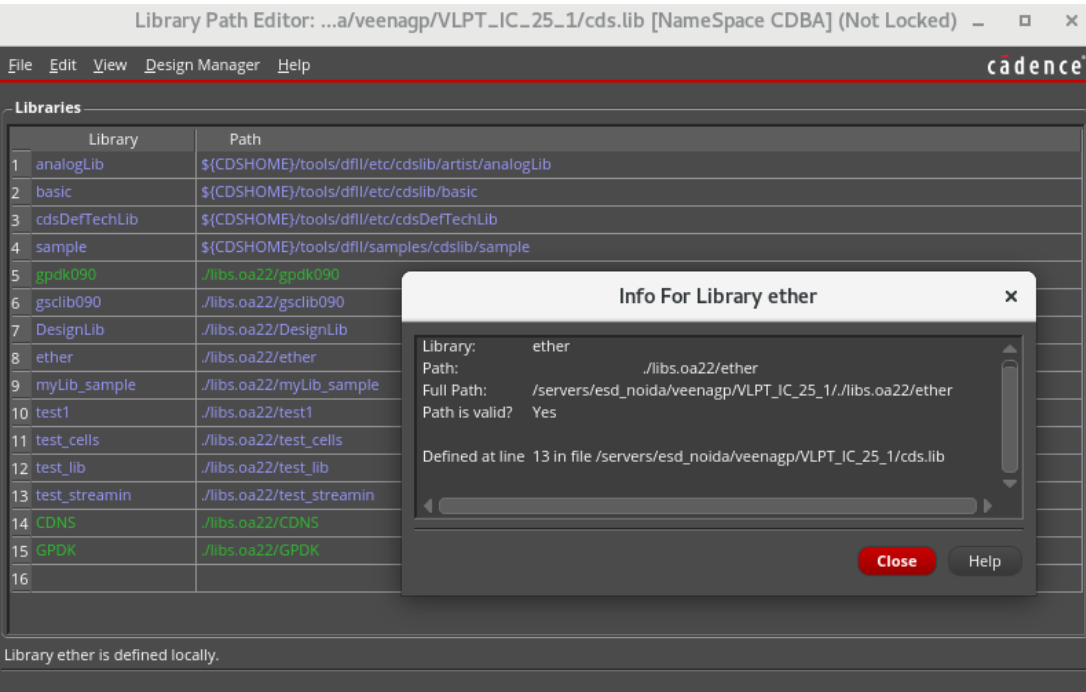
5. Click outside the form to close the **Edit** pull-down menu.

Viewing the Library Path Information

View the exact locations of the libraries that are defined in the *cds.lib* file.

1. In the Library Path Editor form, click the **ether** library.
 - a. In the Library Path Editor form, choose **View – Library Info**.

A text box appears with the library name, path, and where the path is defined in the *cds.lib* file.

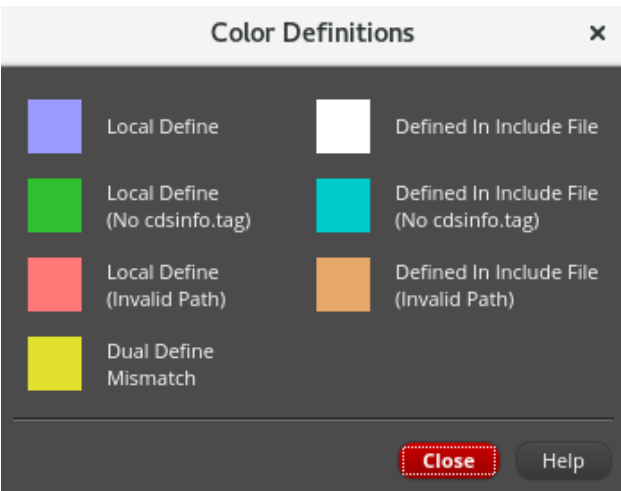


- b. Click **Close** in the text box.

Viewing the Color Definitions Information

1. In the Library Path Editor form, choose **View – Color Legend**.

The Color Definitions form opens.



The colors designate the ways the library paths have been resolved. Red, yellow, and orange are paths that have problems and need attention.

- a. Close the Color Definitions form.

2. In the Library Path Editor form, select **Edit** and deselect **Exclusive Lock**.
If you made any changes, you could save the changes in the *cds.lib* file.
3. From the Library Path Editor form, choose **File – Exit**.
4. Close the text file without saving and keep the Virtuoso session running.

Summary

In this lab, you learned how to

- ◆ View the status and where to make edits to the *cds.lib* file with the Library Path Editor



Lab 2-3 Using the Library Manager

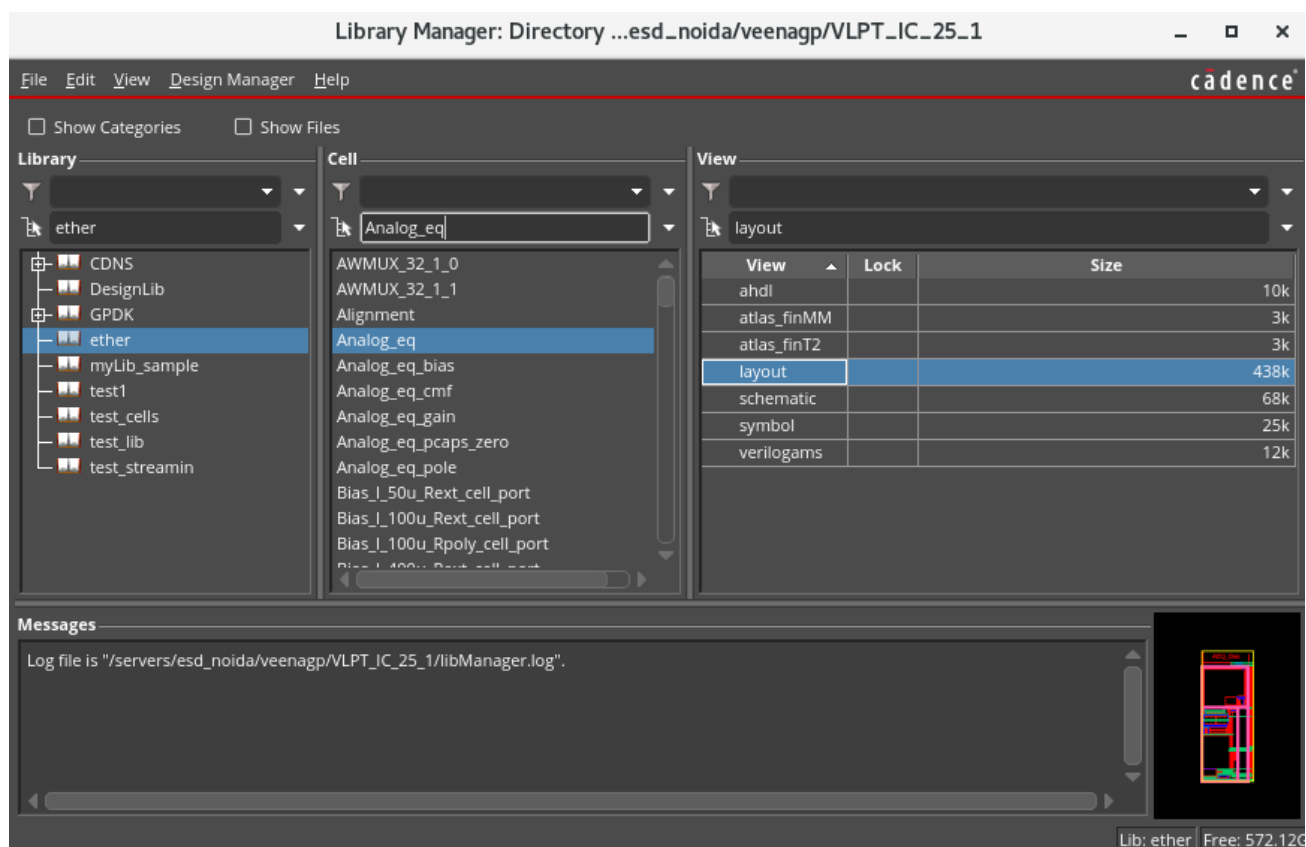
Objective: To create, add, copy, delete, and organize the cells and libraries.

With the Library Manager, you can open a design and edit the information in your library. You can use the Library Manager to create, add, copy, delete, and organize the cells and libraries you use in a design project.

Invoking the Library Manager

1. To open the Library Manager, choose **Tools – Library Manager** in the CIW.

The Library Manager window appears.



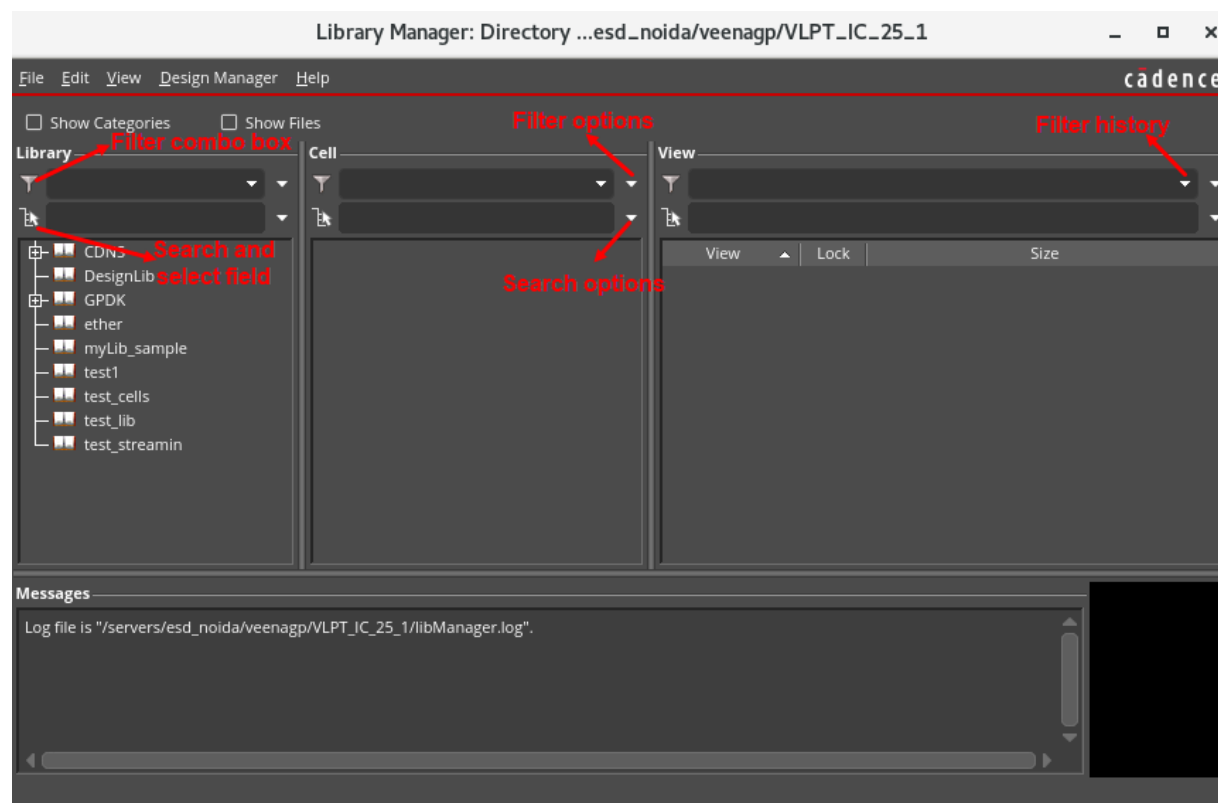
2. Resize the Library Manager window so that you can view all fields clearly.

Using the Filter and Search Options

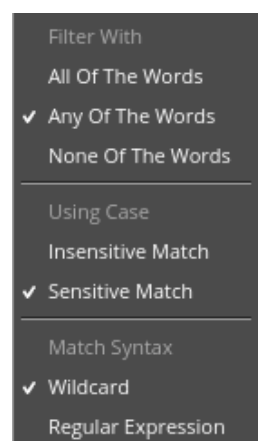
In the Library Manager window, the *Filter* combo box and the *Search and select* fields are provided at the bottom of the *Library*, *Cell*, and *View* list boxes.

Use the *Filter* combo box to specify a filter pattern and accordingly display only a subset of libraries, cells, or views.

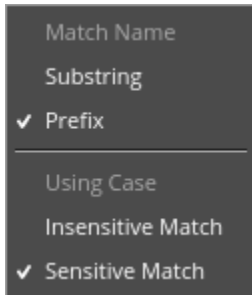
The *Search and select* field lets you search and select a particular library, cell, or view by specifying a search string.



You can set the filter criteria by selecting the required options from the *Filter options* menu, as shown here:



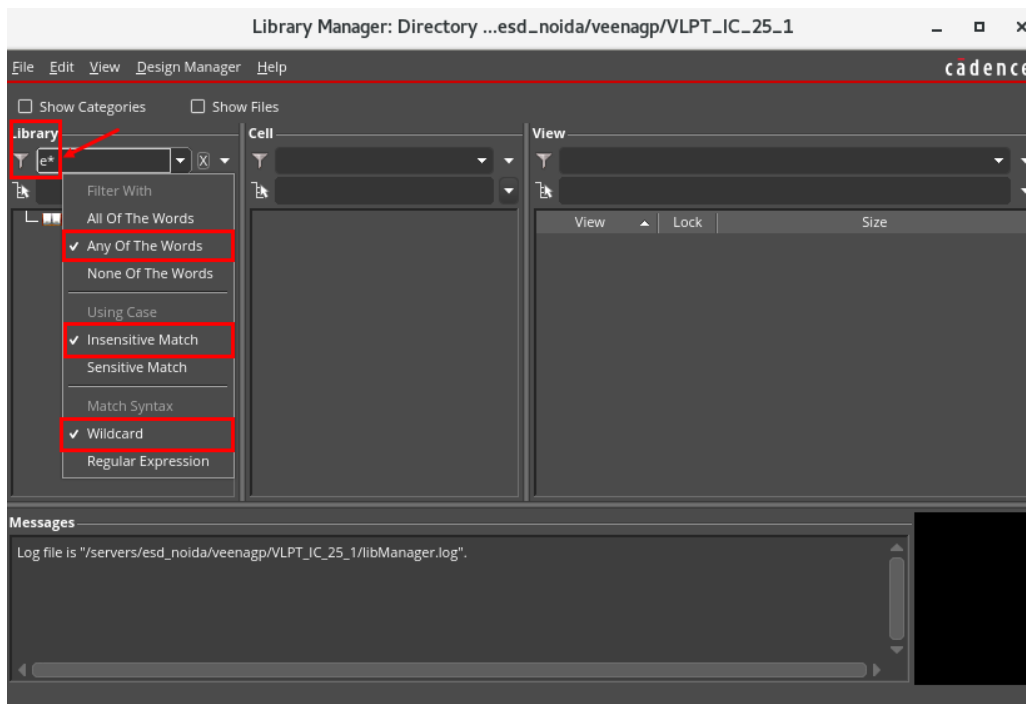
You can set the search criteria by selecting the required options from the *Search options* menu, as shown here:



Using the Filter Combo Box to View Information

To learn more about the cells that are in the design library, use the *Filter* combo box option to find specific cell information.

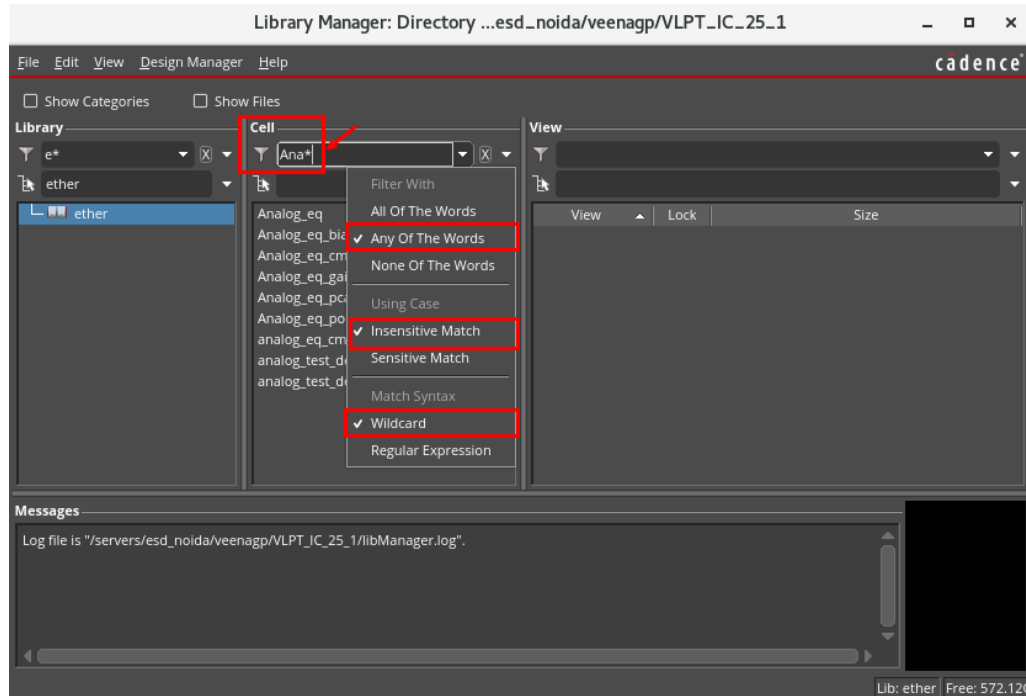
1. In the Library Manager, in the *Filter* combo box, enter the values, as following:
 - a. In the Library *Filter* combo box, enter **e*** and have the *Filter options*, as shown here:



The Library *ether* matching with the *Filter options* is filtered.

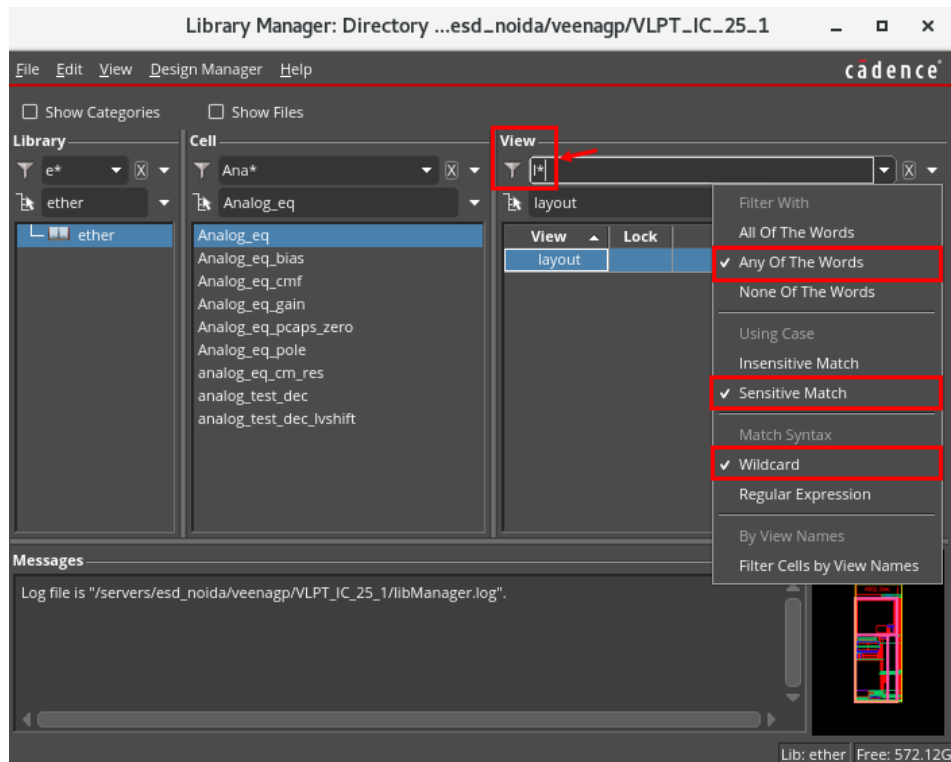
- b. Click the **ether** Library.

- c. In the *Cell Filter* combo box, enter **Ana*** and have the *Filter options*, as shown here:



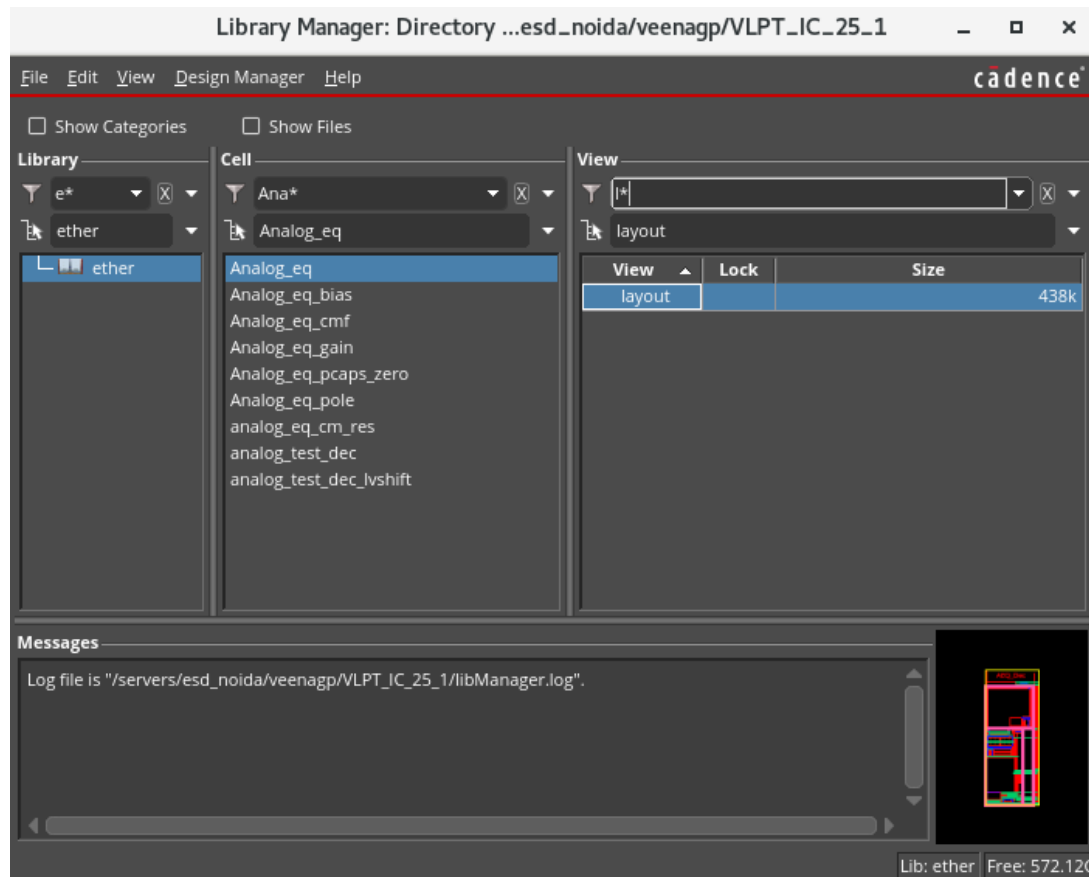
All the cells matching with the *Filter options* are filtered.

- d. Click the **Analog_eq** cell to expand the views.
- e. In the *View Filter* combo box, enter **l*** and have the *Filter options*, as shown here:



The layout view matching with the *Filter options* is filtered.

You can see that the *ether/Analog_eq/layout* cellview is filtered, as shown here:

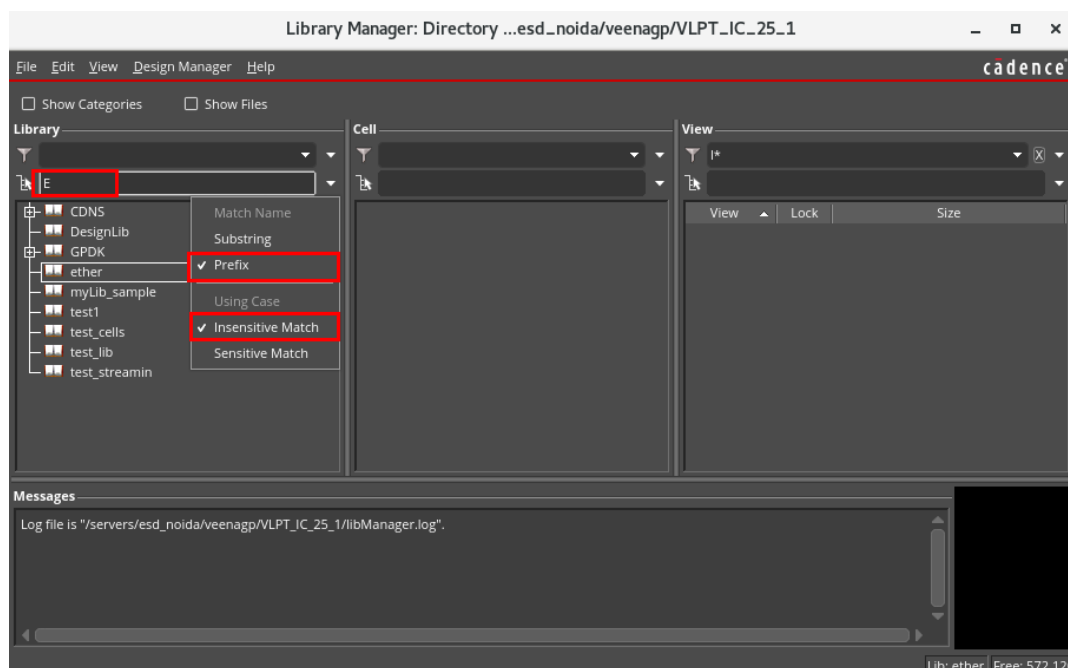


- f. Remove the texts from the Library, Cell, and View *Filter* combo boxes to remove the filter.

Using the Search Option

To search for a specific library, cell, or view, specify a text string in the respective *Search and select* field.

1. In the Library Manager, in the *Search and select* field, enter the values, as following:
 - a. In the Library *Search and select* field, enter **E** and have the *Search options*, as shown here:



Note: Even though **E** (capital) is specified in the *Search and select* field, since the Insensitive Match option is selected in the Using Case section, the *ether* library is searched.

- b. Use the Cell and View fields and explore the Search option as time permits.

Viewing the Contents of the *ether* Library with the Show Categories Option

1. In the Library Manager, click the *ether* library and select the **Show Categories** option.

Notice that the *Category* field is updated. Categories are a way of grouping cells together. If you have many cells in your library, you may want to put cells that are of a similar type in a category.

 - a. In the *Category* field, click **Uncategorized**.

All cells that are not in categories are shown.

- b. In the *Category* field, click **Everything**.
All cells are shown even if they are in a category.
- c. In the *Category* field, click **ADC**.
All *adc* cells are shown.
Notice the cell slider changes size as each of these categories is selected.

Creating a New Category

You create a category in which the **AWMUX** cells are placed.

1. In the Library Manager, choose **Edit – Categories – New**.

The New Category form appears. Enter the values, as shown here:

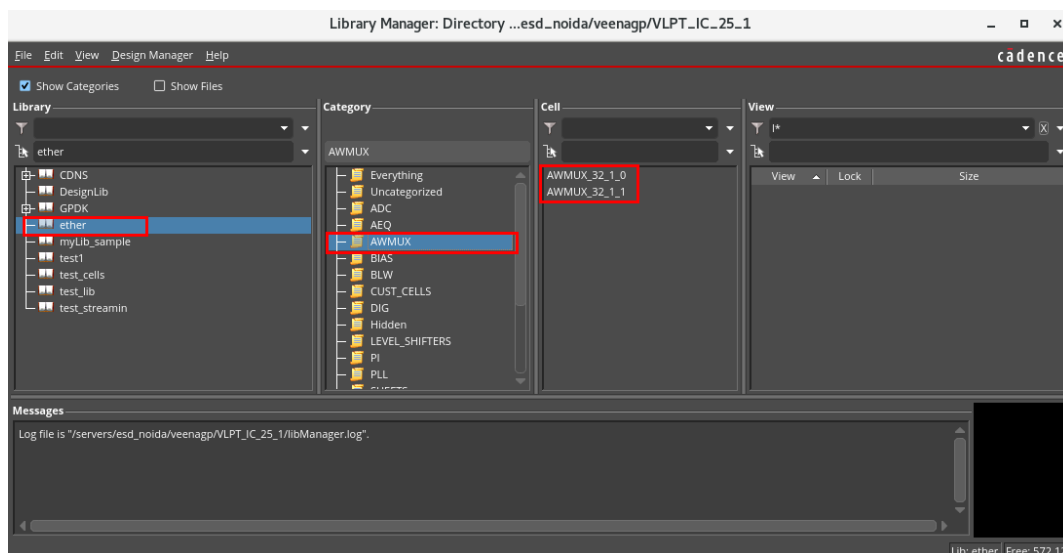
The screenshot shows the 'New Category' dialog box with the following details:

- Category Name:** AWMUX
- Cells:**
 - Not In Category:** 100tx_div2, AEQ_Dec, AEQ_lvlsht_1p0_to_2p5, Alignment, Analog_eq
 - In Category:** AWMUX_32_1_0, AWMUX_32_1_1
 - Buttons:** A right arrow button is highlighted between the two lists.
 - Input fields:** Below each list is a text field with an asterisk (*) and a 'Select' button. The 'Not In Category' text field contains 'AW*'.
- Sub-Categories:**
 - Not In Category:** ADC, AEQ, AWMUX, BIAS, BLW, CUST_CELLS
 - In Category:** (Empty)
 - Buttons:** Similar right and left arrow buttons are present.
 - Input fields:** Similar text fields with asterisks and 'Select' buttons are present.
- Bottom Buttons:** OK, Apply, Cancel, Help. The 'OK' button is highlighted.

- a. In the *Category Name* field, enter the new category name **AWMUX**.
- b. In the *Not In Category* **Select** field, enter **AW***.
- c. In the *Not In Category* field, select the **AWMUX** cells.
- d. Click the right arrow to move the **AWMUX** cell names to the *In Category* field.
- e. Click **OK** in the form.

Note: If you get pop-up that Category “AWMUX” already exists, then choose another name, for example, “AWMUXNEW”.

- In the Library Manager, choose **View – Show Categories** if it is not already selected. Notice the new category containing the **AWMUX** cells.



- In the Library Manager form, deselect the **Show Categories** option.

Creating a New Library

You create a new library, for example, **myLib**.

- In the Library Manager form, choose **File – New – Library**.

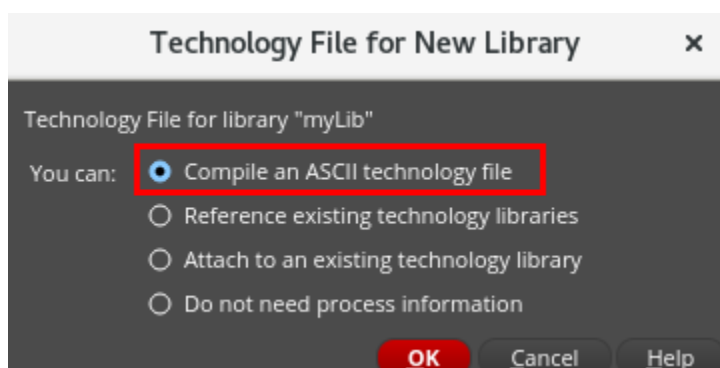
The New Library form appears.

- In the *Name* field of the New Library form, enter **myLib**.

Note: If the name already exists, enter another name, say, for example, **myLib1**.

- Click **OK**.

The Technology File for New Library form opens, as shown here:

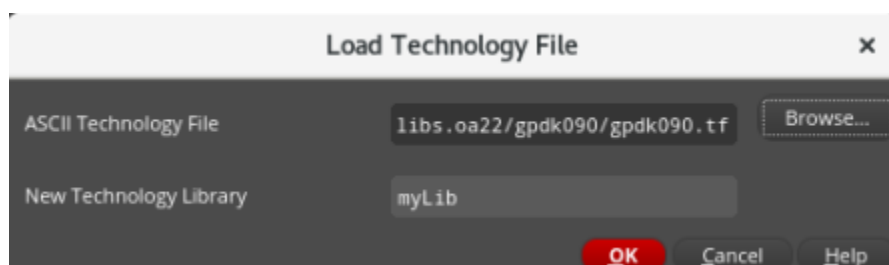


When creating libraries for physical designs, you must provide a technology file (techfile) describing the technology.

4. Select **Compile an ASCII technology file** (the default).

5. Click **OK**.

The Load Technology File form opens, as shown here:



6. Enter (or browse to) **libs.oa22/gpdk090/gpdk090.tf**.

7. Click **OK**.

8. Close the pop-up form showing the technology file loaded successfully.

The new library **myLib** is added to the Library Manager.

9. In the Library Manager, click the **myLib** library, and view the list of cells.

These are the symbolic devices created from the technology file information.

Note: A sample library **myLib_sample** is available for reference.

Copying a Cell

Use the Library Manager to copy the *Bias_bdgp_amp* cell from the *ether* library to *myLib*, keeping the name *Bias_bdgp_amp*.

1. In the Library Manager, deselect the **Show Categories** option, if it is not already done.
2. **Middle-click** the *Bias_bdgp_amp* cell from the **ether** library, and select the **Copy** option from the pop-up menu.

The Copy Cell form opens.

The screenshot shows the 'Copy Cell' dialog box with the following settings:

- From:** Library: ether, Cell: Bias_bdgp_amp
- To:** Library: myLib, Cell: Bias_bdgp_amp
- Options:**
 - ☒ Copy Hierarchical
 - ☒ Skip Libraries
 - Library list: CDNS DesignLib GPDK MyLib analogLib basic cdsDefTechLib gpdK090 gscLib090 myLib myLib_sample sample test1 test_cells test_lib test_streamin
 - ☐ Exact Hierarchy
 - ☒ Views To Copy
 - ☒ All Views
 - Names: layout layout_backup layout_complete schematic symbol
 - Types: None
- ☐ Update Instances: Of Entire Library
- Database Integrity:**
 - ☐ Re-reference customViaDefs
 - ☐ Check existence in technology database
 - ☐ Update technology database
- ☐ Add To Category: Cells *
- Buttons:** OK (highlighted), Cancel, Apply, Help

3. Set the **To Library** name to **myLib**, as shown here.
4. Select the **Skip Libraries** option to keep the referenced cells from being copied out of their source libraries.
5. Keep the **Views To Copy** option, as shown here.

Note: The Library Manager *Copy* functionality now provides a way to select view types to copy. You can specify view names or view types or a combination of both in copy form.

6. Click **OK** to begin the copy.

A form that shows any files that need attention opens. The *data.dm* file needs permission to be overwritten.

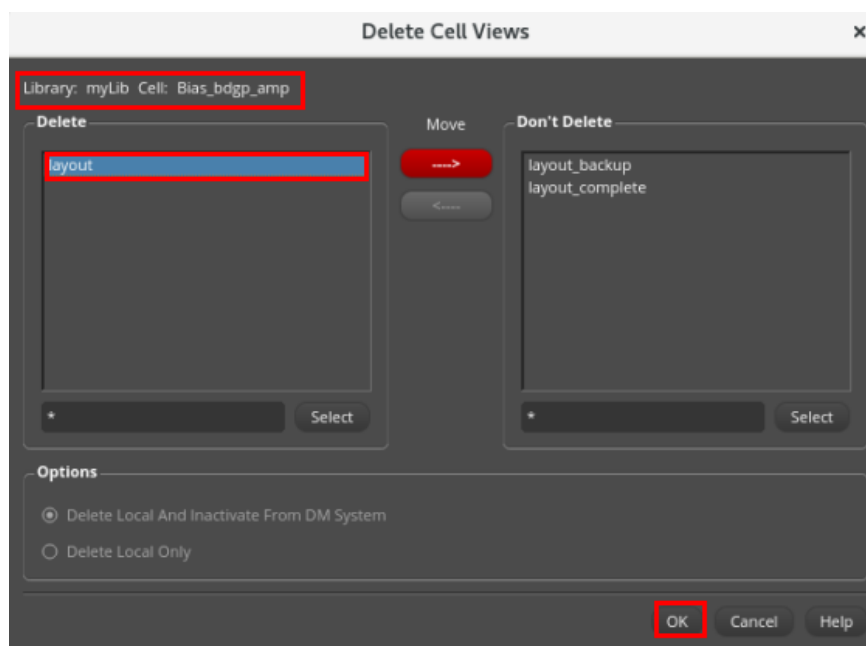
- a. In the new form, click **Overwrite All**.
- b. Click **OK**. Observe the copied cell in the Library Manager.

Deleting a Layout/Cell View

You delete a layout view from the library *myLib*.

1. In the Library Manager, select the **Bias_bdgp_amp** cell from the library *myLib*.
2. Click the **Bias_bdgp_amp layout** view.
3. Choose **Edit – Delete** from the Library Manager pull-down menu.

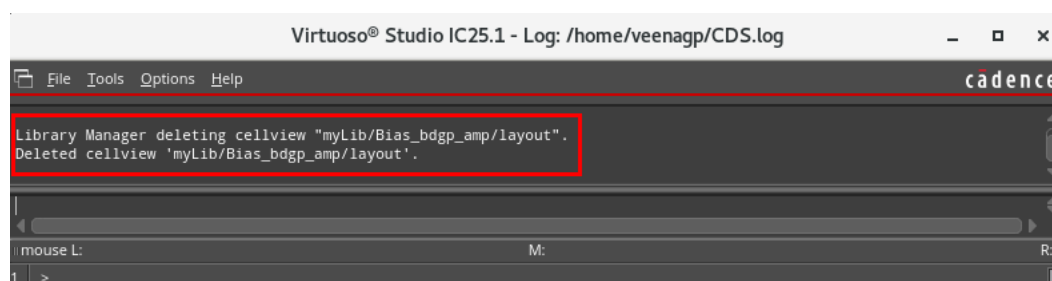
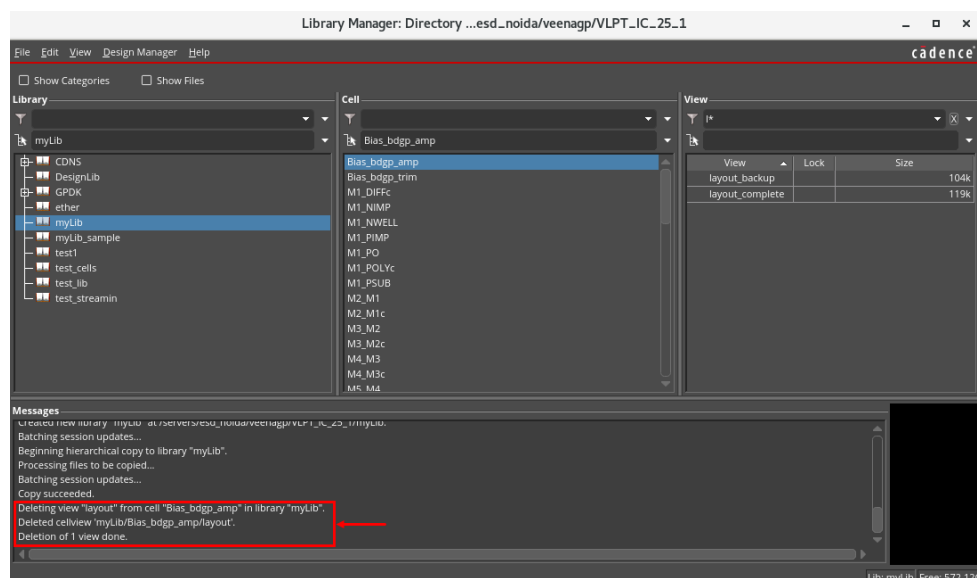
The Delete Cell Views form opens. The *layout* view is in the *Delete* field.



4. Click **OK**.

- Click **Yes** to confirm the deletion.

There are messages in the Library Manager output field, and also in the CIW output pane confirming the action, as shown here:



You can see in the Library Manager that the `myLib/Bias_bdgp_amp/layout` cellview is deleted.

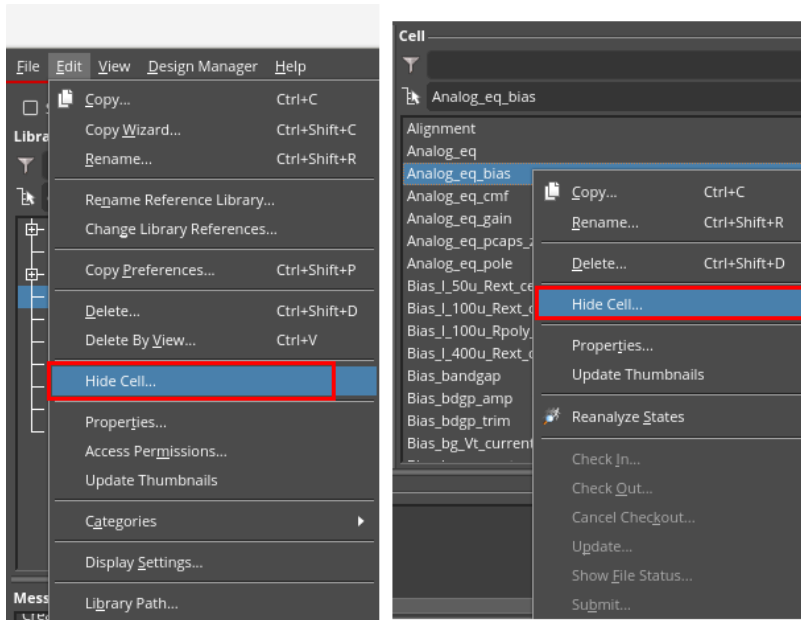
Hiding a Cell

You can hide a cell from the *Cell* list box of the Library Manager window.

You hide the *Analog_eq_bias* cell from the library *ether*.

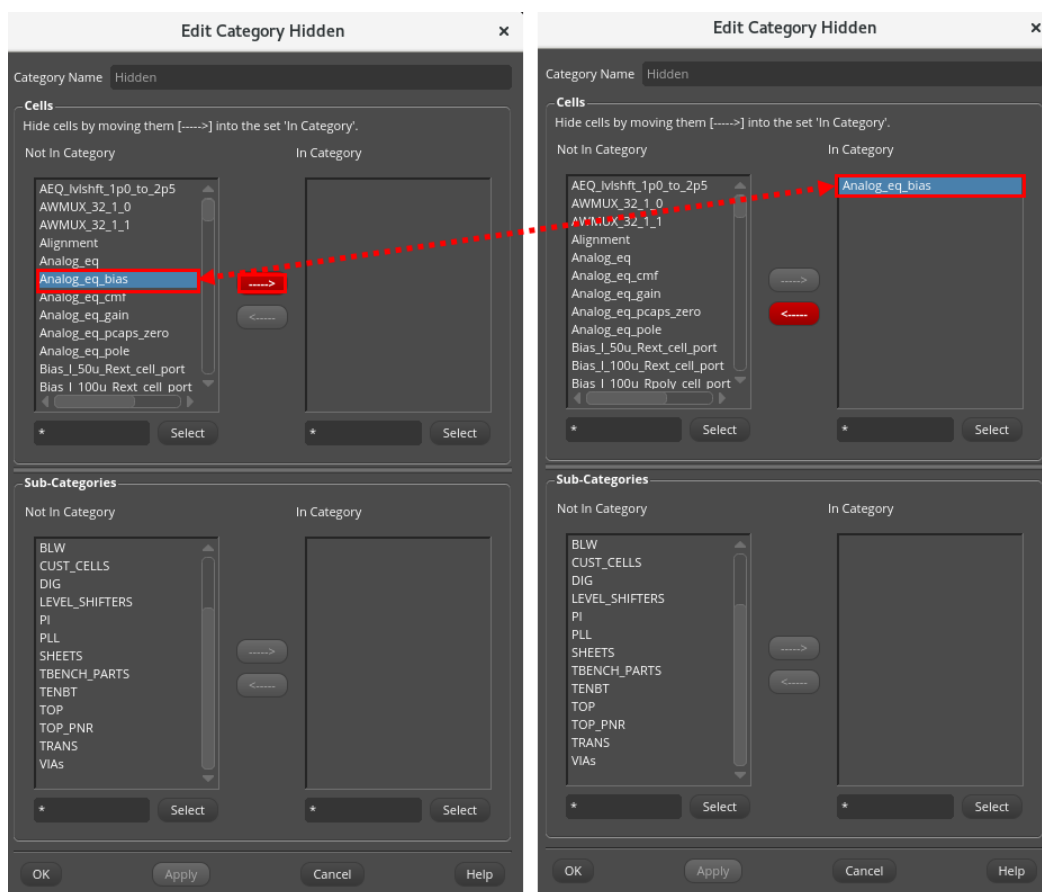
- In the Library Manager, click the **ether** library and select the **Show Categories** option.
- In the Library Manager, select the **Analog_eq_bias** cell from the library *ether*.

3. Choose **Edit – Hide Cell** from the Library Manager pull-down menu or **right-click** the cell and select the **Hide Cell** option from the context-sensitive menu.



The Edit Category Hidden window is displayed.

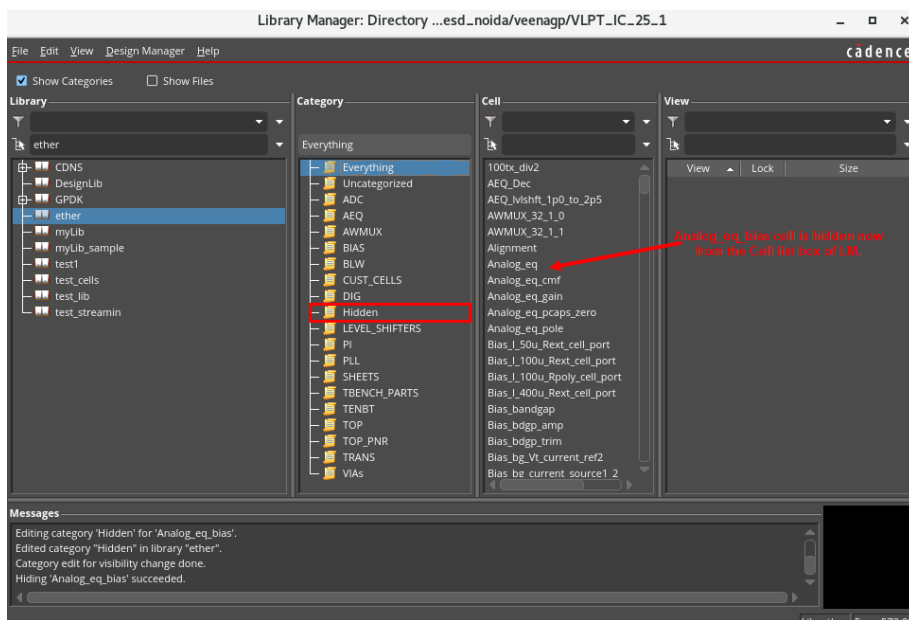
4. In the Cells section of this window, move the cell from the *Not in Category* list box to the *In Category* list box using the **right** arrow button, as shown here:



5. Click **OK**.

The new *Hidden* category is created in the *Category* list box of the Library Manager window, as shown here.

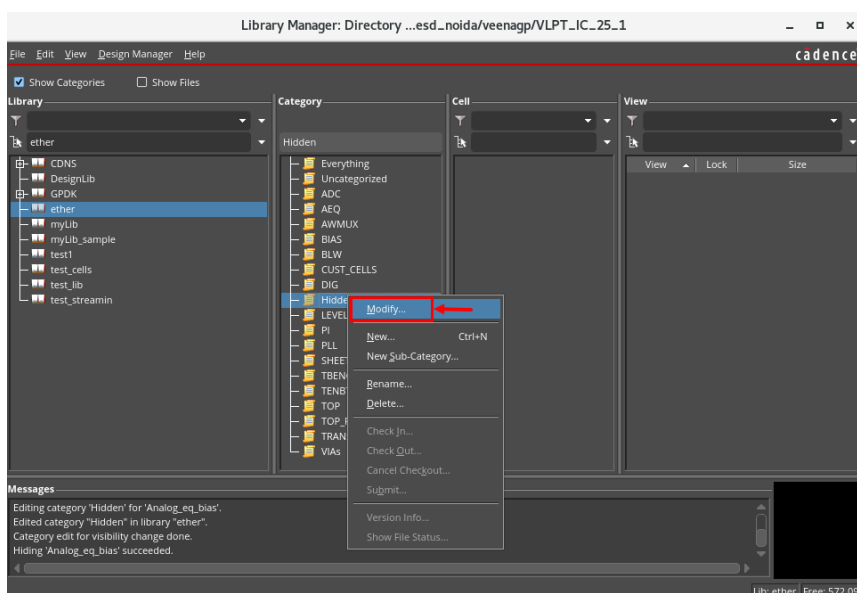
6. You can observe that the *Analog_eq_bias* cell is hidden now from the library *ether* in Library Manager.



Showing the Hidden Cell

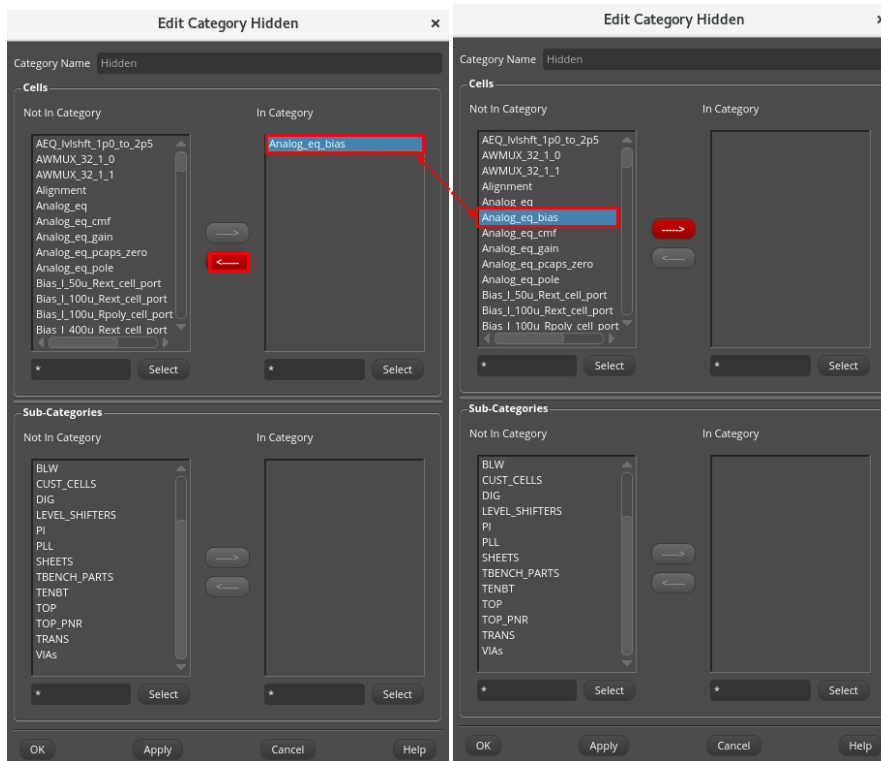
If you have the required permissions, you can show the hidden cells in the Library Manager.

1. **Right-click** the Hidden category and select **Modify** from the context-sensitive menu in the Library Manager.

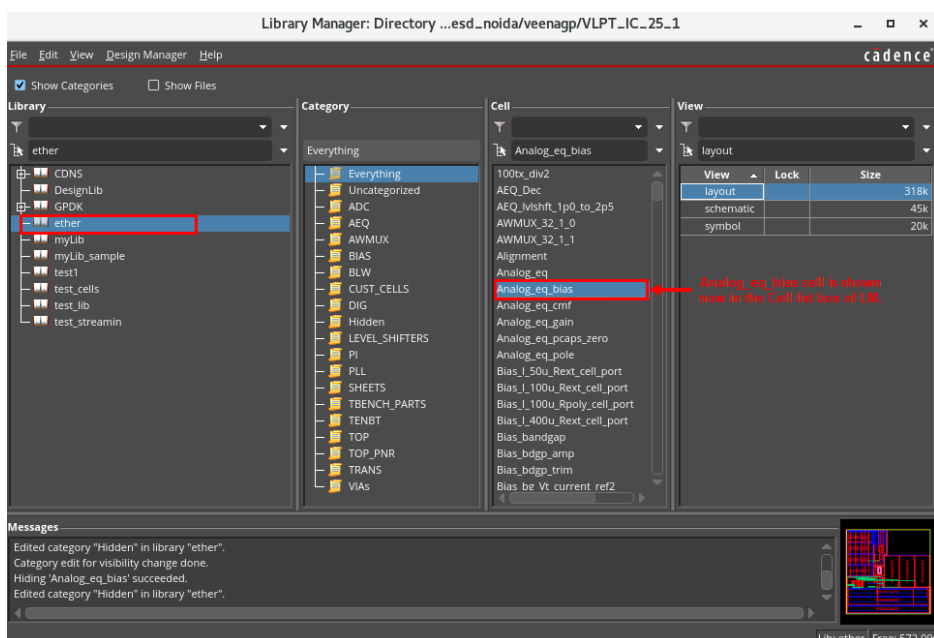


The Edit Category Hidden window is displayed.

2. In the Cells section, move the cell from the *In Category* list box to the *Not In Category* list box using the **left** arrow button.



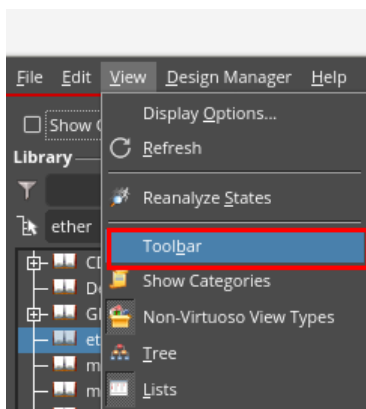
3. Click **OK**.
4. You can observe that the hidden cell *Analog_eq_bias* is shown in the library *ether* in the Library Manager.



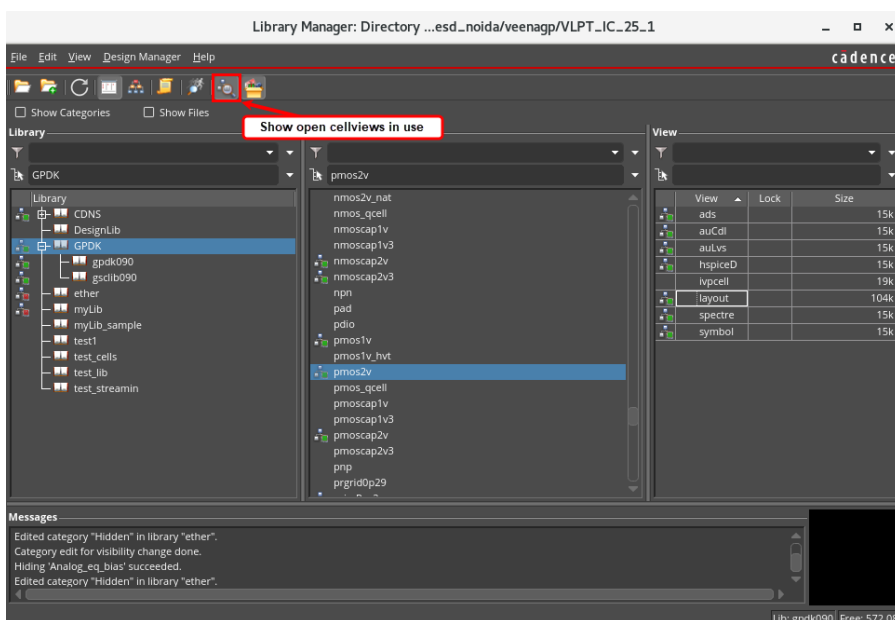
Viewing the Current Cellview Status

You can view the current cellview status by clicking the *Show open cellviews in use* button in the Library Manager toolbar.

1. In the Library Manager form, deselect the **Show Categories** option.
2. In the library manager, click on the **View – Toolbar** to view the toolbars of the library manager.



3. Click the **Show open cellviews in use** button in the Library Manager toolbar, as shown here:



The lists in the window are reloaded with icons next to names. The icons indicate whether a library, cell, or view is opened in read mode (green icon ) or edit mode (red icon ).

4. Click the toggle button *Show open cellviews in use* to hide the status icon indicators.

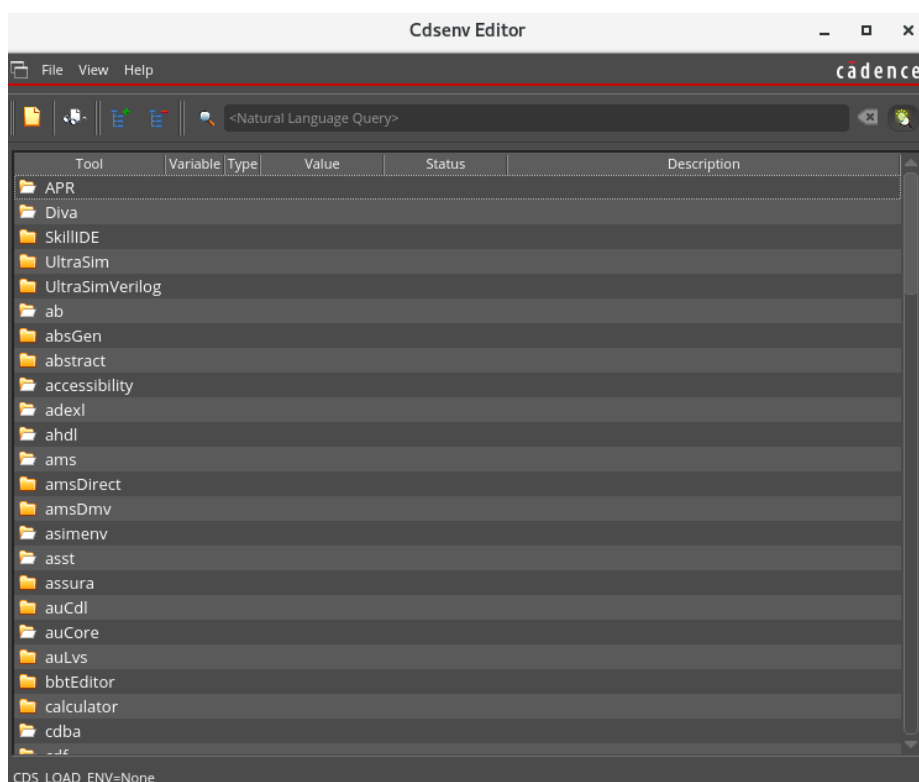
Using the Cdsenv Editor (Optional Exercise)

Support to edit *.cdsenv* files from GUI is available from IC618 version.

The Cdsenv Editor enables you to search for environment variables across multiple *.cdsenv* files belonging to different tools. The editor allows you to change values of environment variables and save the changes to the specified file.

1. From the CIW, choose **Options – Cdsenv Editor**.

The Cdsenv Editor form opens.



Note: The tools with the open folder icon have been loaded into Virtuoso, and the tools depicted with a closed folder icon are tools that have not been loaded.

2. Explore editing the environment variables in *Cdsenv Editor* as time permits and close the *Cdsenv Editor*.
3. Leave the Virtuoso session open for the next lab.

Summary

In this lab, you learned how to

- ◆ Invoke the Library Manager
- ◆ Filter cells in the design library
- ◆ Search cells in the design library
- ◆ Create a category
- ◆ Create a library
- ◆ Use the Library Manager to copy a cell
- ◆ Delete a cellview from a library
- ◆ Hide a cell
- ◆ Show the hidden cell
- ◆ View the current cellview status
- ◆ Use the Cdsenv Editor



Lab 2-4 Setting User Preferences

Objective: To set the window controls.

In this lab, you explore how to set User Preferences using the CIW window.

Setting the Window Options

1. From the CIW, choose **Options – User Preferences**.
The User Preferences form appears.
2. In the CIW Controls section at the bottom of the User Preferences form, follow these steps:

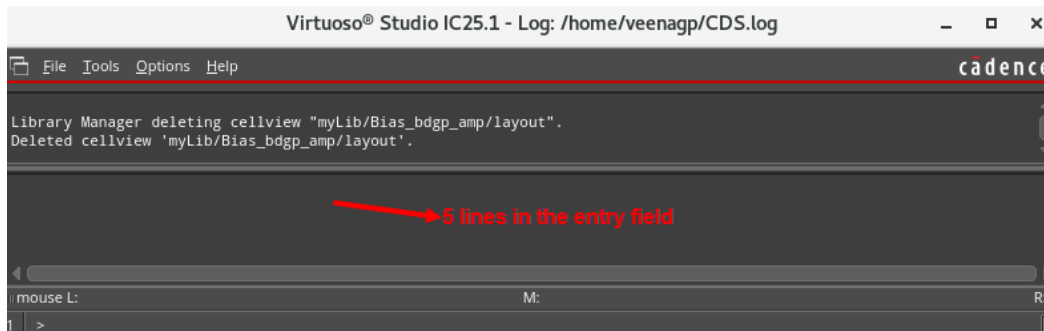
The screenshot shows the 'User Preferences' dialog box with the following sections and settings:

- Appearance**
 - Color Theme: Dark Gray
- Window Controls**
 - Mouse Prompts: Bottom
 - Side Dock Tabs: Bottom
 - Focus Policy: CanvasDelay
 - Focus Delay (ms): 200
 - Default Editor Background Color: [Black swatch]
 - ☐ Create New Window When Descending
 - ☐ Place Manually
 - ☐ Scroll Bars
 - ☒ Tear-Off Menus
 - ☒ Menu Shortcuts
 - ☒ Notifications [Options...]
- Command Controls**
 - ☐ Infix (No Click Necessary for First Point)
 - ☐ Options Displayed When Commands Start
 - Web Browser: firefox
 - Undo Limit: 128
 - Nest Limit: 5
 - Double Click Time (ms): 200
 - Beep Volume: 0
- CIW Controls** (highlighted with a red box and an arrow pointing to the Input Area Lines field)
 - Input Area Lines: 5
 - Output Wrap Mode: Word or Anywhere
 - ☐ Retain Unique Commands
 - Output Area Lines: 1000
 - ☒ Syntax Highlighting
 - ☒ History In Place
 - Tab Stop: 4
 - ☒ Enter Key Executes Command
 - Raise CIW on: ☐ Warning ☐ Error
- License Activity**
 - ☒ Display Indicators
 - Activity: CIW+Session

Buttons at the bottom: OK, Cancel, Defaults, Apply, Help.

- a. Set the *Input Area Lines* field to 5.

- b. Click **OK** in the form.



The CIW window changes to have five lines in the entry field available for SKILL code editing in the entry field.

Changing the Window Controls

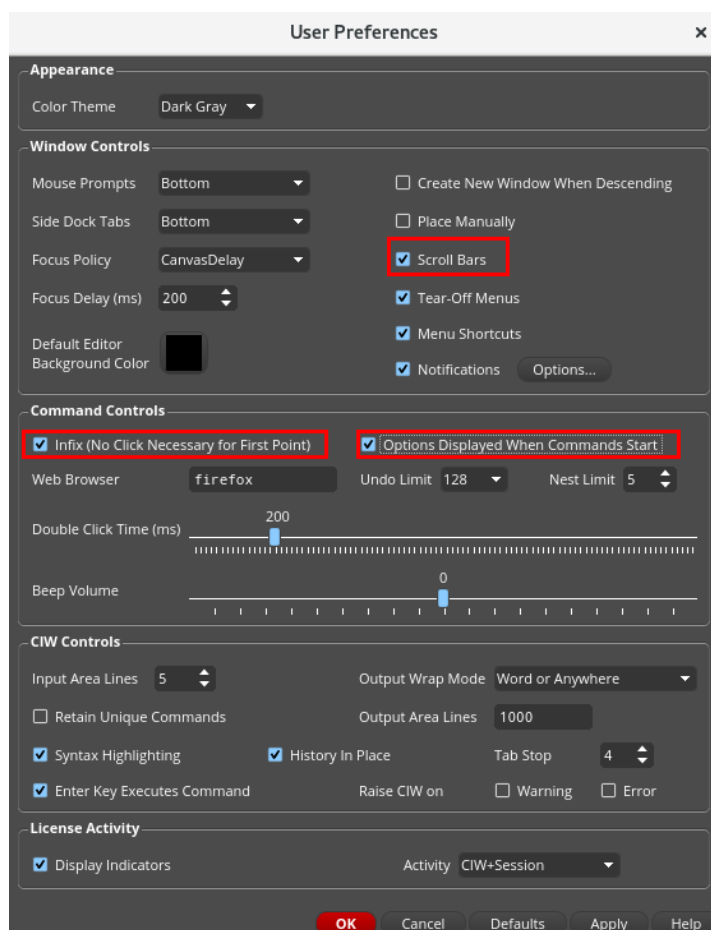
- From the CIW, choose **File – Open**. Enter the following values in the Open File form:

Library	ether
Cell Name	adc_cascode_opamp
View Name	layout

- Click **OK**.

The *ether/adc_cascode_opamp/schematic* and *ether/adc_cascode_opamp/layout* cellviews open in EXL mode. Minimize the schematic view and maximize the layout view.

- From the CIW, choose **Options – User Preferences** to open the User Preferences form.
- In the User Preferences form, do the following:
 - Select **Scroll Bars**.
 - Select **Infix** under Command Controls.

c. Select **Options Displayed When Commands Start**.d. Click **Apply**.

e. Close the layout design window.

Note: You must close the design windows and open them again, for the changes to take effect.

- From the schematic window, choose **Launch – Layout EXL** to open the *ether/adc_cascode_opamp/layout* cellview in EXL mode again.

The design window is now updated with the preferences you have chosen.

Notice the scroll bars are available now in the layout canvas.

- Click in the layout cellview window to make it active.
- Place your cursor near the left side of the *adc_cascode_opamp* instance at approximately **X10 Y10**. This is the beginning point of a zoom-in rectangle.
Because *Infix* is enabled, when you use a bindkey, the location of the cursor when you press the bindkey is the first coordinate or key click of the command.

8. Press the bindkey **Z** to start the **Zoom To Area** command.

Notice the cursor now has the first point of the zoom-to-area window, and a selection window rubber band is ready for the second coordinate of the zoom-to-area window.

9. Click the coordinates **30:30** to complete the **Zoom To Area** command.

The **Zoom To Area** command is completed using the bindkey **Z** and **Infix** option to establish the first coordinate.

10. Select the bindkey **F** to fit the data extents, and then use the **Zoom To Area** command to experiment again, with **Infix** option.

11. Choose **Edit – Move** from the cellview pull-down menu or press the bindkey **M**.

Notice the Move form opens because you selected Options Displayed (when starting a command) in the User Preferences form.

12. **Cancel** the Move command.

13. In the User Preferences form, click the **Defaults** button to get the default user preferences, and click **OK** to close the form.

14. Close the schematic and layout cellviews by choosing **File – Close**.

15. Leave the Virtuoso session running for the next lab.

Summary

In this lab, you learned how to

- ◆ Change the number of lines in the input area
- ◆ Change the window controls
- ◆ Enable/Disable scroll bars
- ◆ Use Infix to save key clicks
- ◆ Enable the command options form when starting a command



Lab 2-5 Using History and Bookmarks

Objective: To display the available recently used cellviews for ease of opening frequently used cellviews and adding bookmarks.

In this lab, you learn how to use the History and Bookmarks options. The Virtuoso software keeps a running history of the cellviews you have used recently.

1. From the CIW, choose **File – Open**. Enter the following values in the form:

Library	ether
Cell Name	AEQ_Dec
View Name	layout

2. Click **OK**.

The *ether/AEQ_Dec/schematic* and *ether/AEQ_Dec/layout* cellviews open in EXL mode.

3. Close the cellviews now.

Using the History

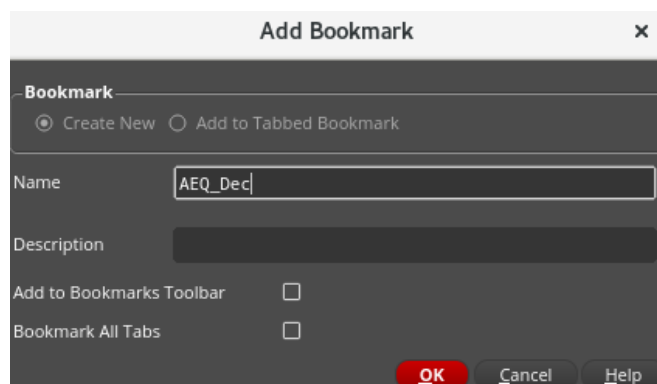
1. In the CIW, choose the **File** menu.

A form appears, showing a list of recent cellviews you have opened. This is the history showing the last 10 cellviews opened.

2. Click the *ether/AEQ_Dec/layout* in the form to open the cellview for editing.

Adding the Bookmarks

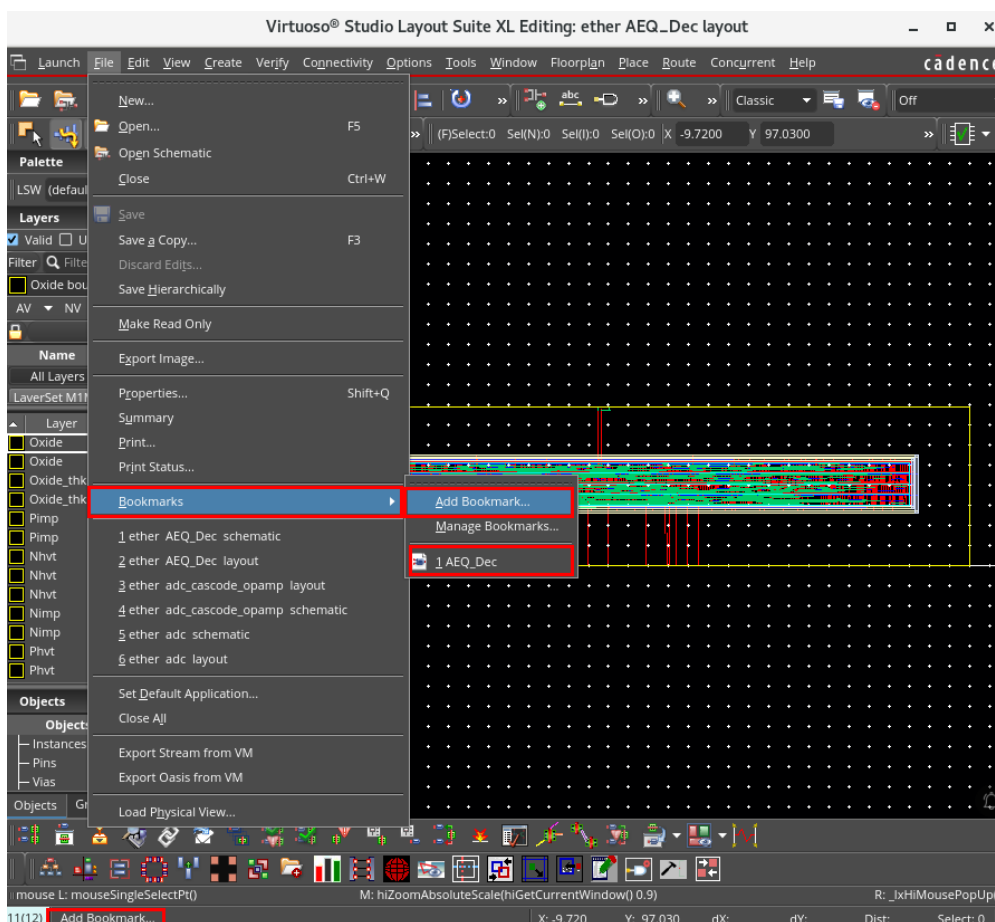
1. In the layout canvas, choose **File – Bookmarks – Add Bookmark**.



- Specify a location and description for the bookmark as per your requirement.
If the cellview has multiple tabs, you can add all tabs to the bookmark.

- Click **OK**.

Note: Refer to the graphic here for the sequence of steps followed.



- Close both the cellviews (no need to save).
- Explore creating more bookmarks as time permits.
- In the CIW, choose **File – Bookmarks**.
- Select the bookmarked entry.
The cellview opens for editing.
This bookmark is persistent and is available in subsequent Virtuoso sessions.
- Close all the cellviews.

9. Keep the Virtuoso session running for the next lab.

Summary

In this lab, you learned how to

- ◆ Use History
- ◆ Create Bookmarks



Lab 2-6 Using World View

Objective: To zoom and pan with the World View Assistant.

In this lab, you learn how to zoom and pan using the World View Assistant.

1. From the CIW, choose **File – Open**. Enter the following values in the form:

Library	ether
Cell Name	top
View Name	layout

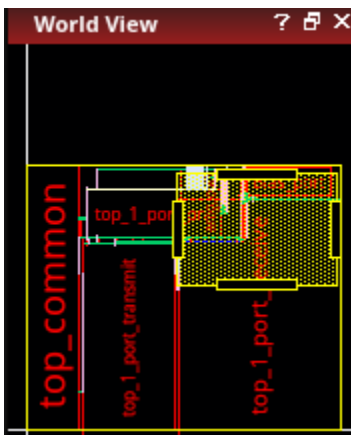
2. Click **OK**.

The *ether/top/schematic* and *ether/top/layout* cellviews open in XL mode, then switch to EXL mode by clicking the **Launch – Layout EXL**. Minimize the schematic view and maximize the layout view.

Using the World View Assistant

1. In the layout canvas, choose **Window – Assistants – World View**.
The World View Assistant opens and docks in the design window.
2. Undock the **World View** Assistant and move it outside the layout canvas.
3. **Right**-click and drag in the layout canvas to zoom in to the top-right area of the canvas.

Observe that the World View now shows the same area as highlighted by the yellow highlight window.



4. Press the bindkey **Shift+F** to see all the levels of the hierarchy in the layout canvas.
5. Select the **yellow zoom and pan** highlight box.
6. Move the box to the lower-left of the **World View** window.
The canvas pans to the new area described by the box.
7. Use the handles on the box to stretch the box larger.
The canvas zooms out.
8. Use the handles on the box to stretch the box smaller.
The canvas zooms in.
9. On the canvas, press the **F** bindkey to fit the canvas.
The World View changes to show the yellow highlight covering the whole cellview.
10. Leave the cellview open for the next lab.

Summary

In this lab, you learned how to

- ◆ Use the World View Assistant



Lab 2-7 Floating the Palettes Away from Cellviews and Saving Workspaces

Objective: To save a nonstandard workspace.

In this lab, you learn how to save the undocked configuration and recall with the Workspace cyclic field.

Saving the Workspace

1. In the *ether/top/layout* cellview, click the **Float** icon at the **upper-right** of the **Palette**.

The Palette disengages from the cellview.

2. Move the **Palette** just outside the cellview.
3. Move the **World View** below the Palette.
4. Click the **Save Workspace** icon in the layout window.



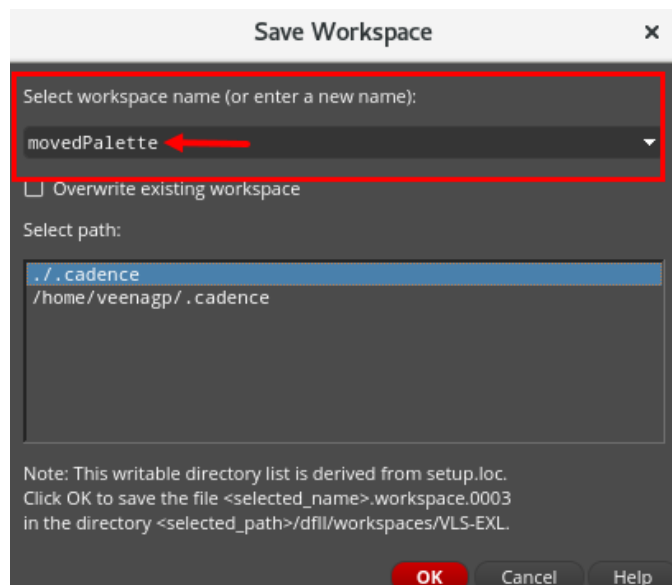
The Save Workspace form opens.

5. Enter a new name for the workspace, say:
`movedPalette`

Note: If the name already exists, enter another name.

6. Click **OK**.

The workspace definition is stored in the *./cadence* directory.



The *Select workspace name* field shows **movedPalette**.

7. Switch between other workspaces to come back to *movedPalette* to see the feature.
8. In the cellview window, choose **File – Close** to close the cellviews.
9. Close the Virtuoso session.
10. In the shell, enter the following command:

```
cd ~
```

This command places you in your home directory.

Summary

In this lab, you learned how to

- ◆ Save undocked configuration and recall with the *Workspace* cyclic field



Module 3: User Interface

Lab 3-1 Using the Layers, Objects, and Grids Assistants

Objective: To use the Layers, Objects and Grids Assistants.

The **Layers**, **Objects** and **Grids** panels are dockable Assistants. You can configure their placement and usage. In this lab, you explore these three panels.

Lab Setup

Follow the steps below to set up the environment and perform the lab:

1. If you continue from the previous lab, move to the sub-heading: [Starting the session](#). Otherwise, proceed to the next step.
2. Update the following line in the /VLPT1_IC_25_1/.cshrc_VLPT1 file with the appropriate valid local tool paths for Virtuoso.

```
setenv CDSHOME /ccstools/cdsind1/Software/IC251Base_lnx86
```

3. Save and close the file.
4. Source the updated class setup file.

```
Source .cshrc_VLPT1
```

Starting the session

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

```
cd ./M03_VLPT1_UsrInterface
```

2. In the terminal window, enter this command:

```
setenv CDS_PALETTE_TYPE MultiAssistant
```

3. Invoke the Virtuoso® session in the same terminal window:

```
virtuoso &
```

4. In the CIW, choose **File – Open**. Enter the following values in the form:

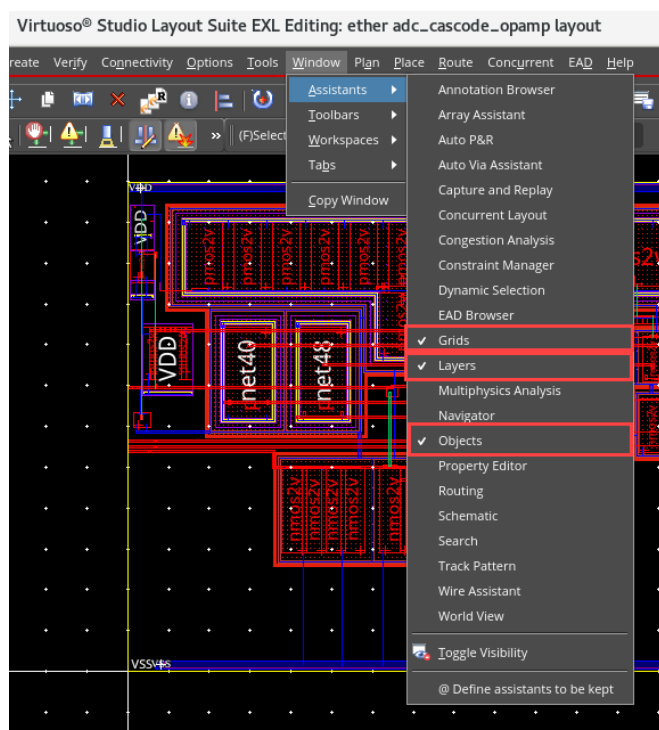
Library	ether
Cell	Analog_eq_bias
View	layout

5. Click **OK**.

The *ether/Analog_eq_bias/schematic* and *ether/Analog_eq_bias/layout* cellviews open in XL mode switch to EXL mode by clicking **launch – Lauout EXL**. Minimize the schematic view and maximize the layout view.

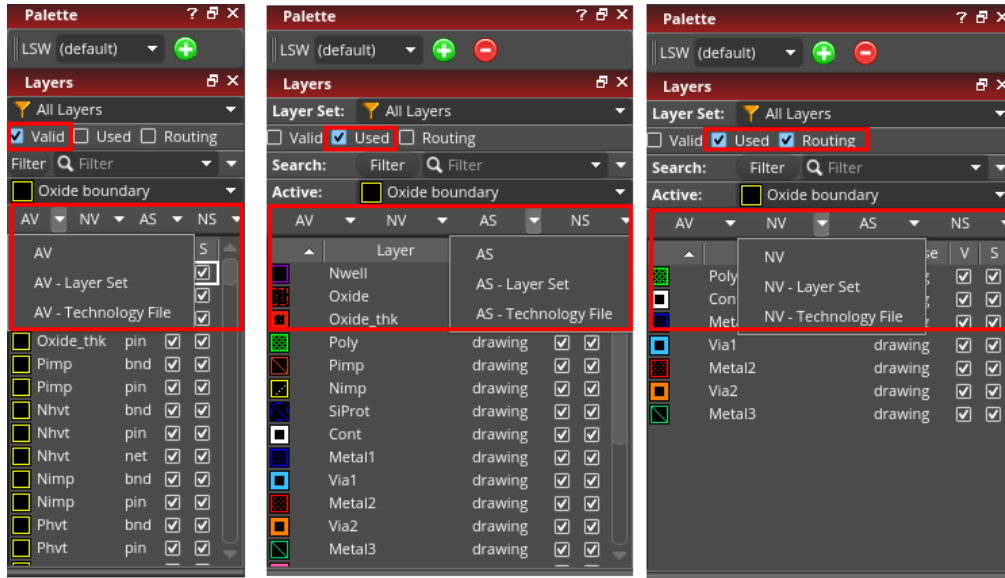
Setting the Current Entry Layer in the Layers Assistant

1. In the layout canvas, from the **Window – Assistants** menu, choose the Palette Assistant, or **right-click** in the layout banner and activate the Palette Assistant from the Assistants menu if not already opened.



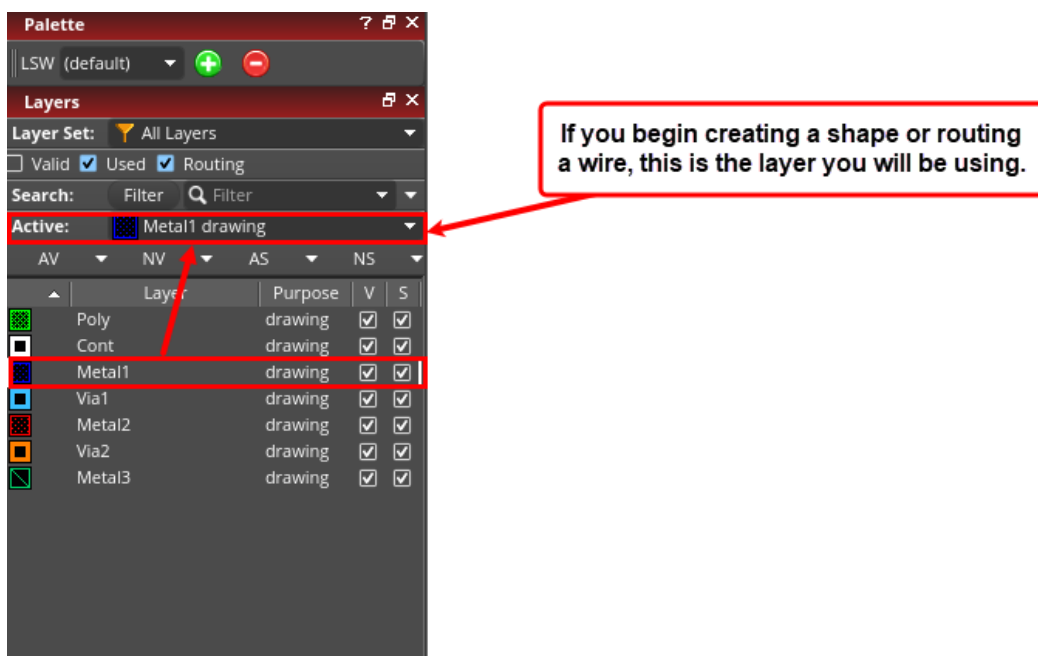
The Layers panel gives you three choices for viewing the layers described in your technology file:

- **Valid** – Displays every layer defined in your technology file, which is defined valid from the techfile.
- **Used** – Displays all layers which currently exist in your design canvas.
- **Used + Routing** – Displays those layers you can route with.



2. Click each option and examine the results.

Note: If you are creating a shape, click on the desired layer in the Layers Assistant to make that layer the active layer. The Active layer is the layer on which you are about to create a shape with.



Making the Layers Visible/Invisible

One of the features of the Layers Assistant is setting visibility.

1. Clear the search field in the Layers Assistant.

This allows the full set of valid layers to be displayed in the Assistant.

2. **Middle**-click the **Metal1** layer in the Layers Assistant.

Note: You can also simply select/deselect the checkboxes for visibility and selectivity in the Assistant.

The current active layer will always be **visible**; however, you can toggle **selectability** in this way.

3. Choose **Metal1** to make it the active layer.

4. Use the **NV (None Visible)** option or uncheck all the other layers in the Layers Assistant.

5. **Left**-click the **AV (All Visible)** option.

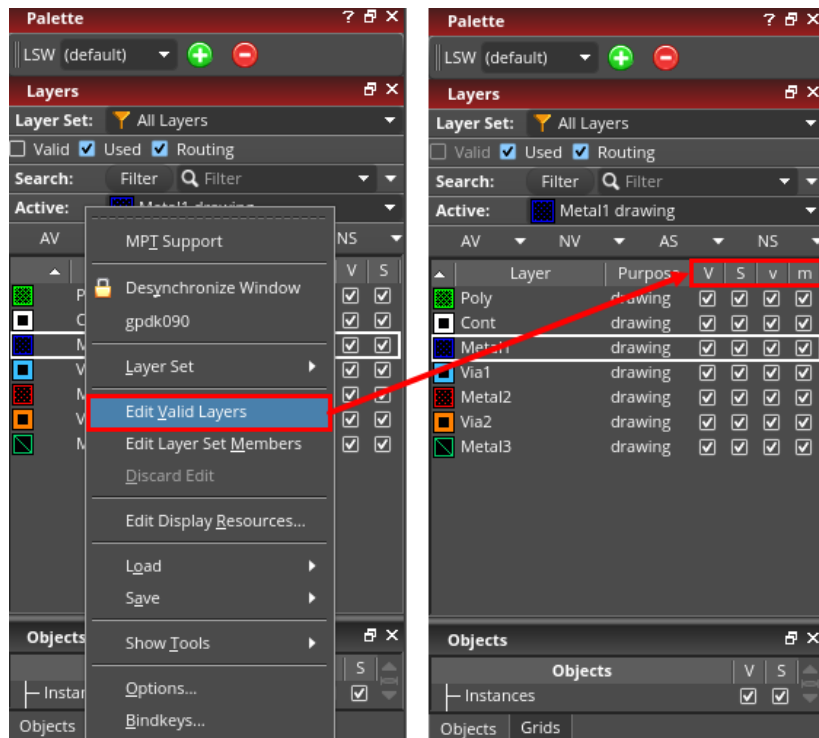
All layers are now visible.

6. Explore the other options in **AV**, **NV**, **AS**, and **NS** as time permits.

Customizing the Valid Layers and Saving the Setting for Later Use

1. In the Layers Assistant, do the following:

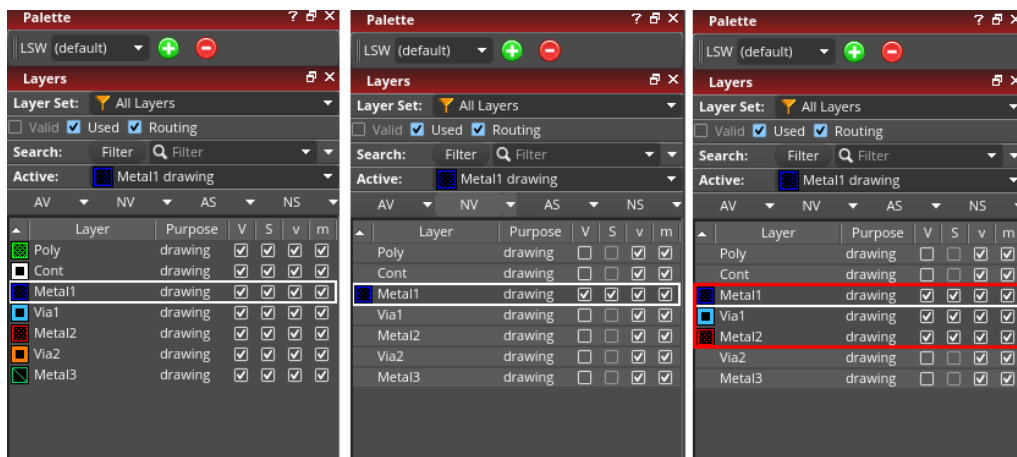
- a. **Right**-click the **Used** option of the Scope section.

b. Choose **Edit Valid Layers**.

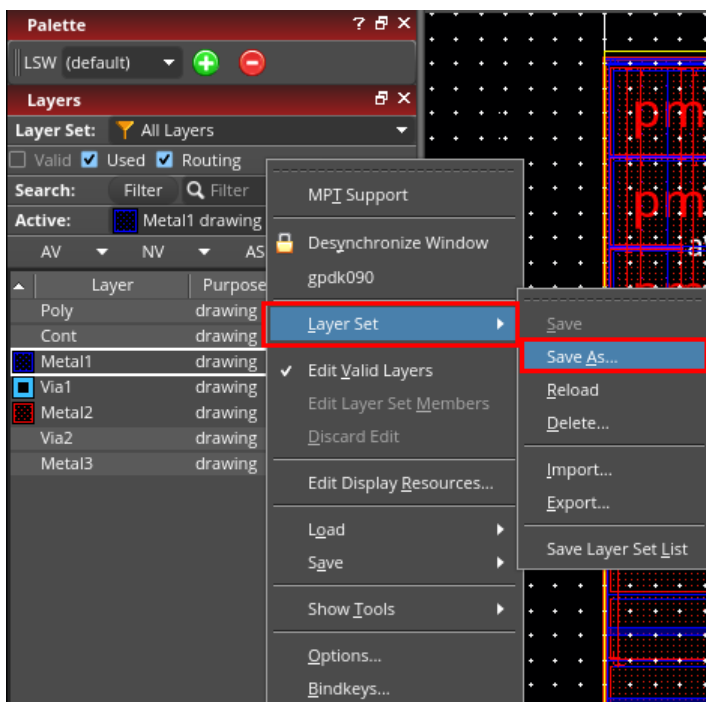
The Layers Assistant now offers you choices as to which layers are Visible (V), Selectable (S), which are valid (v), and have membership (m).

2. Click the **NV** button.

This choice forces all other layers, except **Metal1**, to be invisible and non-selectable while keeping the **V**, **S**, **v**, and **m** options open for **Metal1**.

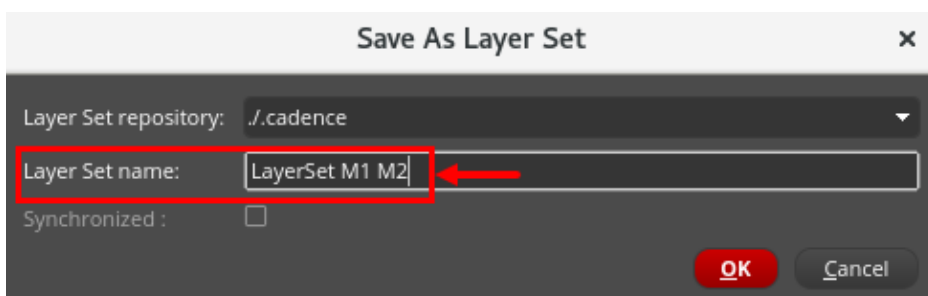
3. Choose the **AV** button to make all the **Used** and **Routing** layers valid again.4. Choose **Metal1**, **Via1**, and **Metal2** active, as shown here:

5. **Right-click** over **Used** and select **Layer Set – Save As**.



The Save As Layer Set form appears.

6. Enter **LayerSet M1M2** in the *Layer Set name* field.



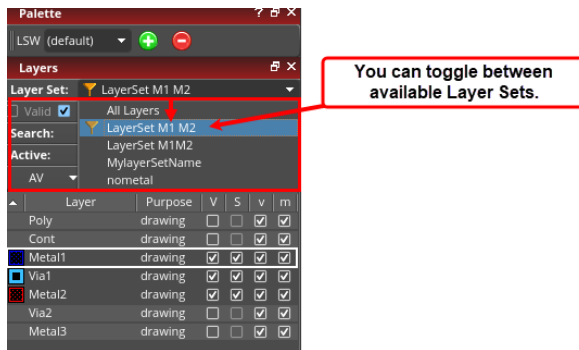
Note: If the name already exists, enter another name.

7. Click **OK**.

Now, this layer set **LayerSet M1M2** has become a choice to switch to in the Layers Assistant.

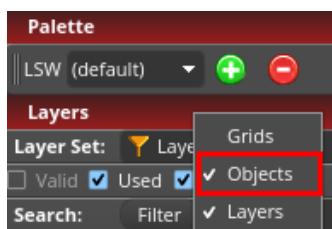
8. In the Layers Assistant, do the following:
 - a. **Right-click** the **Used** option of the Scope section.
 - b. Uncheck **Edit Valid Layers**.

9. You can toggle back and forth between the available Layer Sets, as shown here:

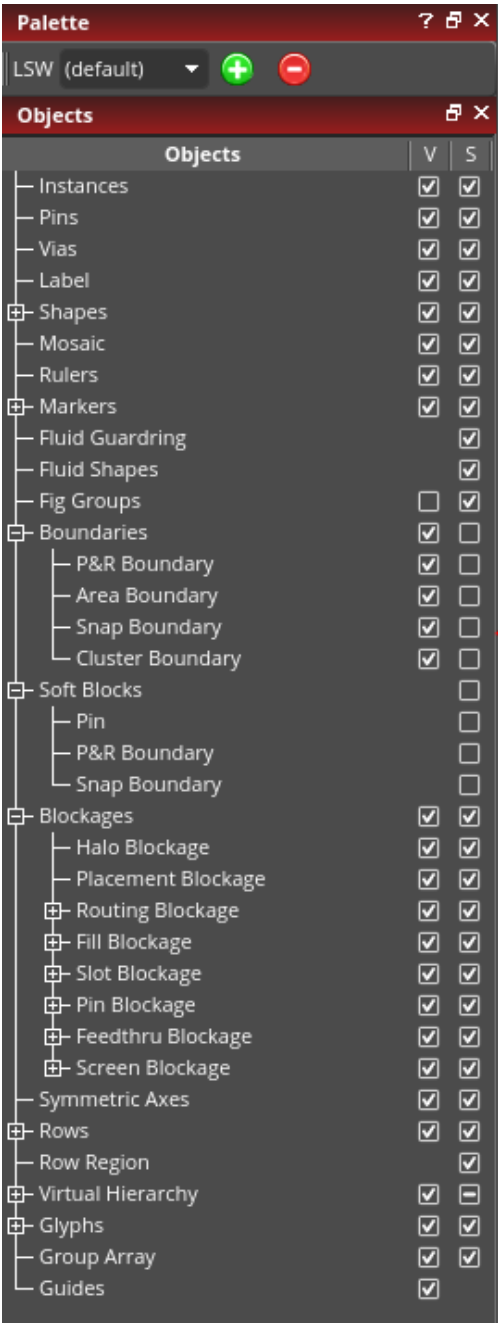


Using the Objects Assistant

1. In the layout canvas, use the **Window – Assistants** menu or **right-click** in the layout banner and activate the Palette Assistant if not already opened.
2. If the Object is not opened, right-click on the Layers to enable the object.



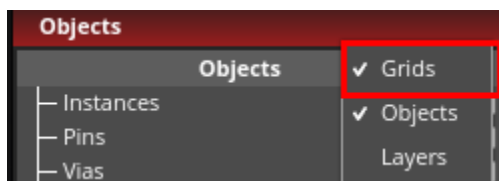
3. Experiment with the selections in this Assistant.



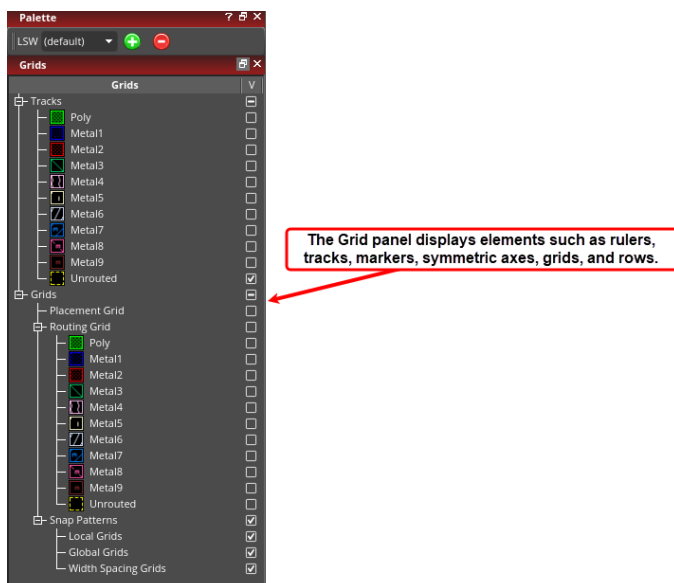
These are the objects you can customize to be either Visible and/or Selectable.

Using the Grids Assistant

1. In the layout canvas, use the **Window – Assistants** menu or **right-click** in the layout banner and activate the **Palette** Assistant if not already opened.
2. If the Grid is not opened, right-click on the object to enable the object.



3. Experiment with the selections in this Assistant.



4. Close the schematic and layout cellviews by choosing **File – Close**.
5. Leave the Virtuoso session open for the next lab.

Summary

In this lab, you learned how to

- ◆ Set/change the current drawing layer (entry layer)
- ◆ Set visibility and selectability
- ◆ Set valid layers for the Layers Assistant and save them to a file
- ◆ Use the Objects Assistant
- ◆ Use the Grids Assistant



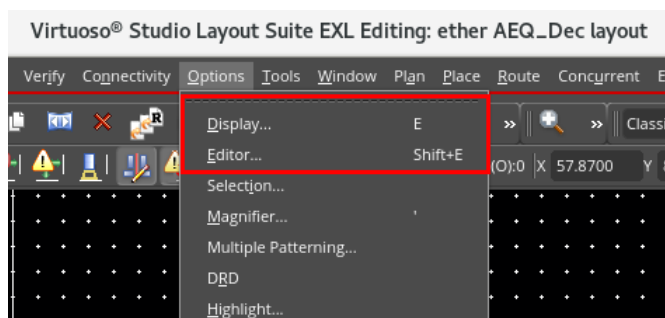
Lab 3-2 Loading and Verifying the Bindkeys

Objective: To use the bindkey user interface to load and verify the bindkeys.

Bindkeys are a convenient way to invoke commands. They can be used instead of pull-down menus. The common commands are assigned to keyboard keys for quick action.

Viewing the Loaded Bindkeys in the Layout Canvas

1. In the Library Manager, select the **ether/AEQ_Dec/layout** cell.
2. In the layout view, **middle-click** and select **Open** from the pop-up menu.
The *ether/AEQ_Dec/schematic* and *ether/AEQ_Dec/layout* cellviews open in XL mode, switch to EXL mode by clicking on the Launch – Layout EXL in the layout.
3. To see a couple of bindkeys listed in a pull-down menu, choose the **Options** pull-down menu from the cellview banner as shown here:



Important: Notice the **Display** bindkey is shown as uppercase **E** and the **Editor** bindkey as **Shift E**. This is to indicate the keyboard key literally as seen on the keyboard.

Loading the Bindkey Files from the *.cdsinit* Startup File

You see the load command in the *.cdsinit* file.

1. In the terminal window where you started the Virtuoso software, enter:

```
more .cdsinit (or) vi .cdsinit
```

Here are the relevant lines of the *.cdsinit* file:

```
; load bindKeys
load(prependInstallPath("/samples/local/schBindKeys.il"))
load(prependInstallPath("/samples/local/leBindKeys.il"))
load(prependInstallPath("/samples/local/lxBindKeys.il"))
```

The three files contain the bindkey assignments for the Schematic, Layout Suite EXL, and Layout Suite MXL.

2. Close the **.cdsinit** file.

Verifying the Bindkey Actions

Check some bindkeys to make sure they have been loaded.

1. Click in the layout canvas to make the window active.
2. Press **E** to bring up the Display Options form.
Note: Alternatively, you can also choose **Options – Display** to bring up the form.
This confirms that the bindkey has been loaded and does the function as expected.
3. Click **Cancel** in the Display Options form.
4. Use the four arrow keys to pan the layout canvas in the direction of the arrows.
5. In the CIW, choose **Options – Log Filter**.
The Set Log File Display Filter form opens.
6. Enable all the buttons in the form to view all the CIW output messages.
7. Set the amount, the layout canvas scrolls, when you use the arrow keys.
 - a. To get the current value, enter this command in the CIW input pane:

```
geGetScrollPercent()
```
 - b. View the return value in the CIW output pane.
 - c. To set the value, enter:

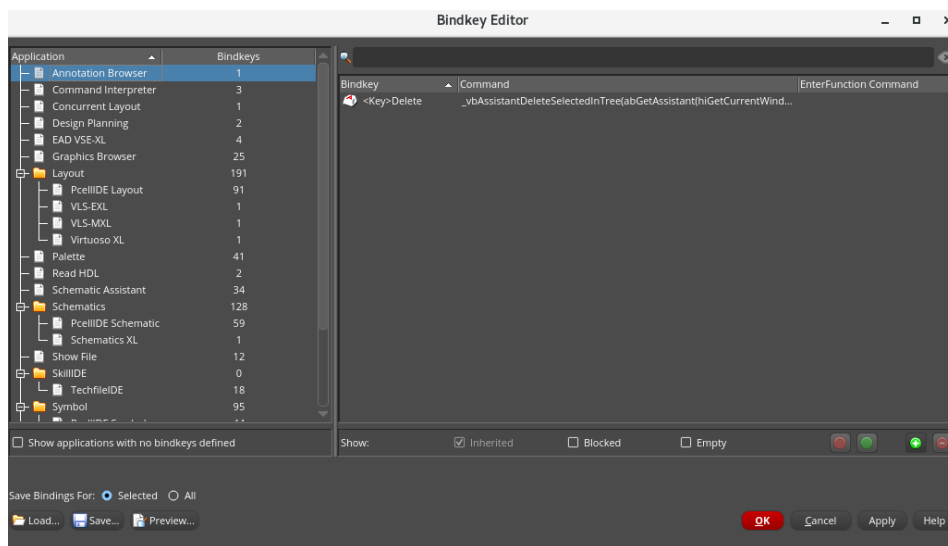
```
geSetScrollPercent(50)
```
8. Use the arrow keys and see the difference in the scroll action.

Viewing the Bindkey Definitions in Bindkey Editor

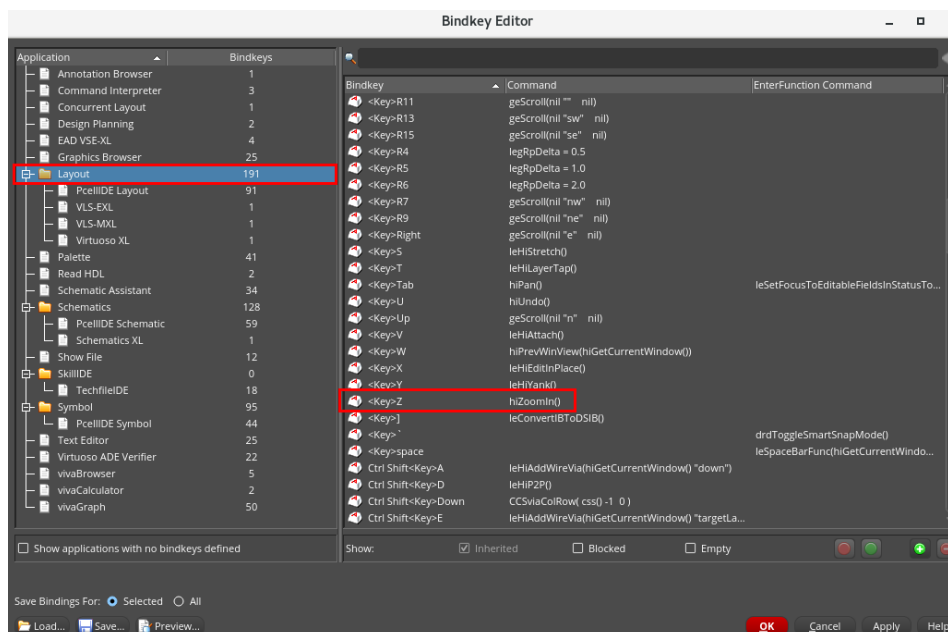
View the bindkey aliases using the *Bindkey Editor*.

1. In the CIW, choose **Options – Bindkeys**.

The Keys or Mouse Binding form, Bindkey Editor, opens.



2. Select **Layout** for the *Application Type Prefix*.



3. Look down the list and find the <Key>Z command, *hiZoomIn()*.

4. Click **Cancel** in the Bindkey Editor form.

5. Click in the layout canvas, and press **Z**.

Notice the prompt in the lower left of the layout canvas:

Enter the first corner of the box you wish to enlarge.

The *hiZoomIn()* command is echoed in the CIW output pane.

Note: This is because you have enabled all the buttons in the **Set Log File Display Filter** form to view all the items in the CIW output area.

6. Click the corners to designate the area to zoom in.

The zoom-in command was executed using the bindkey.

7. Close both the schematic and layout windows by choosing **File – Close**.

8. Leave the Virtuoso session running for the next lab.

Summary

In this lab, you learned how to

- ♦ View the loaded bindkeys in the layout canvas
- ♦ Load the bindkeys using the *.cdsinit* file
- ♦ Verify the bindkey actions
- ♦ Locate the bindkeys
- ♦ Use the Log Filter
- ♦ Locate a particular key and the command associated with it



Lab 3-3 Saving the Defaults

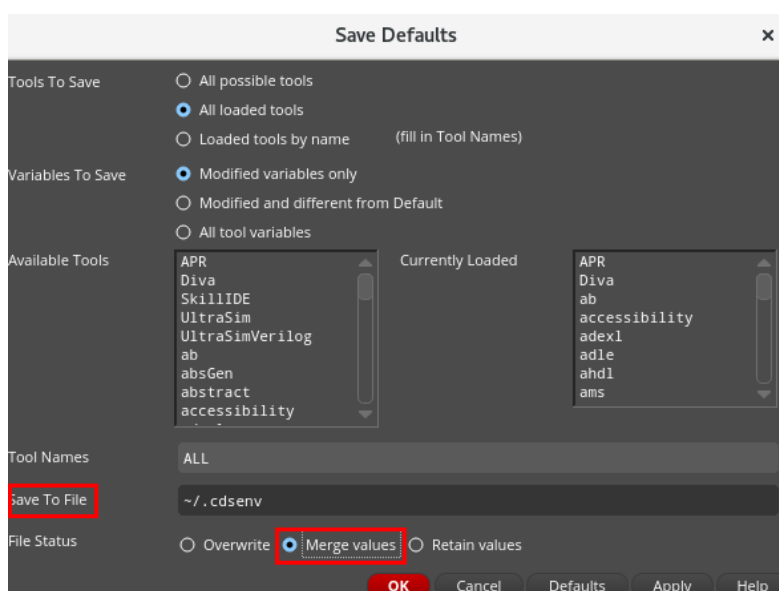
Objective: To store the default settings for the tools.

The tools which are loaded will bring in their default settings. You may change them and then store them for loading next time the tool initializes. The settings are usually stored in the `.cdsenv` file.

Saving the Defaults Settings for the Tools

1. In the CIW, choose **Options – Save Defaults**.

The Save Defaults form opens.



Any changes you have made to the **User Preferences**, **File Preferences**, or other forms which affect the `.cdsenv` variables will be saved into the `.cdsenv` file.

2. Click **OK**.

Important: The new values will be merged into the `.cdsenv` file while protecting other existing values which have not been changed.

3. Close any open forms and cellviews.
4. Leave the Virtuoso session running for the next lab.

Summary

In this lab, you learned how to

- ◆ Store the default settings for the tools



Lab 3-4 Using the Toolbar Manager

Objective: To hide icons in existing toolbars and create new toolbars.

In this lab, you explore the Toolbar Manager, which has the ability to hide icons in toolbars and to create new toolbars.

You hide the *Flip Vertical* and *Flip Horizontal* icons from the layout window and create three new toolbars for the *leHiCreateRuler*, *leHiCreatePolygon*, and *leHiCreateRect* SKILL® functions.

- 1. In the CIW, choose **File – Open** and enter the following values in the form:

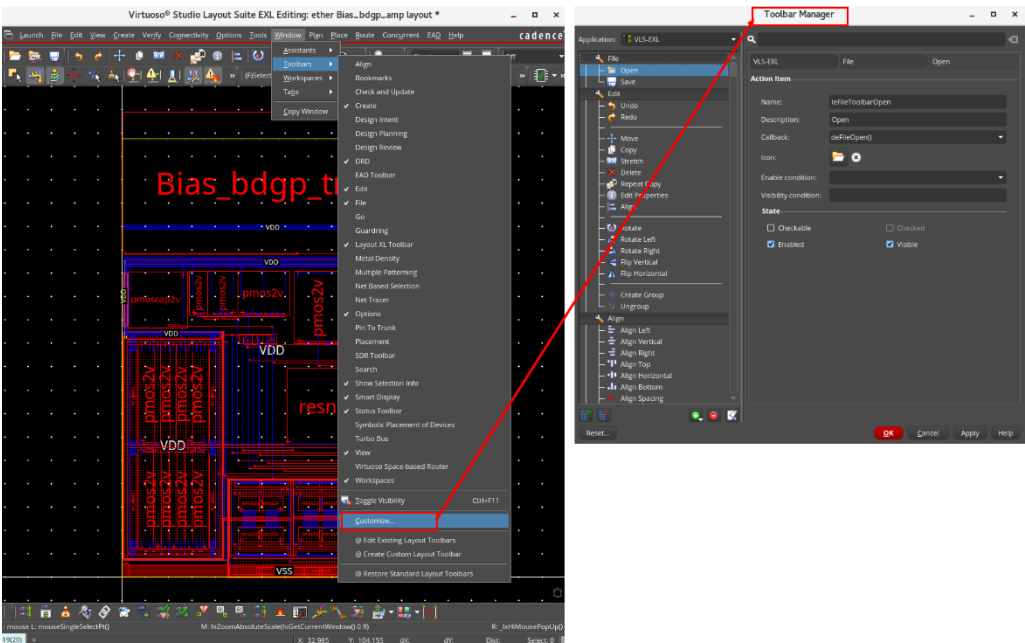
Library	ether
Cell	Bias_bdgp_amp
View	layout

- 2. Click **OK**.

The *ether/Bias_bdgp_amp/schematic* and *ether/Bias_bdgp_amp/layout* cellviews open in XL mode, switch to the EXL mode by clicking the Launch -Layout EXL. Minimize the schematic view and maximize the layout view.

- 3. In the layout window, choose **Window – Toolbars – Customize** or from **CIW – Options – Toolbars** to display the Toolbar Manager form.

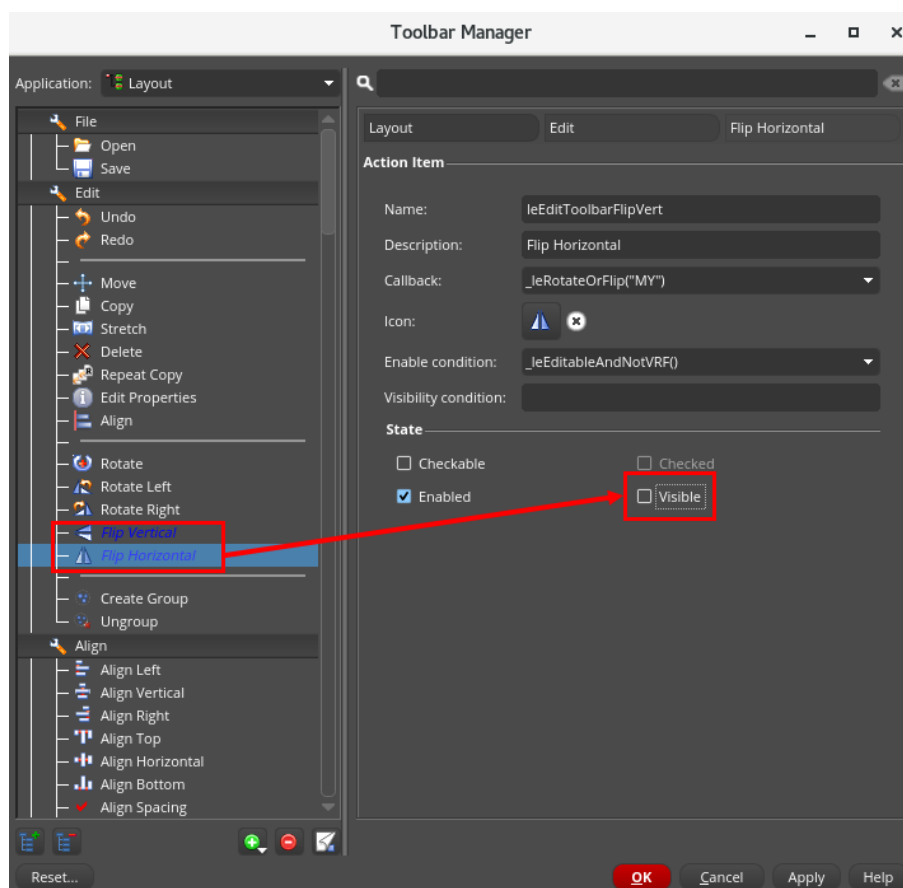
The Toolbar Manager form opens.



Hiding an Icon in a Toolbar

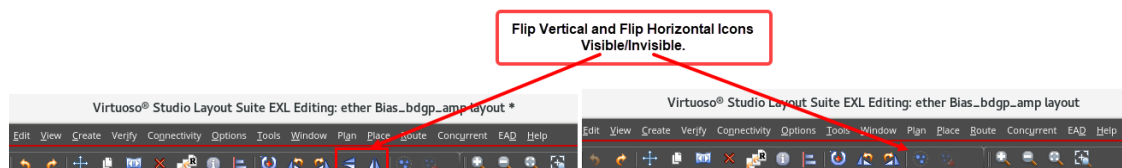
1. Select **Layout** from the *Application* cyclic field in the Toolbar Manager form.
2. Select the **Flip Vertical** icon, **uncheck** the **Visible** button, and click the **Apply** button.
3. Select the **Flip Horizontal** icon and **repeat** the previous step.

Both the *Flip Vertical* and *Flip Horizontal* icons change to italic, as shown in the graphic.



4. Close and reopen the **ether/Bias_bdgp_amp/layout** view.

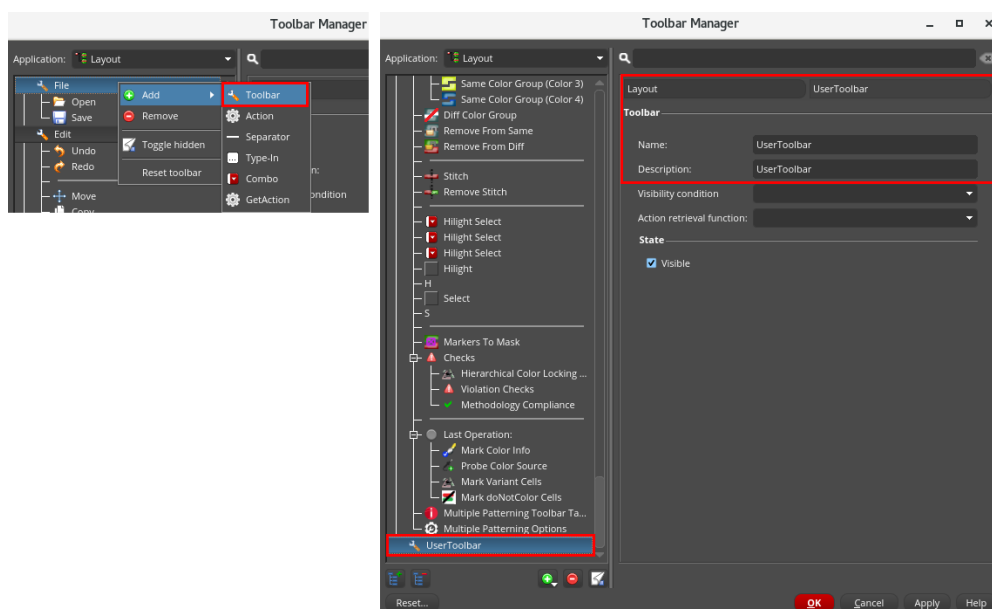
Both the *Flip Vertical* and *Flip Horizontal* icons are not visible now, as shown here:



Creating a New Custom Toolbar

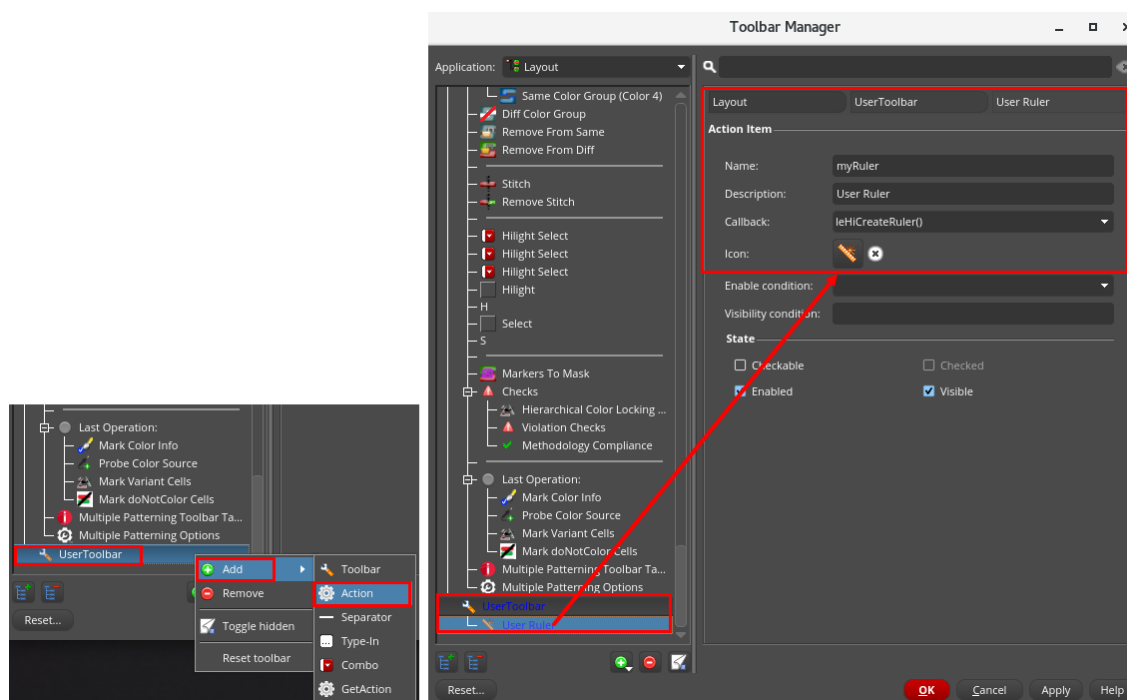
1. Select **Layout** from the *Application* cyclic field in the Toolbar Manager form.
2. **Right-click** the *File* field and choose **Add – Toolbar**.
 - a. In the *Name* field, enter **UserToolbar**.
 - b. In the *Description* field, enter **User Toolbar**.
 - c. Click **Apply**.

User Toolbar is added now in the Toolbar Manager form.



3. **Right-click** the **User Toolbar** and choose **Add – Action**.
 - a. In the *Name* field, enter **myRuler**.
 - b. In the *Description* field, enter the **User Ruler**.
 - c. In the *Callback* field, enter **leHiCreateRuler()**.
 - d. In the *Icon* field, browse and select **ruler-create**.
 - e. Click **Apply**.

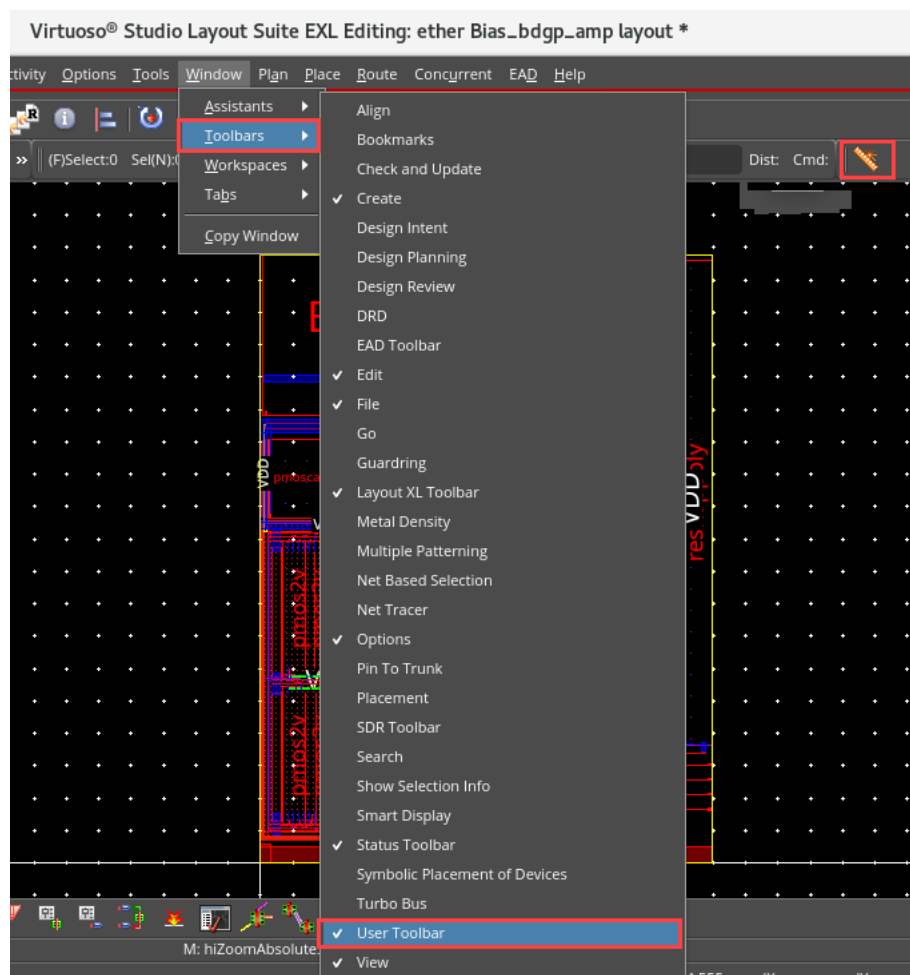
The **User Ruler** icon is added now under **User Toolbar** in the Toolbar Manager form.



Now you close the cellview and reopen again to check the new User Toolbar you have added.

4. Close and reopen a layout, for example, **ether/Bias_bdgp_amp/layout** view to test the new User Toolbar.

You can see that the **Toolbars** menu has the newly added **User Toolbar** entry and the **User Ruler** icon in the layout canvas, as shown here:



Now you have a new toolbar associated with Layout.

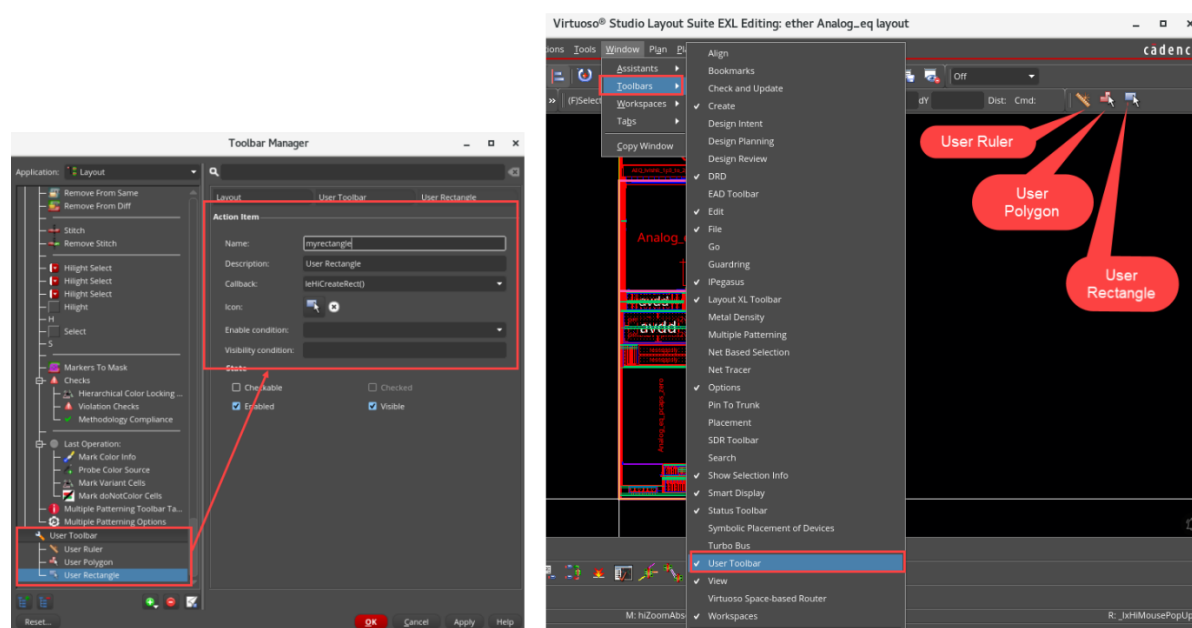
5. Click the **User Ruler** icon and explore using the **Create Measurement** command.

Additional Exercise

If you have time, you can try creating more icons. Use the icons available in the `./icons/` directory.

- ◆ To create an icon for Ruler, use the callback `leHiCreateRuler()` which you tried.
- ◆ To create an icon for Polygon, use the callback `leHiCreatePolygon()`.
- ◆ To create an icon for Rectangle, use the callback `leHiCreateRect()`.

Refer to the new toolbar with the user icons created given here for reference.



Note: A sample toolbar **User Toolbar** is shown here in the Toolbar Manager form for reference. This toolbar has the **User Ruler**, **User Polygon**, and **User Rectangle** icons. If the name already exists, enter another name.

1. Close any open forms and cellviews.
2. Leave the Virtuoso session.
3. In the shell, enter the following command:

```
cd ~
```

This command places you in your home directory.

Summary

In this lab, you learned how to

- ◆ Hide icons in the existing toolbars
- ◆ Create new toolbars



Module 4: Basic Layout Commands

Lab 4-1 Viewing Your Designs

Objective: To view your designs in different ways.

In this lab, you explore how to use the *Pan* command, save and restore a zoomed window, and use the *Zoom* commands.

Lab Setup

Follow the steps below to set up the environment and perform the lab:

1. If you continue from the previous lab, move to the sub-heading: Starting the session. Otherwise, proceed to the next step.
2. Update the following line in the /VLPT1_IC_25_1/.cshrc_VLPT1 file with the appropriate valid local tool paths for Virtuoso.

```
setenv CDSHOME /ccstools/cdsind1/Software/IC251Base_lnx86
```

3. Save and close the file.
4. Source the updated class setup file.

```
Source .cshrc_VLPT1
```

Starting the session

1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

```
cd ./M04_VLPT1_BscLayCmds
```

2. If the Virtuoso session is not running, enter this command in the terminal window:

```
virtuoso &
```

The Command Interpreter Window (CIW) appears.

2. In the CIW, choose **File – Open** and enter the following values in the form:

Library	ether
Cell	Bias_bandgap
View	layout

3. Click **OK**.

The *ether/Bias_bandgap/schematic* and *ether/Bias_bandgap/layout* cellviews open.

Using the Pan Command to Move Around a Design

1. To pan the design, choose **View – Pan** or press the **Tab** key in the layout window.

The CIW prompts: “Point at the center of the desired display”.

2. Click on either of the **Q1** transistors to pan the display.

The display of the cellview changes so that the point you chose becomes the center of the canvas.

Note: Notice in the lower middle of the cellview the **M:hiPan()** mouse button value. The middle mouse button holds the last command executed for easy repetition.

3. To repeat the previous command, press the **middle** mouse button.

The CIW prompt is again asking for a coordinate to center the display.

4. Click another location to center the display again.

5. Press the bindkey **F** to fit the cellview.

Saving and Restoring a Zoomed Window

1. To move back to the previous location, in the layout canvas choose **View – Save/Restore – Previous View** or press the bindkey **W**.

Note: You can move up and down the stack of previous and next windows by using **W** or **Shift+W** bindkeys to move to previous or next views.

2. Choose **View – Save/Restore – Save View**.

3. In the Save View form, enter **Win1** and click **OK**.

You have just saved this window, so you can return to this area later in the editing session.

4. Choose **View – Zoom To Fit All** or press the bindkey **F** to fit the design to the window.

5. Choose **View – Save/Restore – Restore View**.

The Restore View window pops up.

6. Select your entry **Win1** and click **OK**.

The viewport returns to the area you saved as **Win1**.

Note: This is handy when working on large chips where you are moving between a few work areas throughout the day. The windows are persistent during the current editing session.

Moving Around with the Zoom Commands

1. In the layout window, choose **View – Zoom Out** or press the bindkey **Shift+Z**.

The command scales the current window by **0.5**.

2. To return to the original cellview display size, choose **View – Zoom In** or press the bindkey **Ctrl+Z**.

The command scales the current window by **2.0**.

3. To zoom in a specified amount, choose **View – Zoom To Area** or press **Z**.

The CIW prompts: “Enter the first corner of the box you wish to enlarge.”.

4. Click a corner of the area you want to display.

The CIW prompts: “Enter the other corner of the box you wish to enlarge.”.

5. Click the other corner of the area that you want to display.

6. Press the bindkey **F** to fit the full cellview in the design window.

7. Leave the cellviews open for the next lab.

Summary

In this lab, you learned how to

- ◆ Use the Pan command
- ◆ Save and restore a zoomed window
- ◆ Use the Zoom commands



Lab 4-2 Selecting and Deselecting the Objects

Objective: To learn different ways of selecting and deselecting design elements.

In this lab, you explore the different ways of selecting and deselecting the objects.

1. If the cellview is not already open, choose **File – Open** in the CIW window, and enter the following values in the form:

Library	ether
Cell	Bias_bandgap
View	layout

2. Click **OK**.

The *ether/Bias_bandgap/schematic* and *ether/Bias_bandgap/layout* cellviews open.

Using the Mouse to Select Objects in Full Selection Mode

1. Maximize the layout canvas viewport by clicking in the **Maximize** box in the layout canvas viewport's upper right corner.
2. In the layout, choose **Options – Display** or press the bindkey **E**.
 - a. Verify that the **Dynamic Highlight** option is **ON**.
 - b. Click **OK**.

3. Move the cursor over the shapes in the design.

Notice that the shapes are highlighted with a dotted line. This is called dynamic highlighting. The highlight indicates the object or edge you will select when clicking with the left mouse button.

4. Click with the **left** mouse button to select any object.

To show that the shape is selected, the system highlights the shape in bright white. Notice the number of selected objects in the upper center **(F)Select:1**. Here, **(F)** indicates a **Full Select** mode. Full Select mode only lets you to select entire objects. You cannot select just edges or a vertex in Full Select mode.

5. Click in the cellview window banner to be sure that the cursor is focused in the design canvas.

- Choose **Edit – Select – Deselect All** or press the bindkey **Ctrl+D** to deselect all the objects.

The shape is now deselected, and the white selected highlighting is removed.

How many objects are shown as being selected in the window banner now?

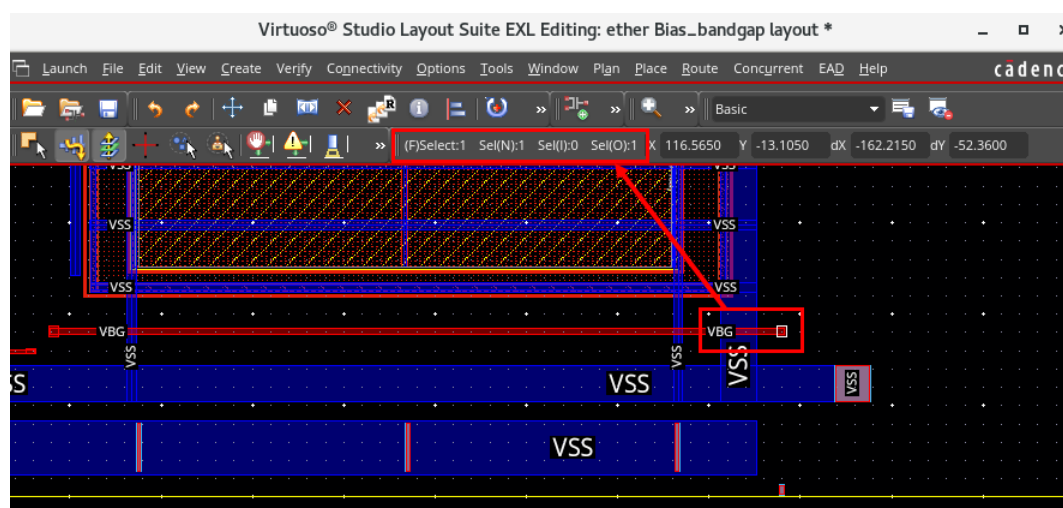
Answer: **(F)Select:0**

Choosing Between Objects

- Press the bindkey **Shift+F** to view all the levels of the hierarchy in the layout canvas.
- In the layout, choose **View – Pan** or press the bindkey **Tab** to pan.
- In the CIW, enter this coordinate for the new center of the view, **128.0:6.3**.

The coordinates are displayed in the top right just above the canvas in X and Y fields.

- Use right mouse drag to create the zoom-in window, as shown here:



- Click to select the **via**, as shown in the graphic.
- Click at the same location again.

How many objects can you select at one time by just clicking?

Answer: You can select only one object at a time with a single click without using the **Shift** key.

- Press the bindkey **Ctrl+D** to deselect all the objects.
- Press the bindkey **F** to fit the design canvas.

Adding or Subtracting Objects to the Selected List

1. Drag with the **right** mouse button to zoom in to the top left of the design so that you can see the power busses.
2. Click to select the power bus contact in the upper left of the design.
Note: You can observe the number of selected objects in the upper center of the layout canvas.
3. To select more objects, press and hold the **Shift** key, and click the **left** mouse button.
Note: You can continue to select objects one at a time using this method.
4. To remove an object from the selected set, press and hold the **Ctrl** key, and click the **left** mouse button on a selected object to deselect.

Using the Drag Box to Select or Deselect Objects

1. To select a group of objects, press and hold the **left** mouse button, and drag the cursor to draw a selection box around the group.
All the previously selected objects are deselected. All the objects fully inside the selection box are now selected.
Note: You can use **Shift** and drag the **left** mouse to add more objects. **Ctrl** and drag using **left-click** will deselect the objects from the selected set.

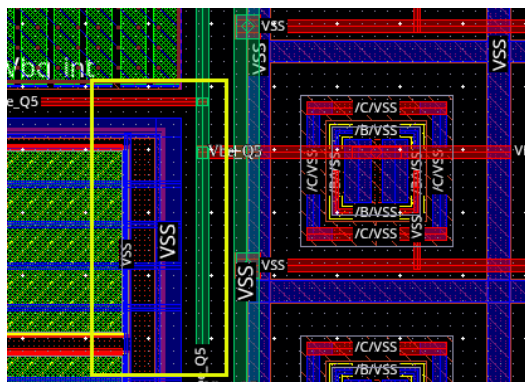
Using the Mouse to Select Objects in Partial Selection Mode

1. Press the bindkey **Shift+F** to view all the levels of the hierarchy in the layout canvas.
2. Click the **Maximize** button to reduce the size of the layout canvas so that you can see the CIW.
3. Press the bindkey **Tab** to pan.

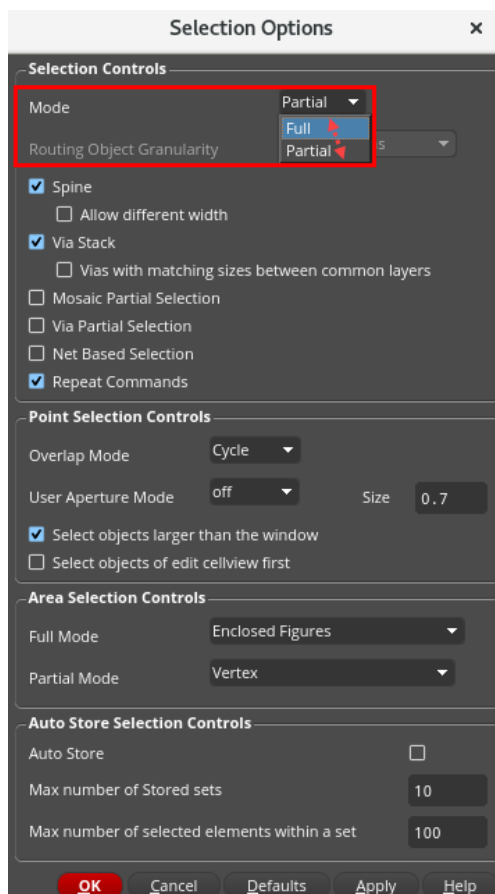
4. In the CIW, enter this coordinate for the new center of the view, **123:67**.

You will find a **Metal1** shape and paths which will demonstrate partial select.

Note: If required, zoom in to see the mentioned objects more clearly as shown here:



5. To select a single vertex, in the layout canvas, choose **Options – Selection** to open the Selection Options form, and in the Selection Controls section, in the Mode field, select **Partial** from the cyclic field and click **OK** in the form or press the bindkey **F4**.



6. Hover on the corner of the blue **Metal1** layer shape. When you see the corner highlighted, you can select just the corner.

7. Click to select the corner.

The **F4** bindkey toggles between full and partial selection. The layout window banner shows **(P)Select:1**.

The banner updates to show **(P)Select:1 Sel(N):1 Sel(I):0 Sel(O):1**. This indicates the Partial Select count is 1. There is 1 Net selected, 0 Instances selected, and 1 Object selected.

8. To select an edge of the **Metal1** rectangle, move the cursor over an edge, and notice the dynamic highlight is just on one edge and two corners. Click on the edge of the **Metal1** layer rectangle.

The previous selected corner is now deselected, and the edge is now selected.

9. To select a group of path ends, draw a selection box around the ends of the **Metal1** paths you see.

10. You can hold the **Shift** key to add to the selected set when dragging to select more path ends.

11. Press the bindkey **Ctrl+D** to deselect all the objects.

12. Close the design windows.

13. Leave the Virtuoso® session running for the next lab.

Summary

In this lab, you learned how to

- ◆ Use the mouse to select objects in full selection mode
- ◆ Choose between objects
- ◆ Add or subtract objects to the selected list
- ◆ Use the drag box to select or deselect objects
- ◆ Use the mouse to select objects in partial selection mode



Lab 4-3 Using the Alignment Commands

Objective: To use the Quick Align Command and the Alignment toolbar.

In this lab, you explore the Quick Align command and the Alignment toolbar.

1. In the CIW, choose **File – Open** and enter the following values in the form:

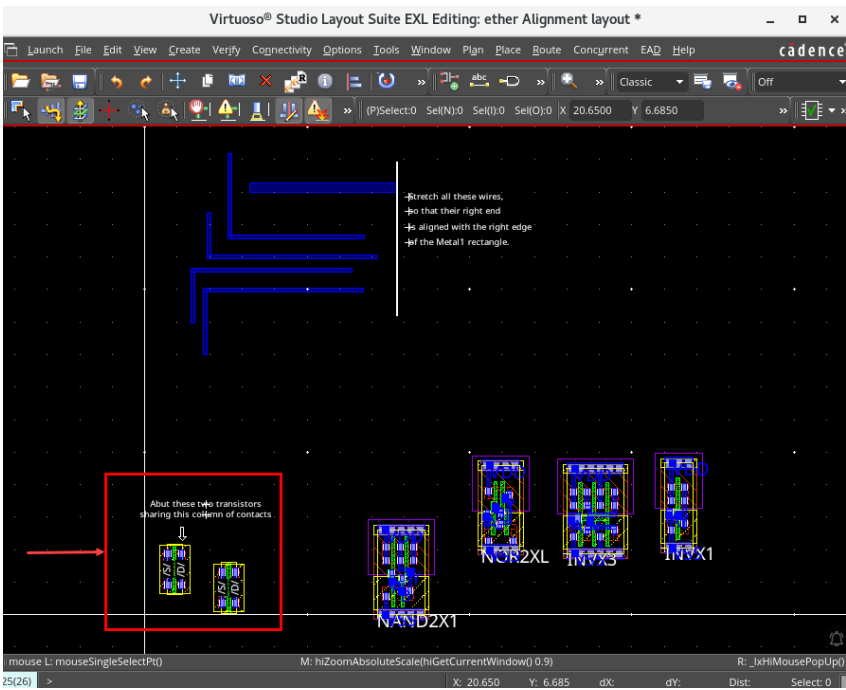
Library	ether
Cell	Alignment
View	layout

2. Click **OK**.

The *ether/Alignment/layout* cellview opens for editing in XL mode.

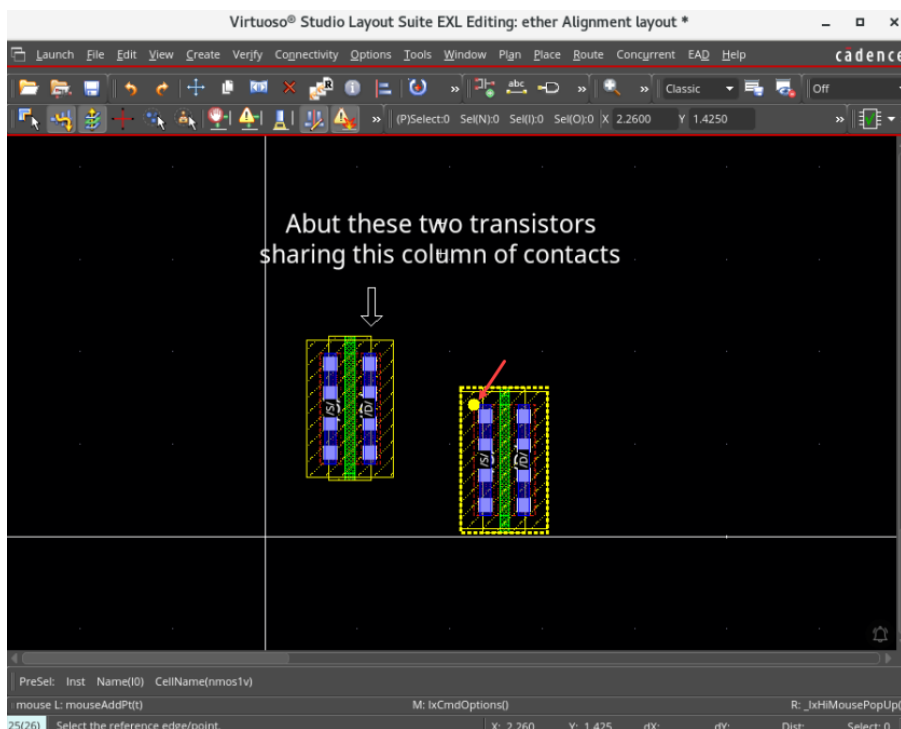
Using Quick Align to Abut Transistors

1. In the layout window, zoom to the two transistors in the **lower-left** corner.
You need to abut them so that they share the “inner” column of contacts.



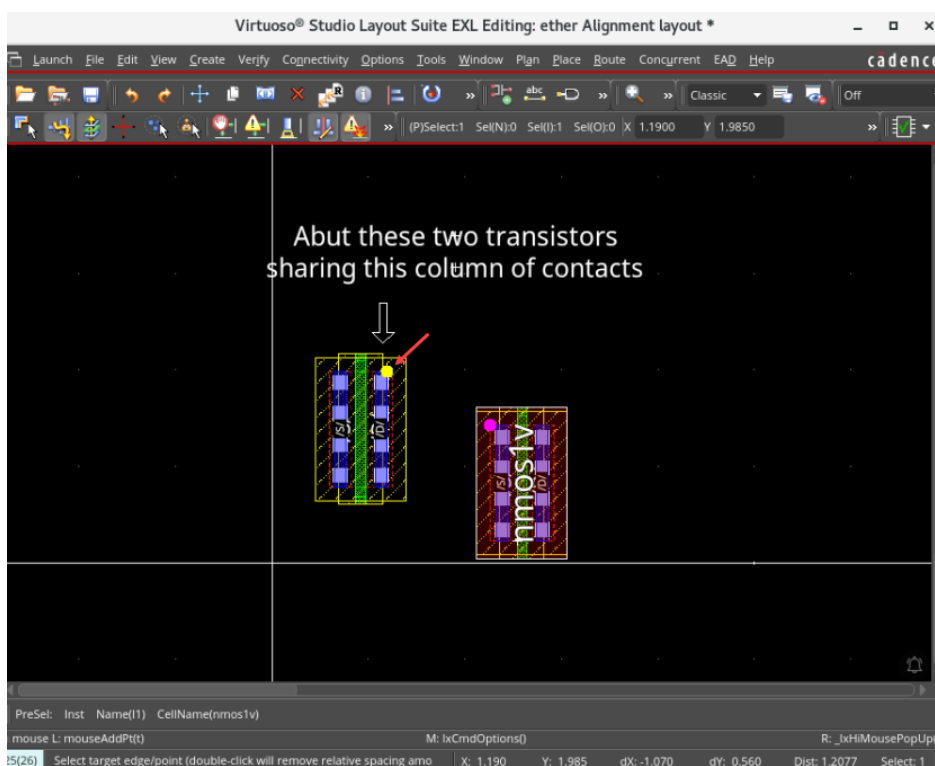
2. In the layout canvas, choose **Edit – Quick Align** or press the bindkey **A**.

3. Select the **upper-left** corner of the **right** transistor *as reference to move*, as shown here:

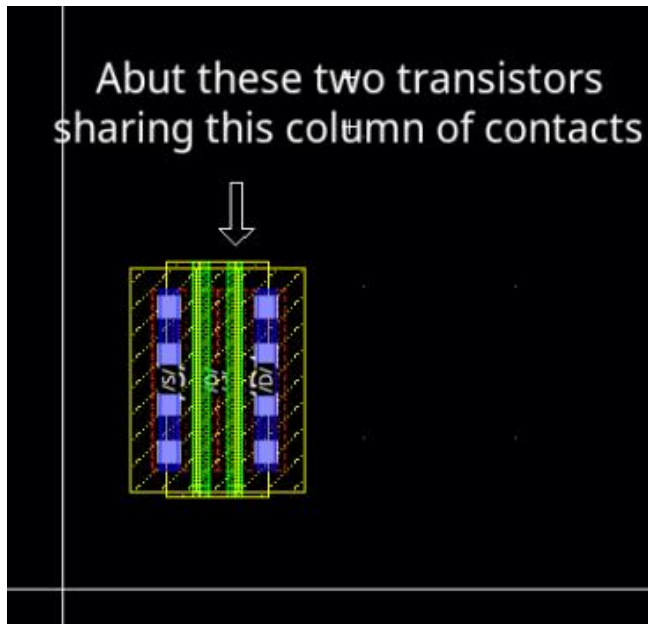


The selected corner turns pink, as shown.

4. Select the **upper-right** corner of the **left** transistor *to be used as reference*, as shown here:

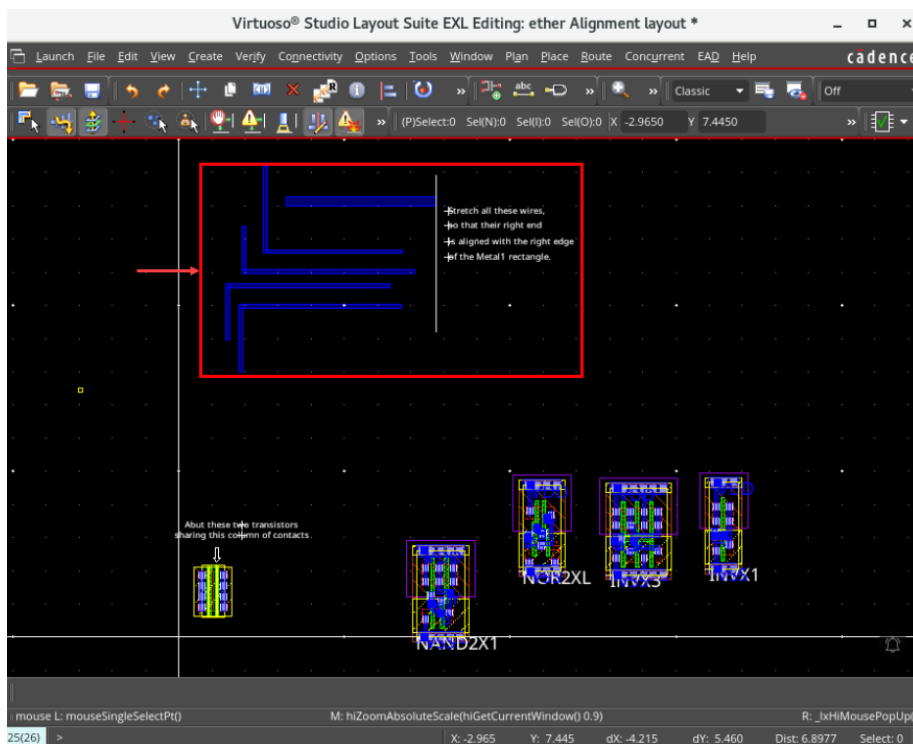


- The two transistors are abutted, as shown here:



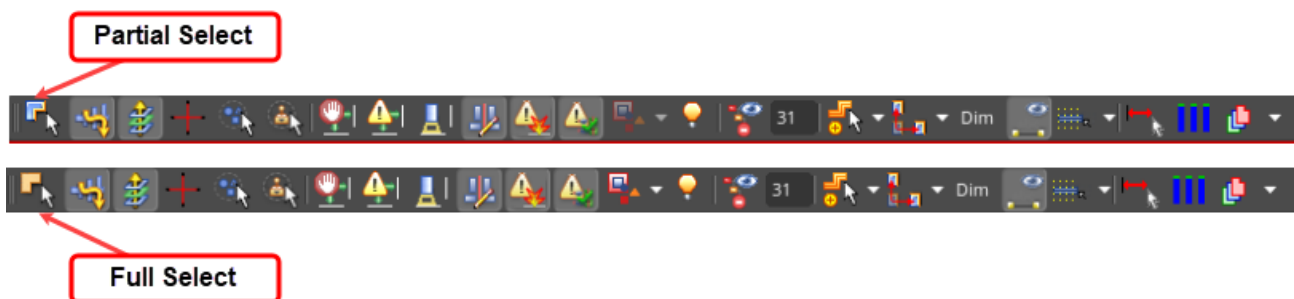
Stretching Wires Using *Partial Select* and *Quick Align*

- Press the **ESC** key to close the **Quick Align** mode if it is active and zoom to the set of **Metal1** wires.



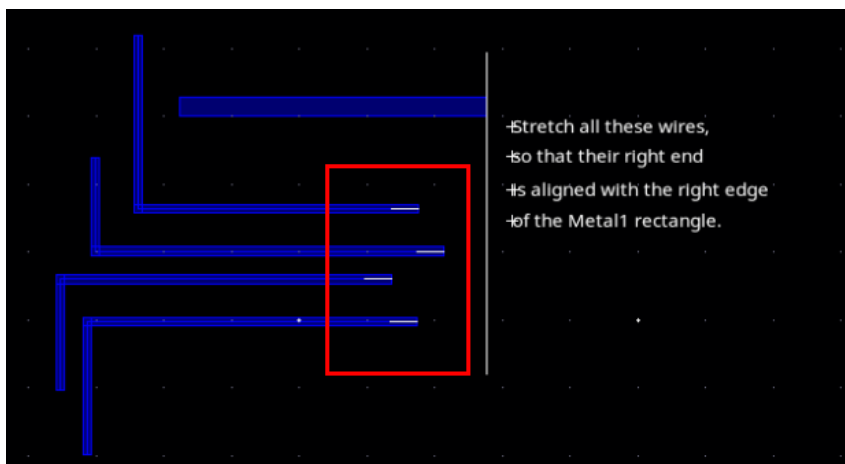
2. Press **F4** to turn on partial selection.

Tip: You can check whether the partial selection mode is active or not by looking at the **Select Mode** icon in the **Options** toolbar.



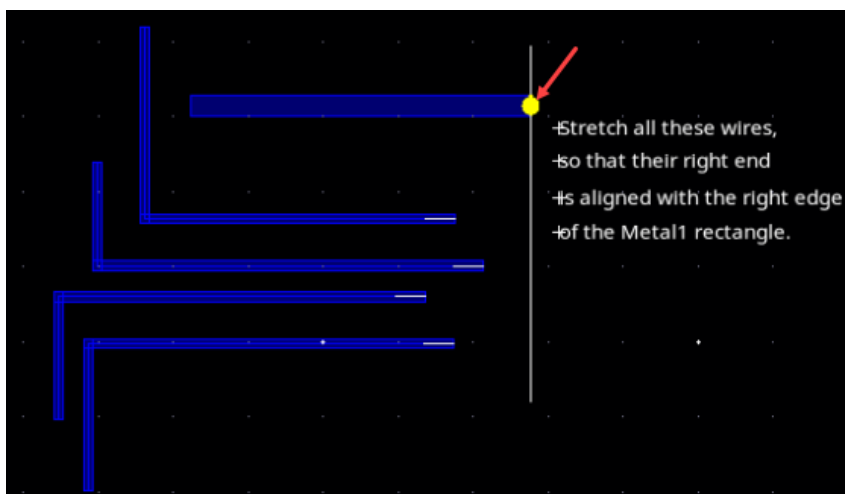
3. In the layout canvas, select the **right** side end-caps of all the wires, as shown here:

Note: You need to be in the *partial selection* mode.

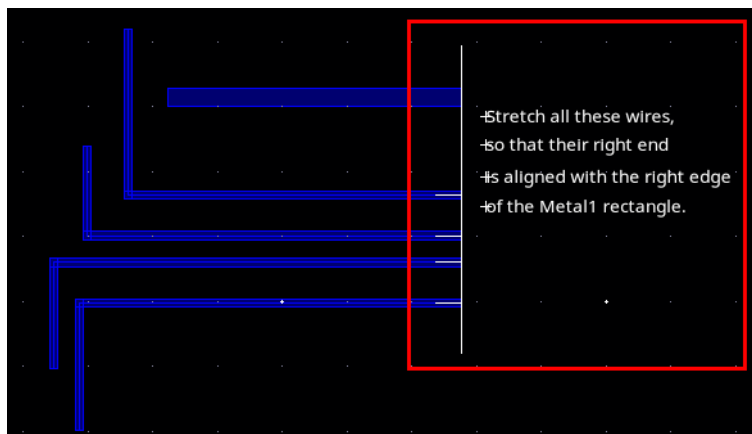


4. In the design canvas, choose **Edit – Quick Align** or press the bindkey **A**.

5. **Double-click** the *reference* object, as shown here:

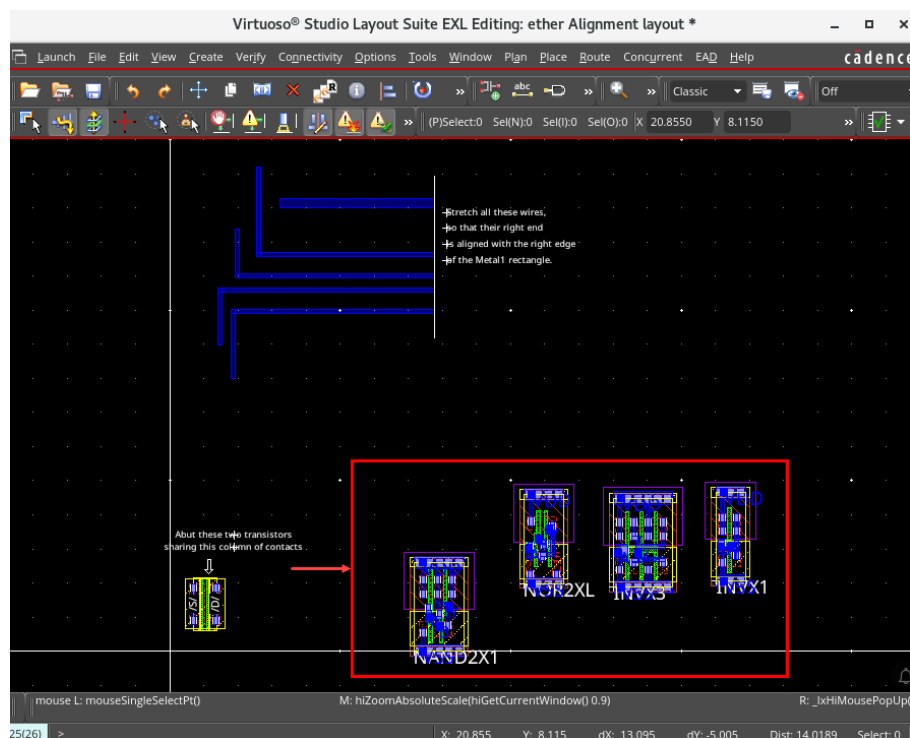


This stretches (aligns) all endcaps (current selected objects) to the reference point, as shown here:



Using the Align Toolbar

1. Press the **ESC** key to exit the **Quick Align** mode if it is active and zoom to the set of standard cells.



2. Choose **Window – Toolbars – Align** to activate the Align toolbar.

Note: This step is needed only if the toolbar is not visible yet.



3. Activate the **Align Spacing** text field and enter **0**, as shown here:

This defines the distance between the aligned objects.

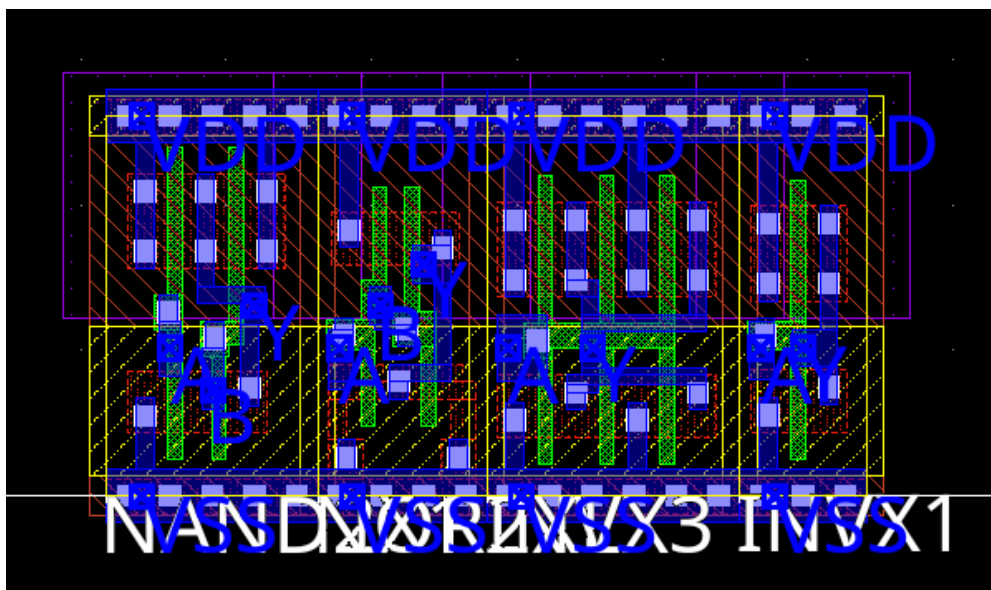
Tip: You might need to rearrange the toolbars to properly see the Align toolbar.



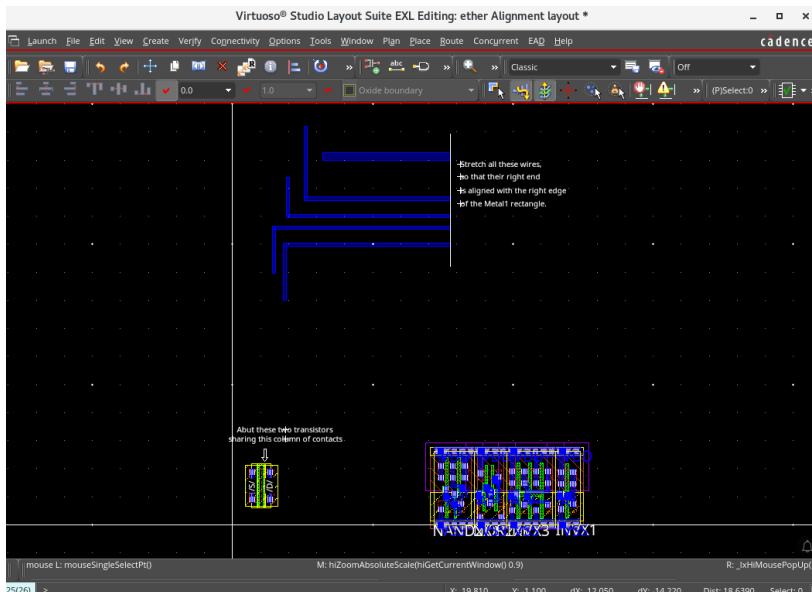
4. Select the stdcells and click the **Align Bottom** toolbar button, as shown here:



The stdcells are aligned with their *instance* layer properly abutted, as shown here:



The layout view with aligned objects is shown here for your reference.



5. Close any open forms and cellviews. Do **NOT** save.
6. Close the Virtuoso platform.

Summary

In this lab, you learned how to

- ◆ Use Quick Align to abut transistors
- ◆ Stretch wires using Partial Select and Quick Align
- ◆ Use the Align toolbar



Module 5: iPegasus DRC and Fill for Virtuoso Studio

Lab 5-1 Setting Up and Running Virtuoso DRD and iPegasus DRC.

Objective: To set up and run Virtuoso DRD and iPegasus DRC.

The iPegasus™ provides the capability to run sign-off quality in-design DRC checks while developing the layout design. This lab will provide a practical overview of the iPegasus DRC package.

Note: This lab was tested and verified with the following software versions:

Virtuoso® : IC25.10
Pegasus™ : PEGASUS24.12

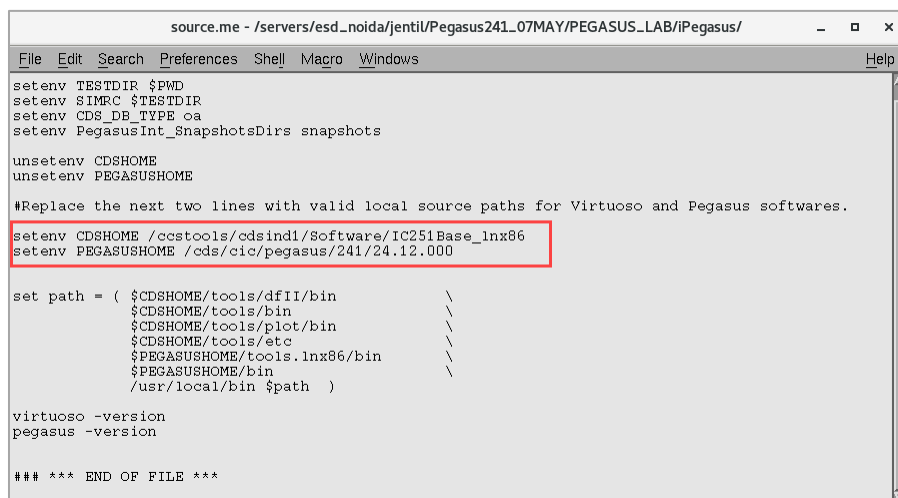
Note: Licenses required for this lab.

Pegasus Licenses needed: Cadence iPegasus DRC for Virtuoso Studio
Virtuoso Licenses needed: Virtuoso Layout Suite XL

Lab Setup

Follow the steps below to set up the lab environment.

1. Move to the `$VLPT1_IC25_1/M05_VLPT1_iPegasus` directory.
 - a. Open the `source.me` file and update the Virtuoso and Pegasus software tool paths with valid local paths.



```
source.me - /servers/esd_noida/jentil/Pegasus241_07MAY/PEGASUS_LAB/iPegasus/
File Edit Search Preferences Shell Macro Windows Help
setenv TESTDIR $PWD
setenv SIMRC $TESTDIR
setenv CDS_DB_TYPE oa
setenv PegasusInt_SnapshotsDirs snapshots

unsetenv CDSHOME
unsetenv PEGASUSHOME

#Replace the next two lines with valid local source paths for Virtuoso and Pegasus softwares.
setenv CDSHOME /cstools/cdsind1/Software/IC251Base_lnx86
setenv PEGASUSHOME /cds/cic/pegasus/241/24.12.000

set path = ( $CDSHOME/tools/dfII/bin \
             $CDSHOME/tools/bin \
             $CDSHOME/tools/plot/bin \
             $CDSHOME/tools/etc \
             $PEGASUSHOME/tools.lnx86/bin \
             $PEGASUSHOME/bin \
             /usr/local/bin $path )

virtuoso -version
pegasus -version

*** END OF FILE ***
```

- b. Save and close the file.
 - c. Source the `source.me` file:

```
source source.me
```

This step sets up the environment variables for the iPegasus DRC and Fill labs and displays the software versions.

Setting Up and Running Virtuoso DRD

DRD is an in-design verification tool that supports and works on technology file constraints. When enabled, DRD provides visual and physical feedback about design constraint violations while you are interactively creating and editing a layout. DRD picks Design constraints from the technology library. DRD can be run in three modes – Notify, Enforce, and Post Edit. We will demonstrate the Notify mode in this section.

1. Start the Virtuoso session from the *iPegasus* directory.

```
virtuoso &
```

2. In the Library Manager, under *FracN_PLL_45* library, open the following:

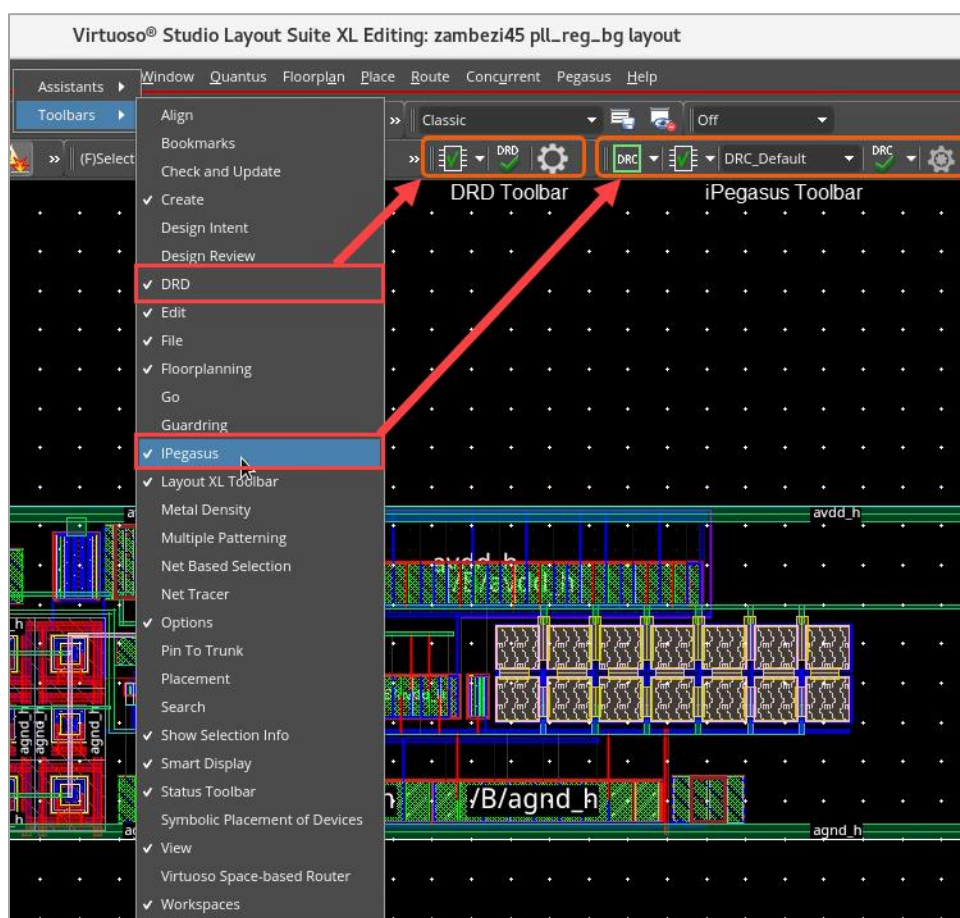
Sub-library *zambezi45*, cell *pll_reg_bg*, view *layout*.

The Virtuoso Layout and Schematic Viewers open and display corresponding cell views.

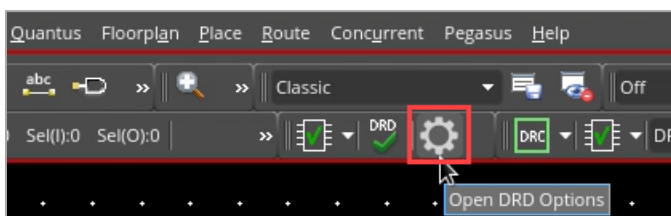
Ignore the warnings.

Once the correct tool versions are sourced, the DRD and iPegasus toolbars are tucked into the Virtuoso Studio Layout Suite.

Note: Alternatively, you can enable both toolbars by right-clicking the empty pane and selecting **Toolbars → DRD/iPegasus**.

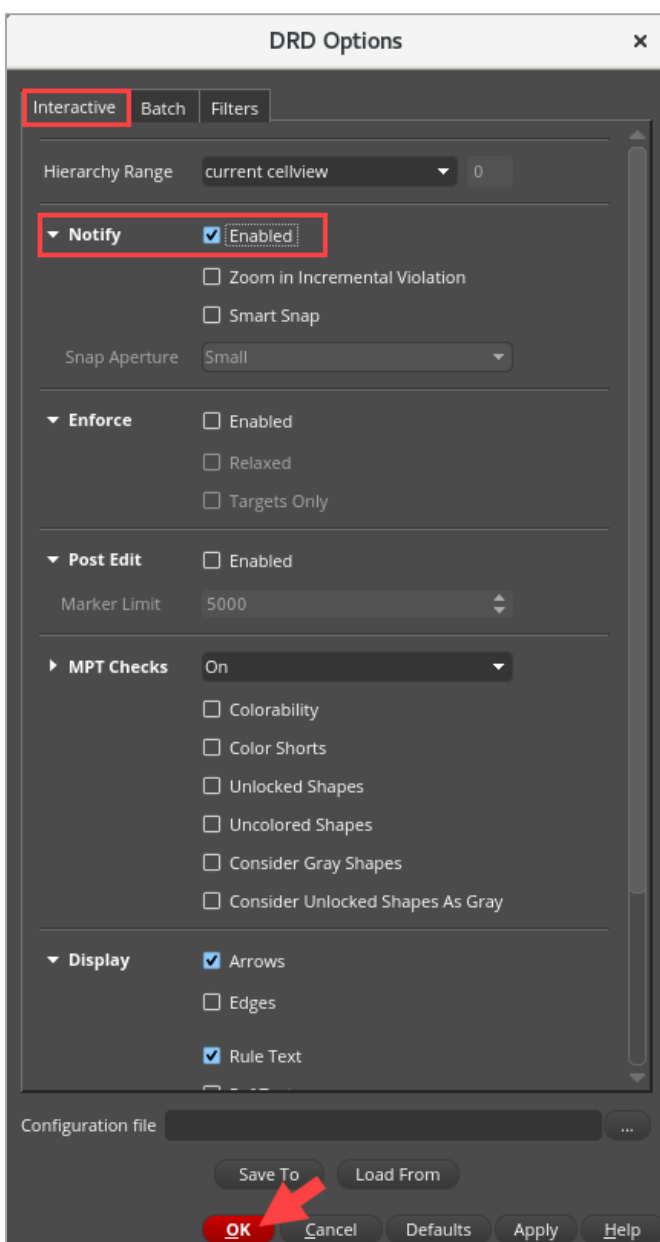


3. In the layout window, click the **Open DRD Options** icon on the DRD toolbar.



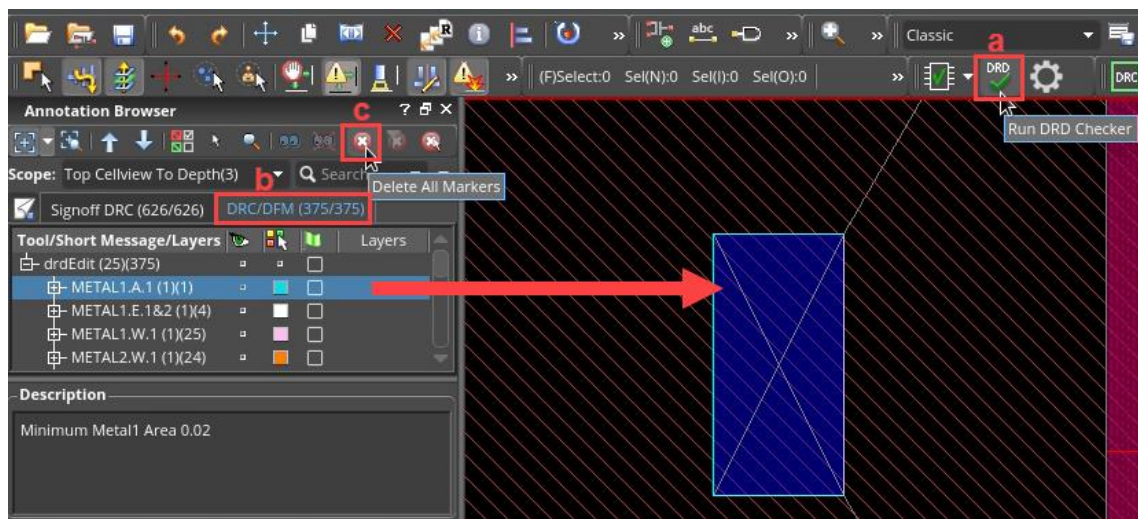
That opens the DRD Options form.

4. In the **DRD Options** form → **Interactive** tab, select the **Notify** → **Enabled** checkbox and click **OK**.



That closes the DRD Options form.

5. Back in the Layout suite, on the DRD toolbar,
 - a. Start the DRD run by clicking the **Run DRD Checker** icon.



The *Run DRD Checker* lets you run interactive DRD on the selected design scope (default Checker Scope: Current Cellview).

At the end of the DRD run, the violations get loaded in the Annotation Browser
→ DRC/DFM tab.

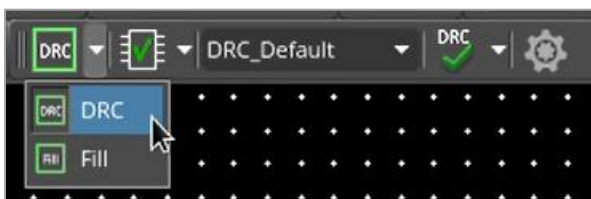
- b. Investigate DRC violations with the Annotation Browser.
- c. At the end, click the **Delete All Markers** icon to clear the DRD markers.

Do not close the Virtuoso session.

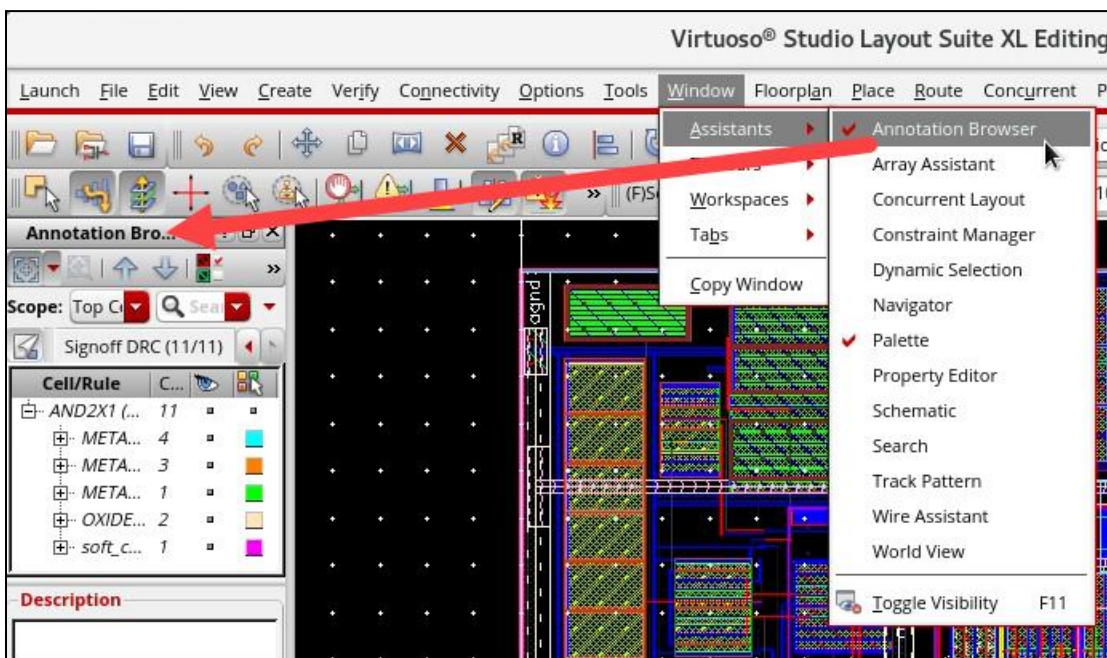
Running iPegasus DRC – XL/EXL/MXL Mode (Limited Mode DRC)

Under iPegasus limited mode DRC, the user can run DRC (with some limitations) without needing an iPegasus license over a restricted area. This area limit is process-node dependent. Refer to the lecture slides for details. Under the limited mode, iPegasus DRC will run with just a Virtuoso XL/MXL/EXL license; no iPegasus license is needed to start DRC checks. However, you will need an iPegasus license for more extensive area checks.

1. Ensure that *zambezi45 | pll_reg_bg | layout* is open in the layout suite. Press **F** to fit the design.
2. Ensure the iPegasus Toolbar is running in DRC mode.

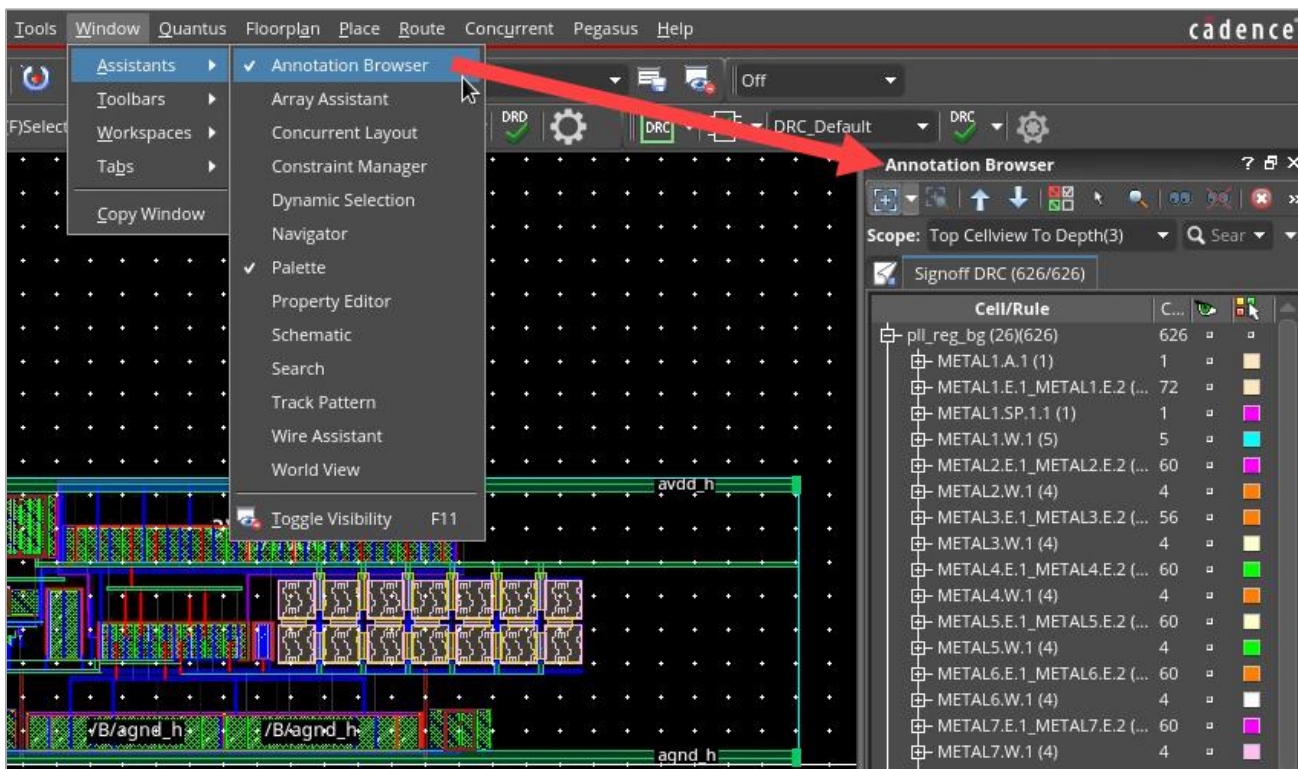


3. Select **Window > Assistants > Annotation Browser** in the Virtuoso Layout suite.



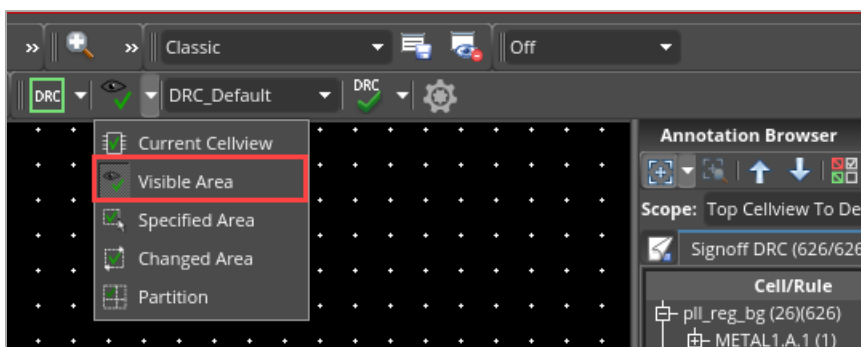
That loads the Annotation Browser to LHS.

4. Drag and drop the Annotation Browser utility to the RHS.



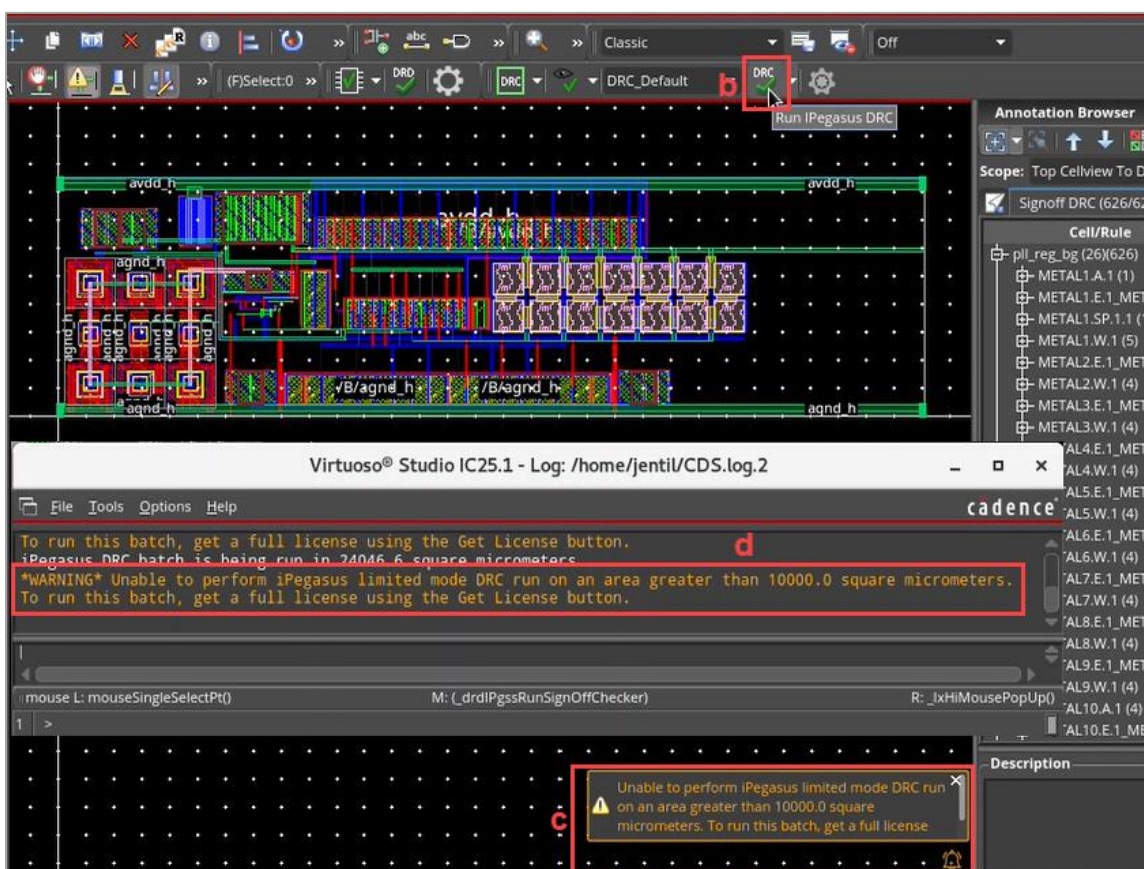
5. In the iPegasus toolbar:

- a. Switch the Checker Scope to *Visible Area*.



With this selection, iPegasus runs DRC checks on the visible area in the layout canvas window. Ensure the whole layout is visible (Fit mode) on the screen.

- b. Click the **Run iPegasus DRC** icon to start the run.

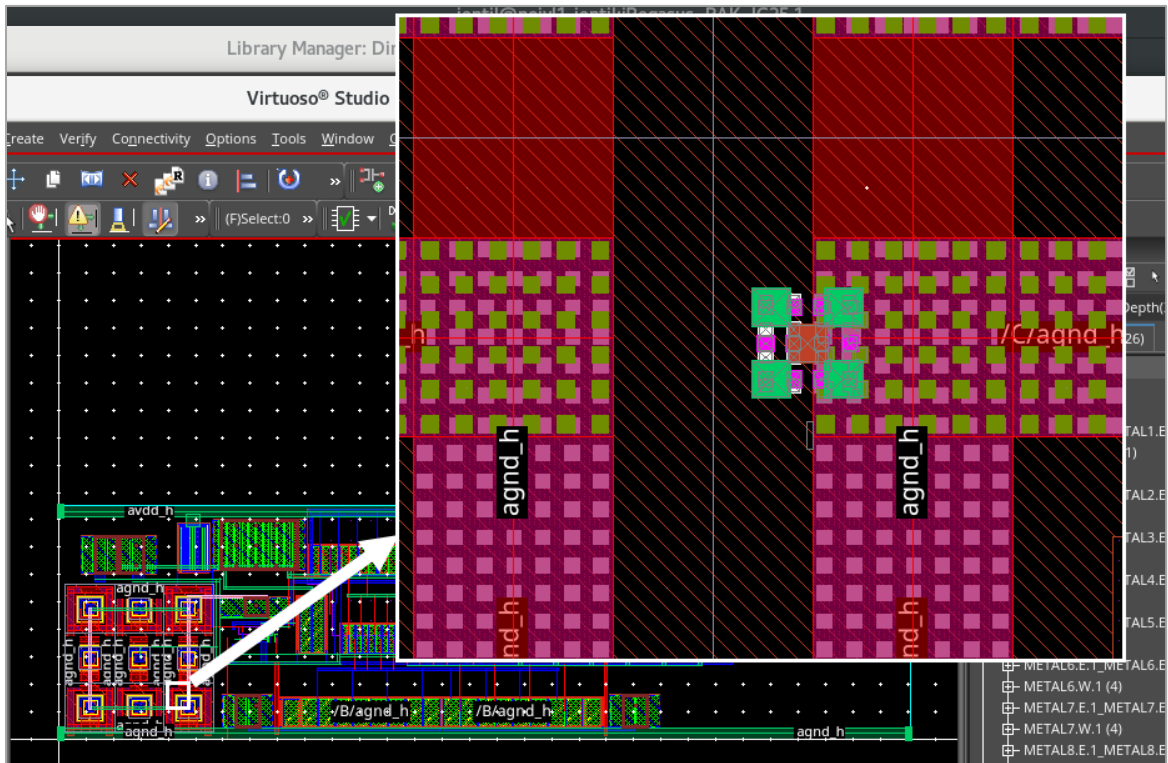


The run fails to take off as the checked area is more than the area allowed for a limited mode DRC run.

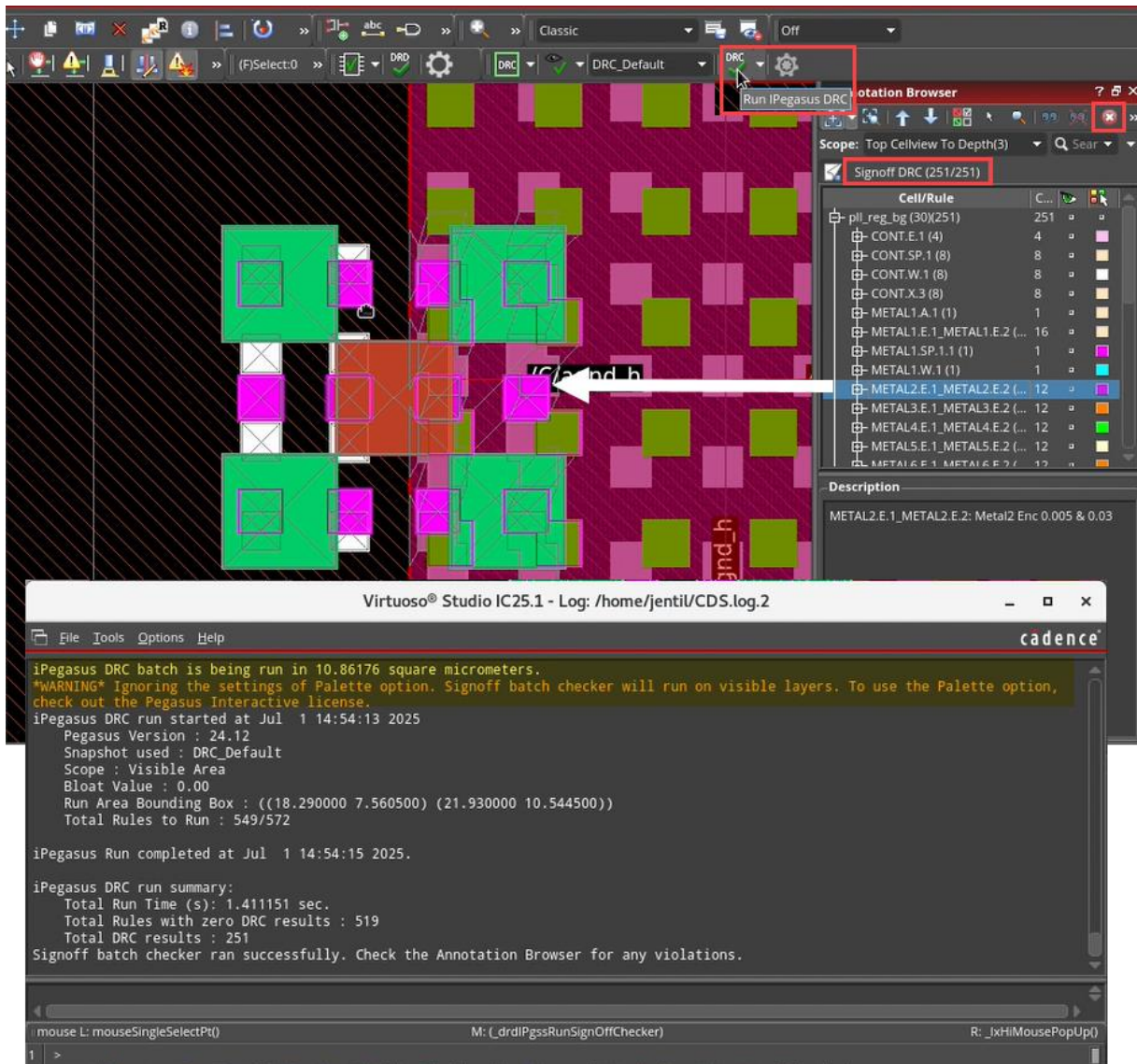
- c. A notification is displayed on the layout suite.
- d. A warning message is displayed on the CIW window.

To proceed with the run, you must obtain a full license using the Get License option under the Run iPegasus DRC drop-down.

- Back in the layout suite, zoom in to a smaller area to qualify for a limited-mode DRC run.



7. Click the **Run iPegasus DRC** icon to start the run.



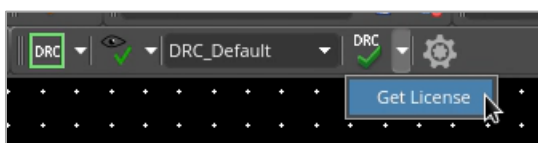
- The iPegasus DRC violations are loaded under the Annotation Browser → Signoff DRC tab. Analyze the results.
- Review the run details summarized in CIW. The checked area is much smaller, so the iPegasus can run DRC without the license.
- CIW message mentions that by default, the Palette feature is ON in limited mode iPegasus DRC runs.
- Click the **Delete All Markers** icon to clear the markers in the Annotation Browser.

Do not close the Virtuoso session.

Running iPegasus DRC – Full Mode

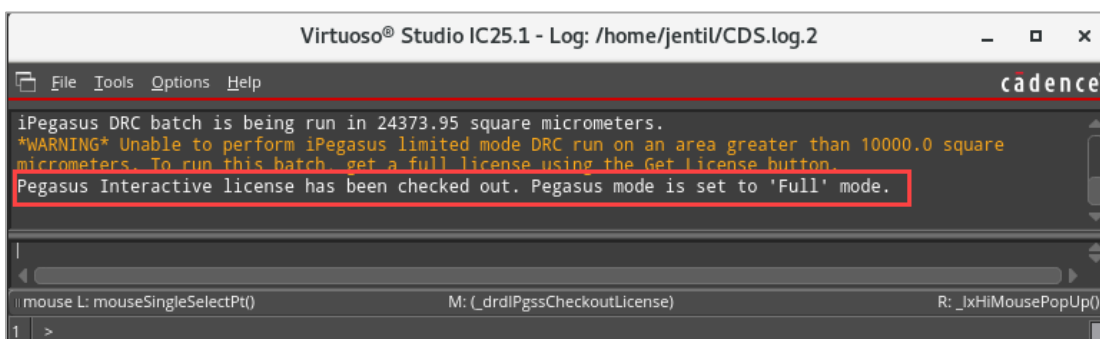
With the Full Mode, which requires the iPegasus DRC license, you can run InDesign DRC on designs regardless of the checked area size.

1. Ensure that *zambezi45 | pll_reg_bg| layout* is open in the layout suite and scope is set to *Visible Area*. Press **F** to fit the design.
2. To run iPegasus DRC on a complete design in full mode, select the *Get License* option from the DRC drop-down menu.



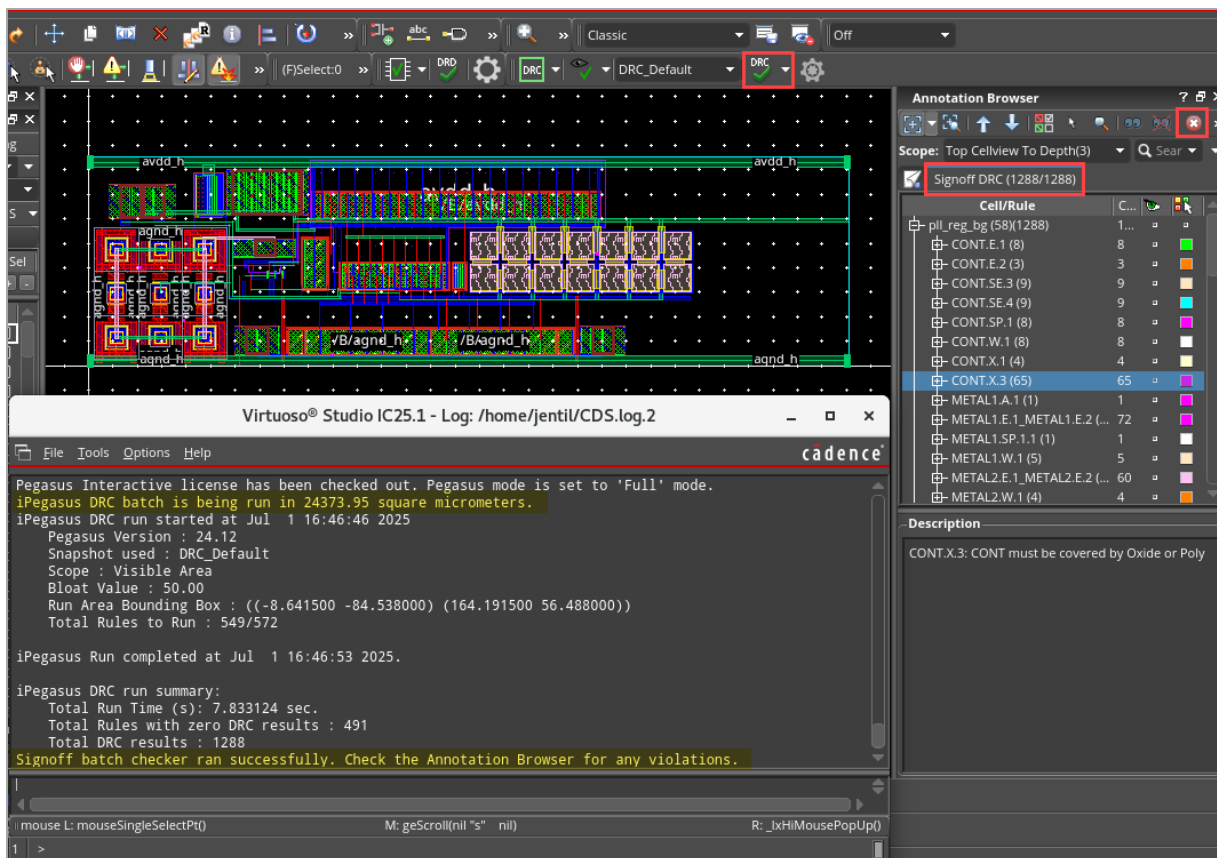
Virtuoso will fetch the iPegasus license.

The CIW log message will confirm that you are in iPegasus full mode.



Note: Once you click *Get License*, you cannot return to the limited mode DRC in the same session.

- Click the **Run iPegasus DRC** icon to start the run.



The iPegasus DRC violations for the design are loaded under the *Annotation Browser* → *Signoff DRC* tab. Analyze the results. The DRC is run with the signoff rule deck. Note that there is a wide range of violations related to metals, vias, contacts, poly, and N-well, to name a few.

- Clear all the markers by clicking the **Delete All Markers** icon in the Annotation Browser.

Do not close the Virtuoso session.

Running iPegasus DRC – With Snapshot

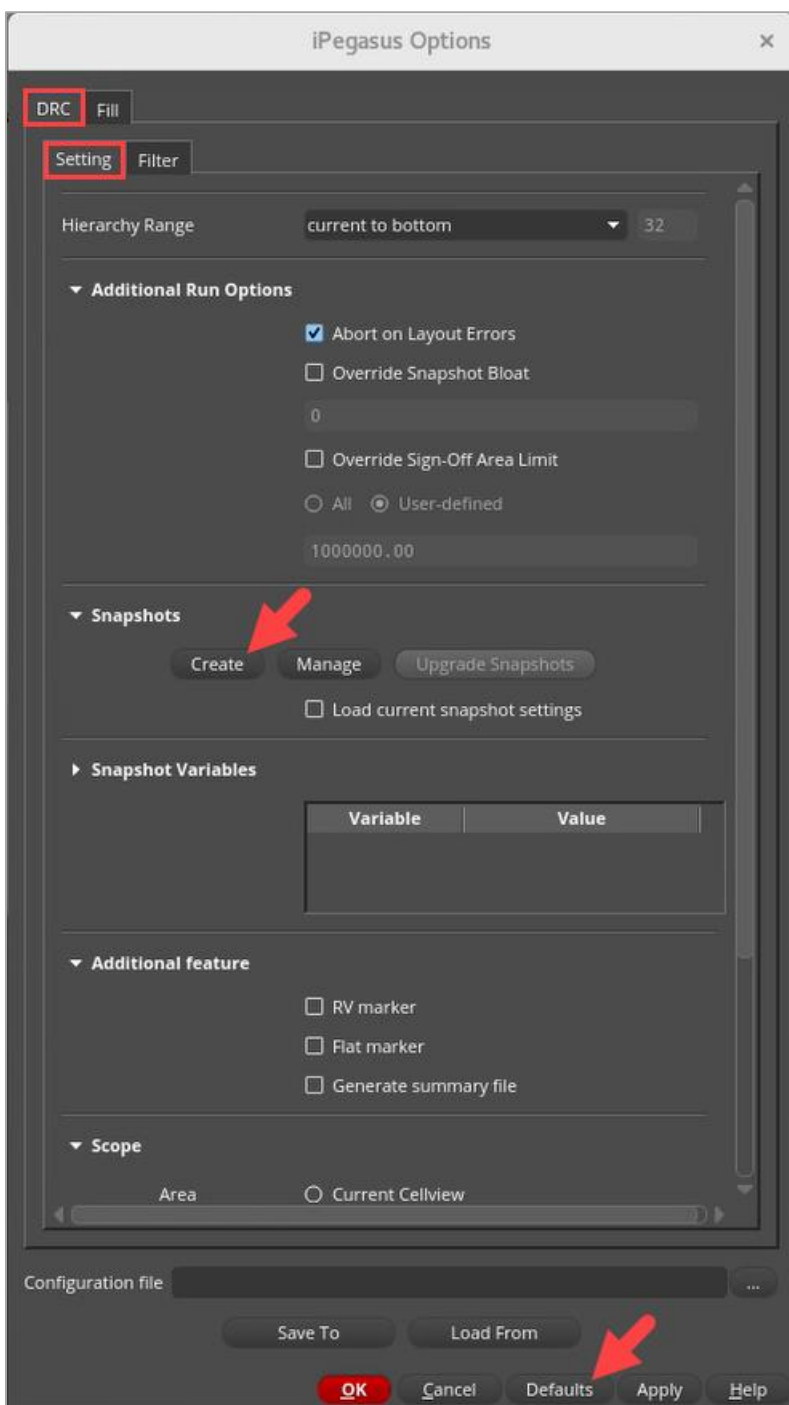
A Snapshot captures design rules in a compiled binary format, which only Pegasus understands. This format allows Pegasus direct access to selected design rules during DRC without reinterpreting them. Let's start with creating a sample snapshot.

- Ensure that *zambezi45 | pll_reg_bg | layout* is open in the layout suite. Press **F** to fit the design.

2. In the iPegasus DRC toolbar, click the **Open iPegasus DRC Options** icon.



That opens the *iPegasus Options - DRC* form.



3. In the **iPegasus Options - DRC** form:

a. Click on the **Defaults** icon to load the default state of the form.

b. In the Settings tab, click **Snapshots → Create**.

That opens the *Pegasus Snapshot Creator* form. We will create a Snapshot that includes only the Metal rules.

4. In the **Pegasus Snapshot Creator** form:

a. Under the **Input** tab, note that Pegasus picked up the layer map file from the technology.

b. Fill in Snapshot Name and Snapshot Comment as shown below:

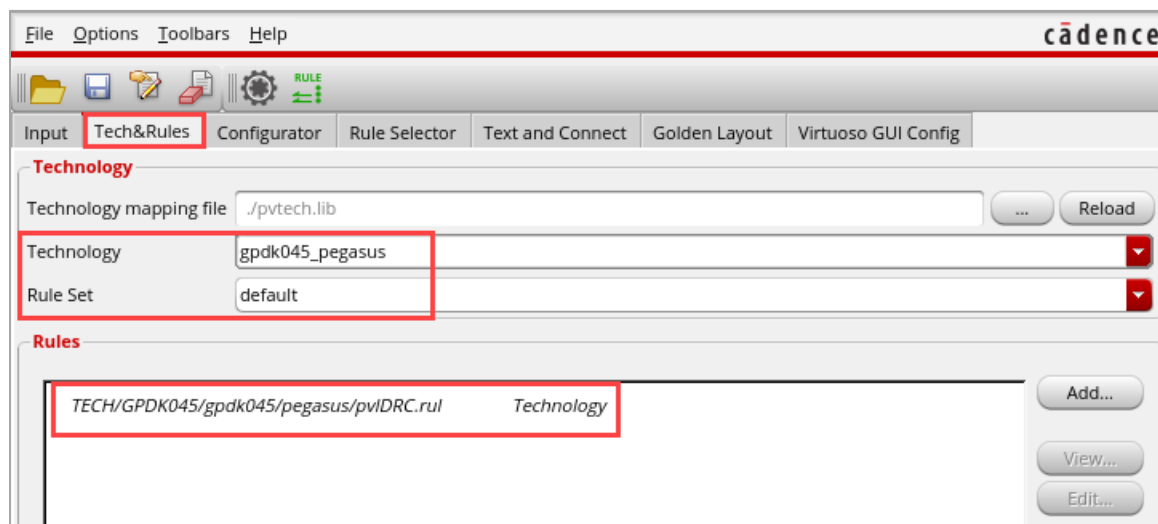
Snapshot Name: *DRC_ALL_METAL*

Snapshot Comment: *Info: DRC rules for all Metal (GPDk045)*

The screenshot shows the 'Pegasus Snapshot Creator' dialog box with the 'Input' tab selected. The 'Layer Map List' section contains a text field with the path 'ew/iPegasus_RAK_IC25.1/TECH/GPDk045/gpdk045_preRel_15DEC11/gpdk045/gpdk045.layermap'. Below this is an 'Object Name Table' and a checked 'Enable Coloring' checkbox. The 'Output' section at the bottom contains fields for 'Snapshot Name' (DRC_ALL_METAL), 'Snapshot Comment' (Info: DRC rules for all Metal (GPDk045)), 'Snapshot Path' (/servers/esd_noida/jentil/Pegasus241_07MAY/PEGASUS_LAB/iPegasus_new/iPegasus_RAK_IC25.1/snapshots), and 'Snapshot Type' (DRC). The 'OK' button is highlighted in red.

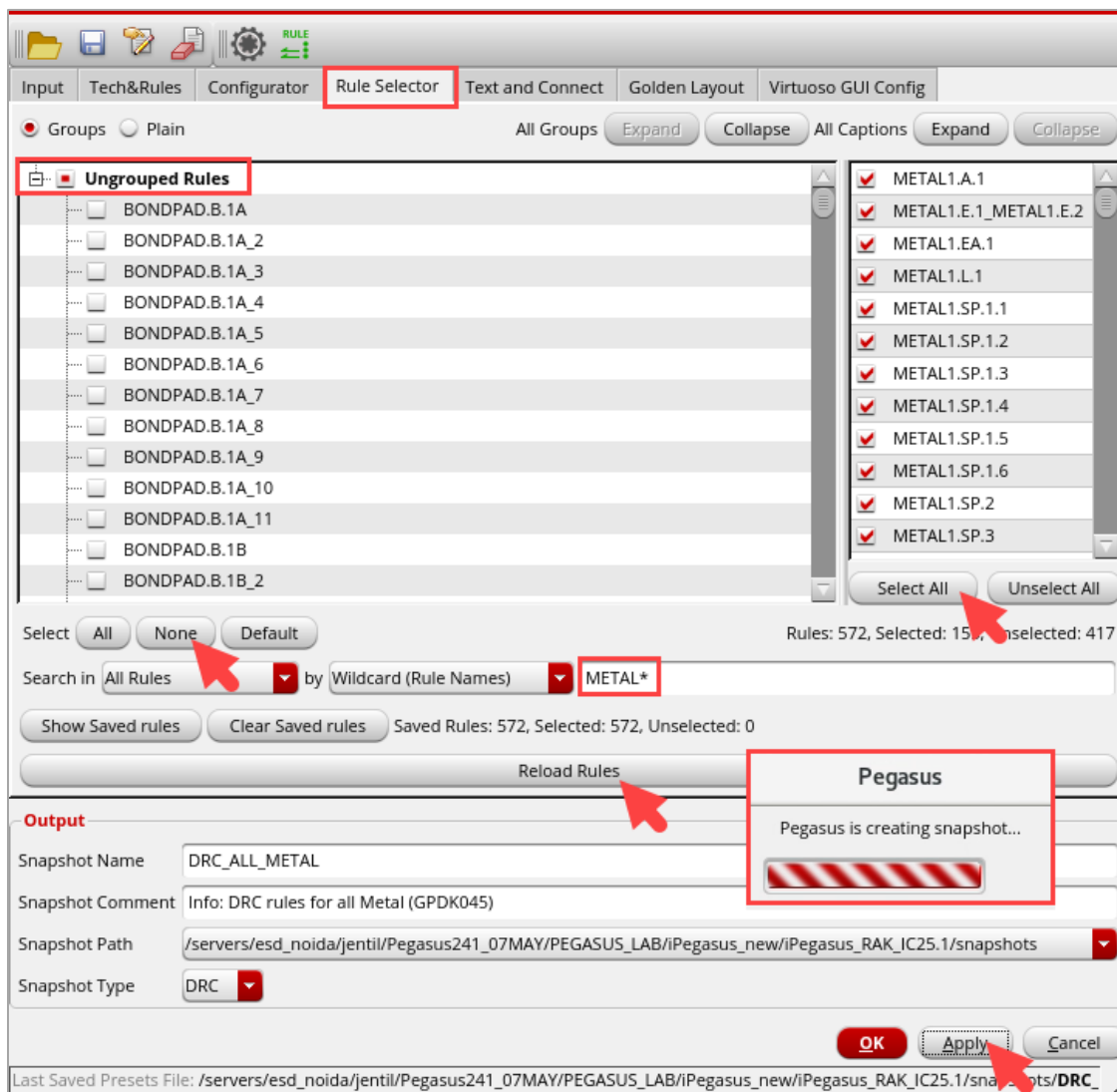
- c. Under the **Tech&Rules** tab, set technology to *gpdk045_pegasus*.

It will automatically pick the DRC rule deck from the technology area.

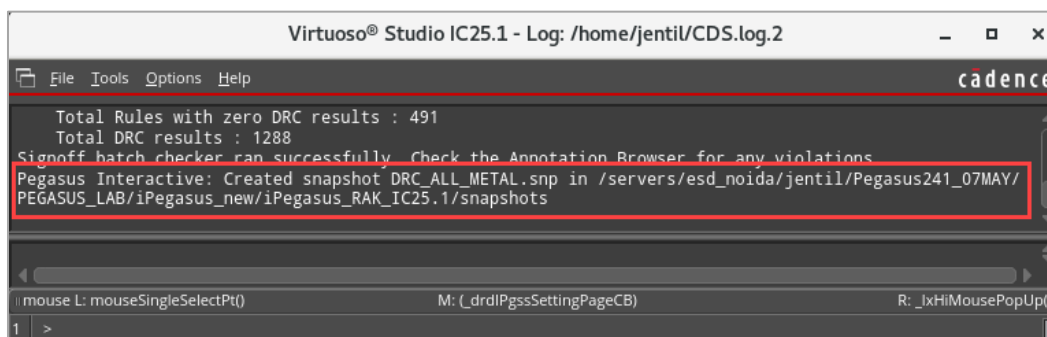


- d. Under the **Rule Selector** tab, click the **Reload Rules** field.
That will load the *Ungrouped Rules* from the active rule deck.
- e. On the *Ungrouped Rules* item, click the plus + sign.
That will expand and display the rules from the pvlDRC.rul file, with count.
- f. Filter only Metals-related rules by searching for *METAL** in the Search field.
It will show all the Metal rules on the right side.

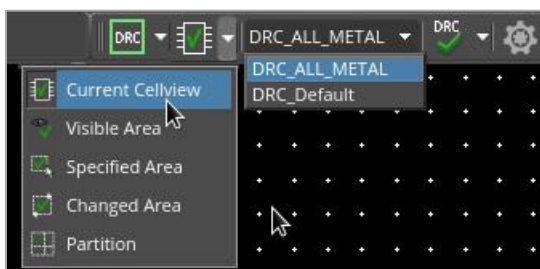
- g. Click **None** from the Select field and then **Select All** on the right-hand wildcard section to select all METAL* rules only.



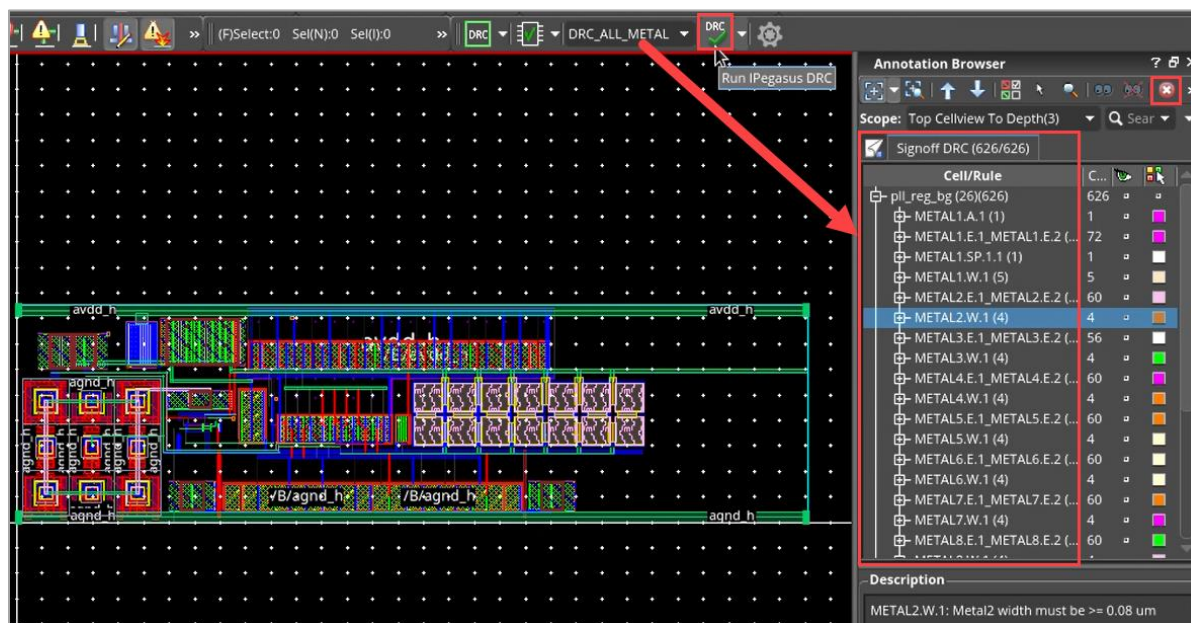
- h. Click **OK** or **Apply** on the Pegasus Snapshot Creator.
- A progress bar will be displayed until snapshot creation is complete.
- In CIW, Pegasus Snapshot Creator outputs the following message:



5. Switch back to the layout window and observe the newly created Snapshot DRC_ALL_METAL, which is populated under the snapshot combo field on the iPegasus DRC toolbar. In the toolbar:



- a. Select the **DRC_ALL_METAL** snapshot.
- b. Switch the Checker Scope to *Current Cellview*.
- c. Click the **Run iPegasus DRC** icon to start the run.



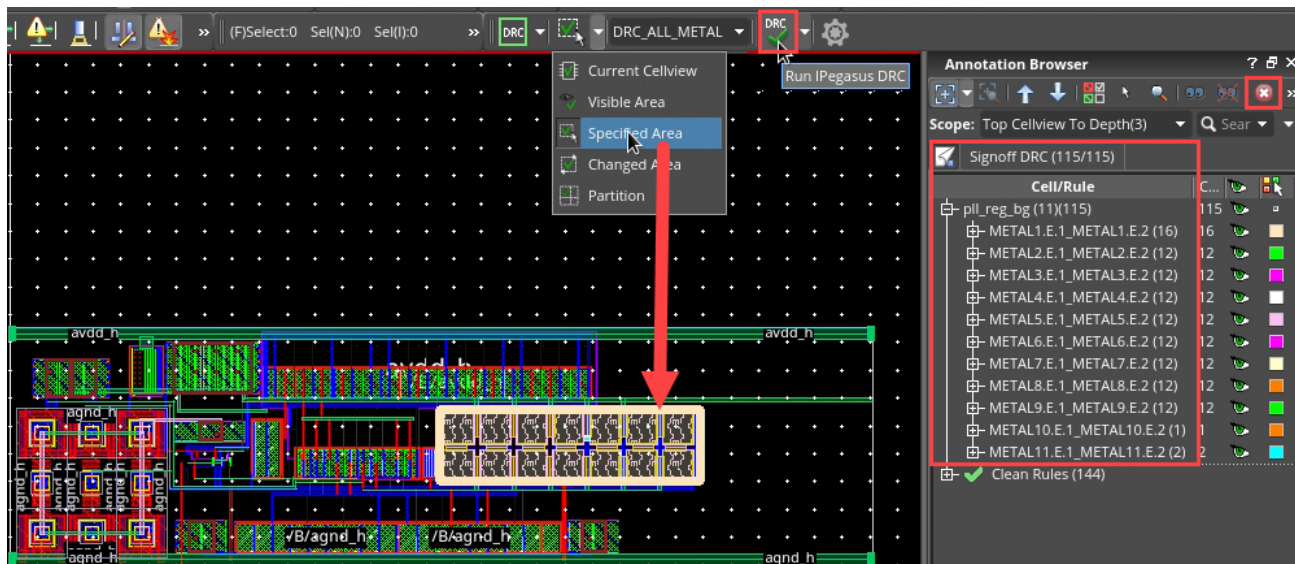
The DRC violations for the whole cell view are loaded under the *Annotation Browser* → *Signoff DRC* tab. Analyze the results. Note that only metal violations have been reported. This is due to the snapshot used for the run, DRC_ALL_METAL, a customized, metal-only-rules deck.

- d. Clear all the markers by clicking the **Delete All Markers** icon in the Annotation Browser.
6. Close the *Pegasus Snapshot Creator* and the *iPegasus DRC Options* forms.

Running iPegasus DRC Within the Specified Area

Use this feature to run iPegasus DRC only on a selected design portion.

1. Ensure that *zambezi45 | pll_reg_bg | layout* is open in the layout suite. Press **F** to fit the design.
2. In the iPegasus DRC toolbar:

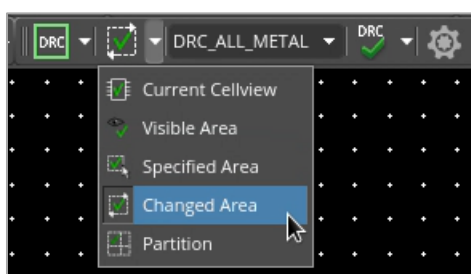


- a. Switch the Checker Scope to the *Specified Area*.
Virtuoso prompts you to draw the area where you want to run the DRC checker.
- b. Select an area as indicated in the picture below.
Click first at the left-top corner of the bounding box, then at the lower-right corner. The selected area is highlighted in a pink box.
- c. Leave the snapshot as *DRC_ALL_METAL*.
- d. Click the **Run iPegasus DRC** icon to start the run.
iPegasus will run the area-based DRC and report violations.
Notice the reduced number of DRC violations in the Annotation Browser.
Here, you have run only Metal DRC checks over a selected area.
- e. Clear all the markers by clicking the **Delete All Markers** icon in the Annotation Browser.

Running iPegasus DRC Only Over the Changed Area

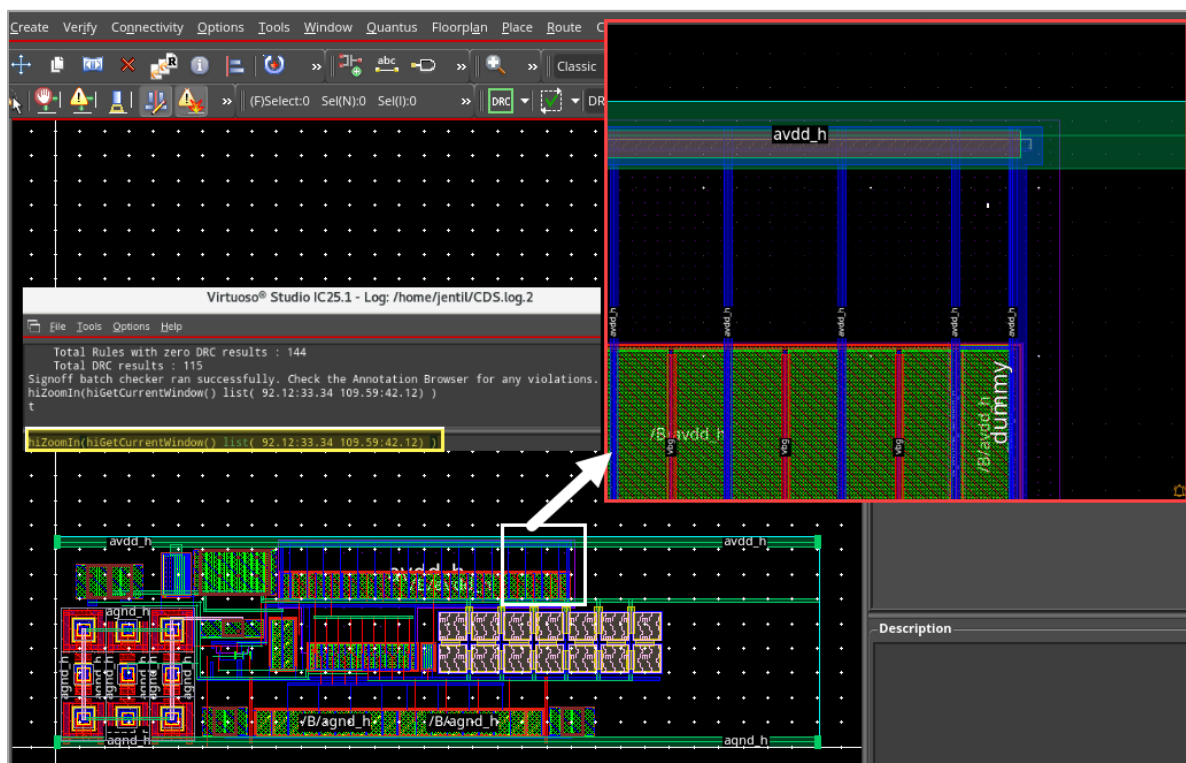
Use this feature to run iPegasus DRC only over the changed portion of the design. In the Changed Area mode, iPegasus tracks the layout editing portion in virtual memory. This area tracking is preserved between each DRC run.

1. Ensure that `zambezi45 | pll_reg_bg | layout` is open in the layout suite. Press **F** to fit the design.
2. In the iPegasus DRC toolbar:
 - a. Switch the Checker Scope to *Changed Area*.



- b. Leave the snapshot as `DRC_ALL_METAL`.
3. In the layout suite, zoom in to the area with the SKILL command

```
CIW> hiZoomIn(hiGetCurrentWindow() list( 92.12:33.34 109.59:42.12) )
```

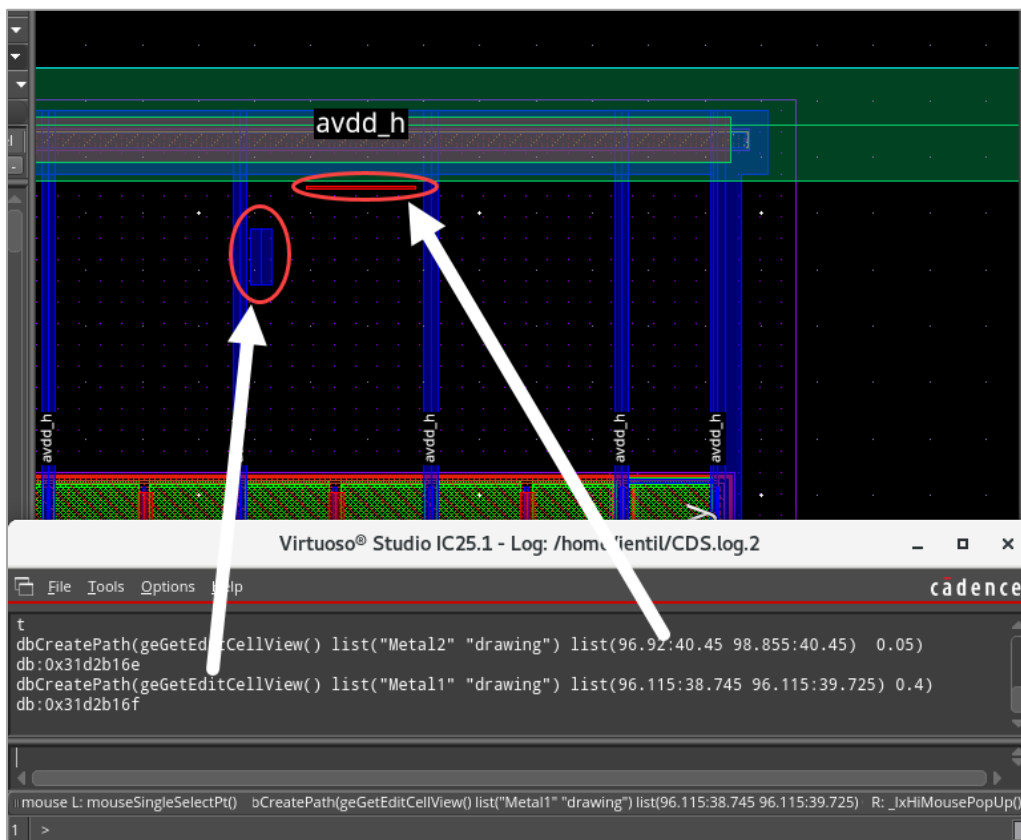


4. In CIW, type the following two commands:

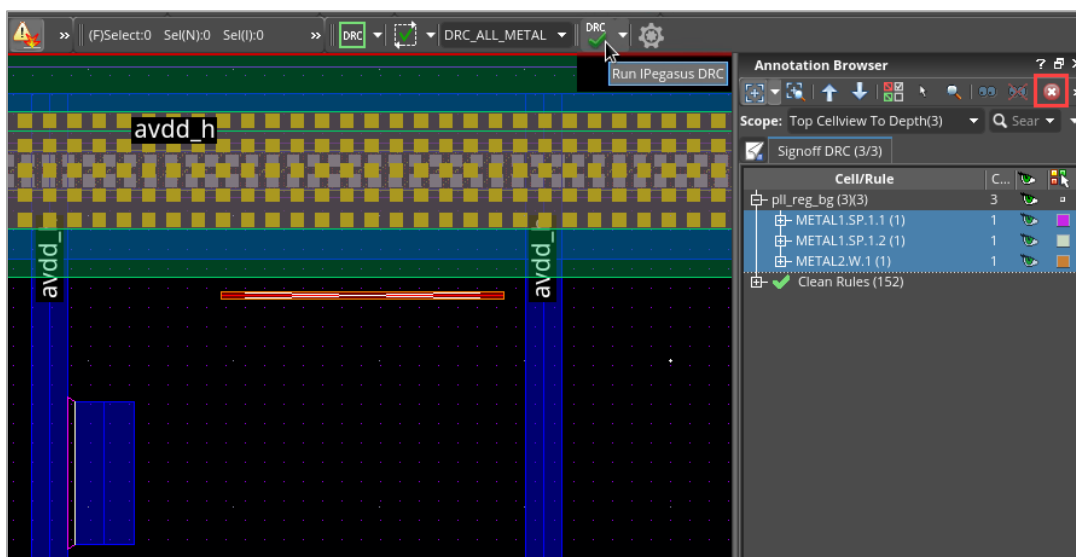
```
CIW> dbCreatePath(getEditCellView() list("Metal2" "drawing")
list(96.92:40.45 98.855:40.45) 0.05)
```

```
CIW> dbCreatePath(getEditCellView() list("Metal1" "drawing")
list(96.115:38.745 96.115:39.725) 0.4)
```

These commands will create a couple of path segments, as shown below:

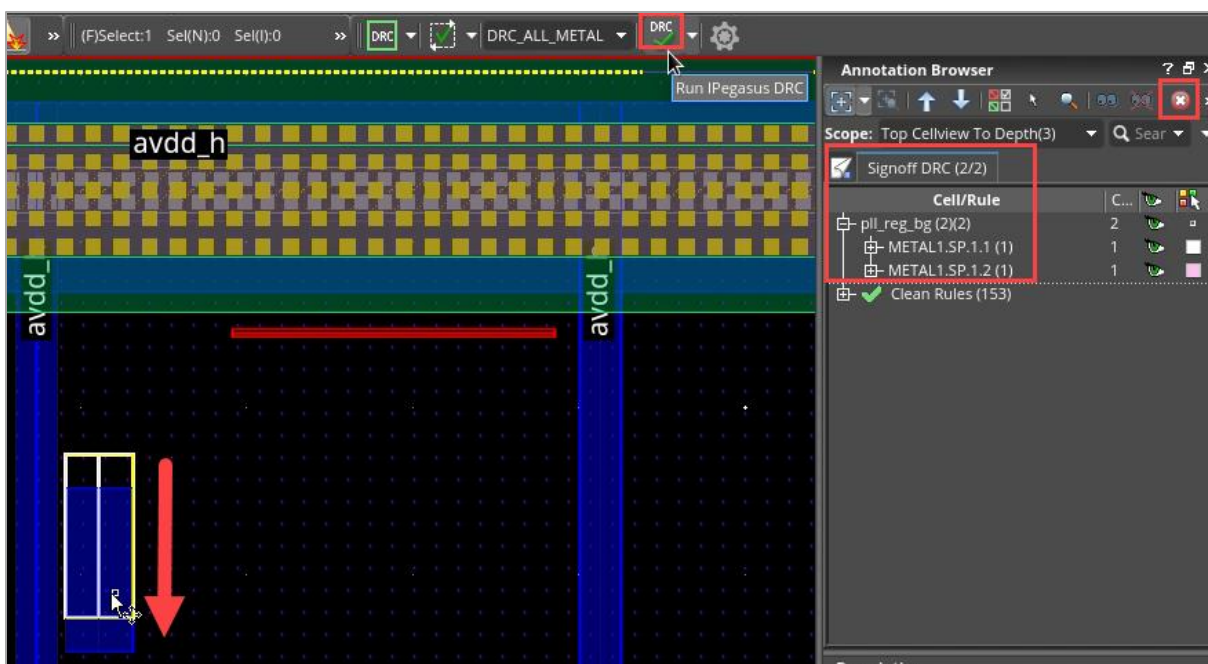


5. Click the **Run iPegasus DRC** icon to start the run.



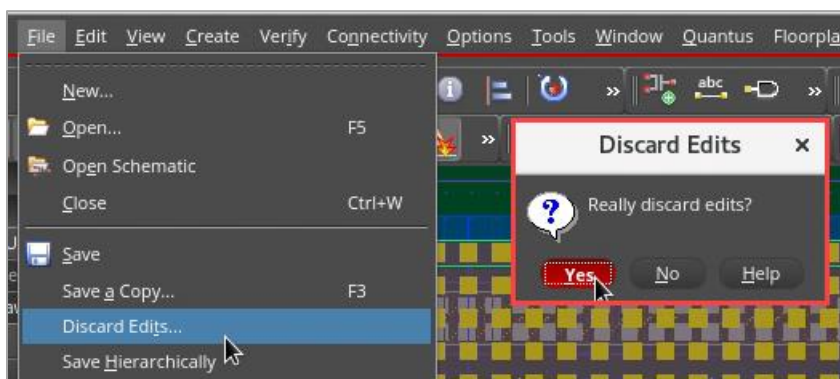
DRC violations on Metal1 and Metal2 are present around the same area where the newly created Metal1 and Metal2 paths are present.

6. Clear all the markers by clicking the **Delete All Markers** icon in the Annotation Browser.
7. Move the Metal1 path a little bit in the vertical direction.



8. Click the **Run iPegasus DRC** icon to start the run.
You will see that only Metal1 DRC violations are reported in the Annotation Browser since we have moved only the Metal1 path horizontally between the DRC runs.
9. Clear all the markers by clicking the **Delete All Markers** icon in the Annotation Browser.
10. In the layout suite, discard the layout edits by selecting **File → Discard Edit**.

Click **Yes** in the pop-up.



Running iPegasus DRC with Rule Filter

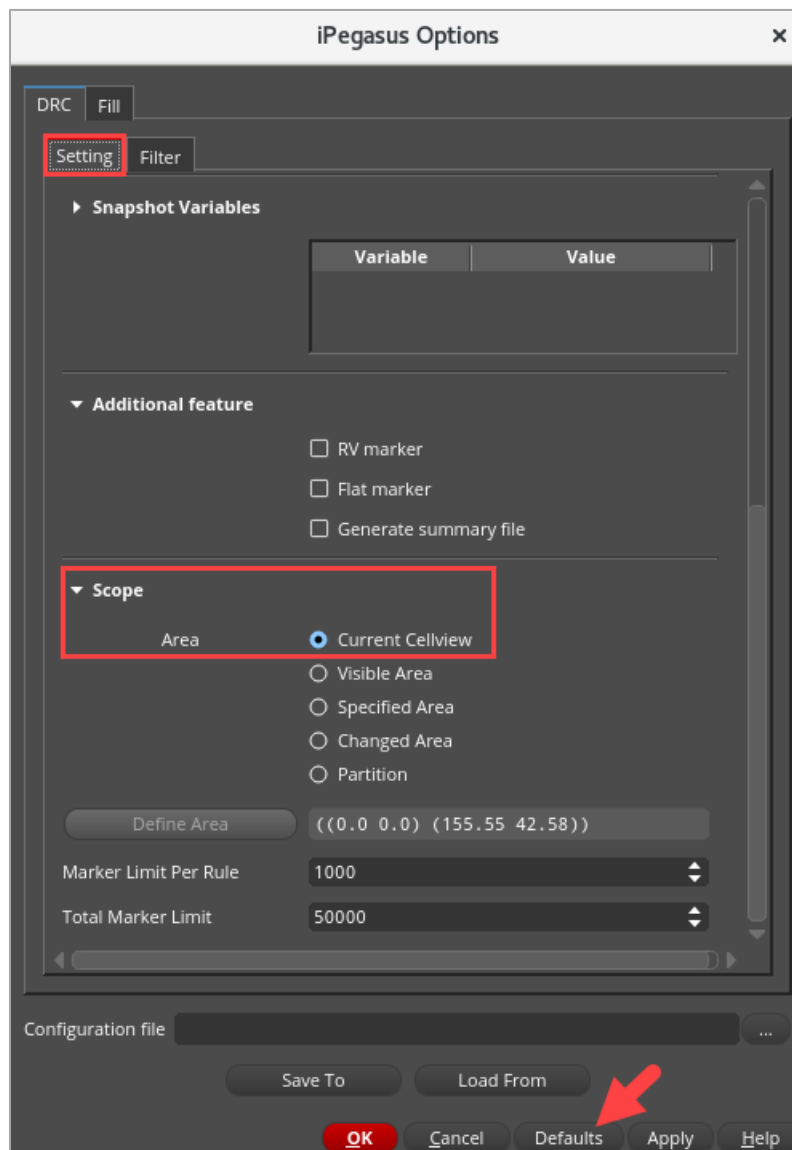
Use this feature to run iPegasus DRC with only a selected set of rules.

1. Ensure that *zambezi45 | pll_reg_bg | layout* is open in the layout suite. Press **F** to fit the design.
2. In the iPegasus DRC toolbar,



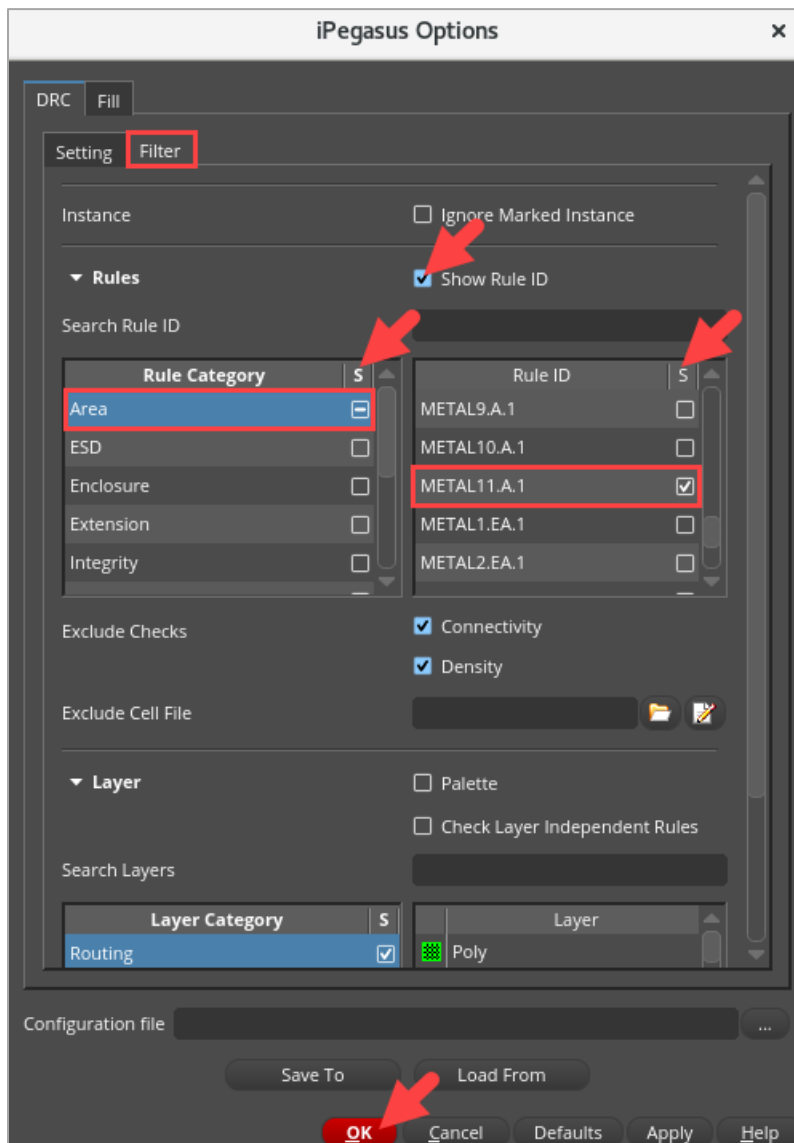
- a. Change the snapshot to *DRC_Default*.
 - b. Click the **Open iPegasus DRC Options** icon.
That opens the *iPegasus DRC Options* form.
3. In the **iPegasus Options - DRC** form:
 - a. Click on the **Defaults** icon to load the default state of the form.

- b. Under the Setting tab, ensure the **Current Cellview** is selected as the checker Scope.



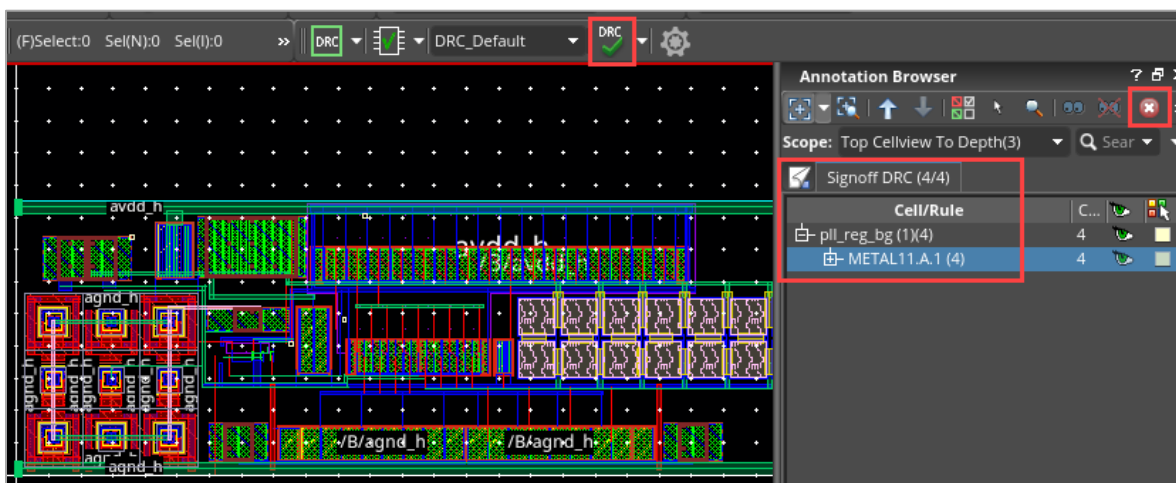
- c. Under the Filters tab, review the *Rules* section.
- d. Click **S** in the header of *Rule Category* to unselect all categories.
- e. Select the checkbox for the **Area** under Rule Category.
- f. Select the checkbox for the **Show Rule ID**.
- Review the *Area* category rules listed under the rules section on RHS.
- g. Click **S** in the header of the *Rule ID* section on RHS to unselect all area rules.
- h. Select **METAL1.A.1** rule only.

- i. Click **OK** on the *iPegasus Options - DRC* form.



4. Click the **Run iPegasus DRC** icon to start the run.

Investigate violations with the Annotation Browser. There are only METAL1.A.1 violations.



- a. Clear all the markers by clicking the **Delete All Markers** icon in the Annotation Browser.

Running iPegasus DRC with Palette Filter

Use this feature to run iPegasus DRC only on layers visible in the Virtuoso Layer Palette.

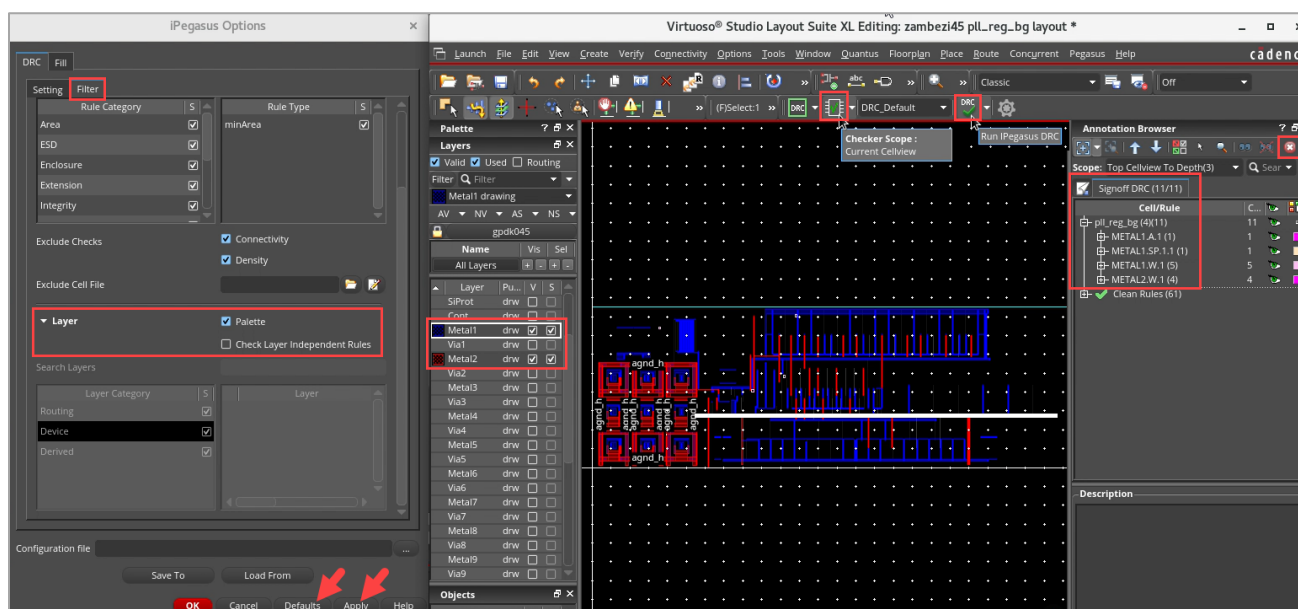
1. Ensure that *zambezi45 | pll_reg_bg | layout* is open in the layout suite. Press **F** to fit the design.
2. In the iPegasus DRC toolbar,



- a. Make sure that the snapshot is set to *DRC_Default*.
- b. Click the **Open iPegasus DRC Options** icon.
That opens the *iPegasus DRC Options* form.
3. In the **iPegasus Options - DRC** form:
 - a. Click on the **Defaults** icon to load the default state of the form.
 - b. Under the Filter tab – Layer section, select the checkbox for **Palette**.

This will deactivate the ‘*Search Layers*’ section in the form and enable the *Palette* feature. The *Palette* feature instructs iPegasus to run DRC on only the visible layer(s) in the *Layers Palette* window.

- c. Note that, by default, the *Check Layer Independent Rules* checkbox is unselected.
 - d. Click **OK** on the iPegasus Options - DRC form.
4. In the *Virtuoso Layout Suite* → *Layers Palette* window, turn on visibility only for the Metal1 and Metal2 drawings.



5. In the iPegasus DRC toolbar:
 - a. Ensure that the Checker Scope is set to *Current Cellview*.
 - b. Click the **Run iPegasus DRC** icon to start the run.
Investigate violations with the *Annotation Browser*. DRC violations for only Metal1 and Metal2 drawings are reported in the *Annotation Browser*.
 - c. Clear all the markers by clicking the **Delete All Markers** icon in the *Annotation Browser*.
 - d. In the *Virtuoso Layout Suite* → *Layers Palette* window, turn on visibility of all layers.

Do not close the Virtuoso session.



Lab 5-2 Setting Up and Running iPegasus Fill

Objective: To set up and use iPegasus to perform signoff quality fill.

The iPegasus Fill feature, powered by the Pegasus sign-off engine, can generate dummy fills using the complete foundry fill decks or a customized set of rules. The iPegasus Fill feature is snapshot-based. The tool automatically maps generated dummy fills into the Virtuoso OpenAccess database. The generated fills are immediately included in the block library-cell-view. There is no need to merge the fills with the design manually.

This lab will provide a practical overview of the iPegasus fill package.

Note: Licenses required for this lab.

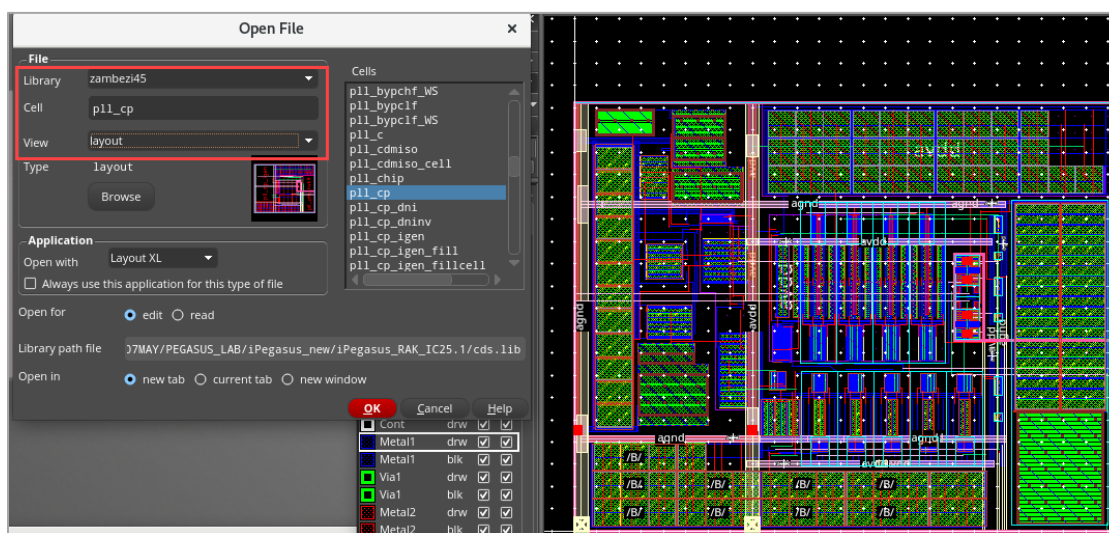
Pegasus Licenses needed: Cadence iPegasus Fill for Virtuoso Studio

Virtuoso Licenses needed: Virtuoso Layout Suite XL

Run iPegasus Fill – With Snapshot

In this section, we will learn how to perform sign-off quality Metal fill using the iPegasus Fill feature with a snapshot.

1. In the Virtuoso Layout Suite, open the following cell: Library *zambezi45*, cell *pll_cp*, view *layout*. Press **F** to fit the design.



2. Change the iPegasus Toolbar to Fill mode.



3. In the iPegasus Fill Toolbar, click the **Open iPegasus Fill Options** icon.

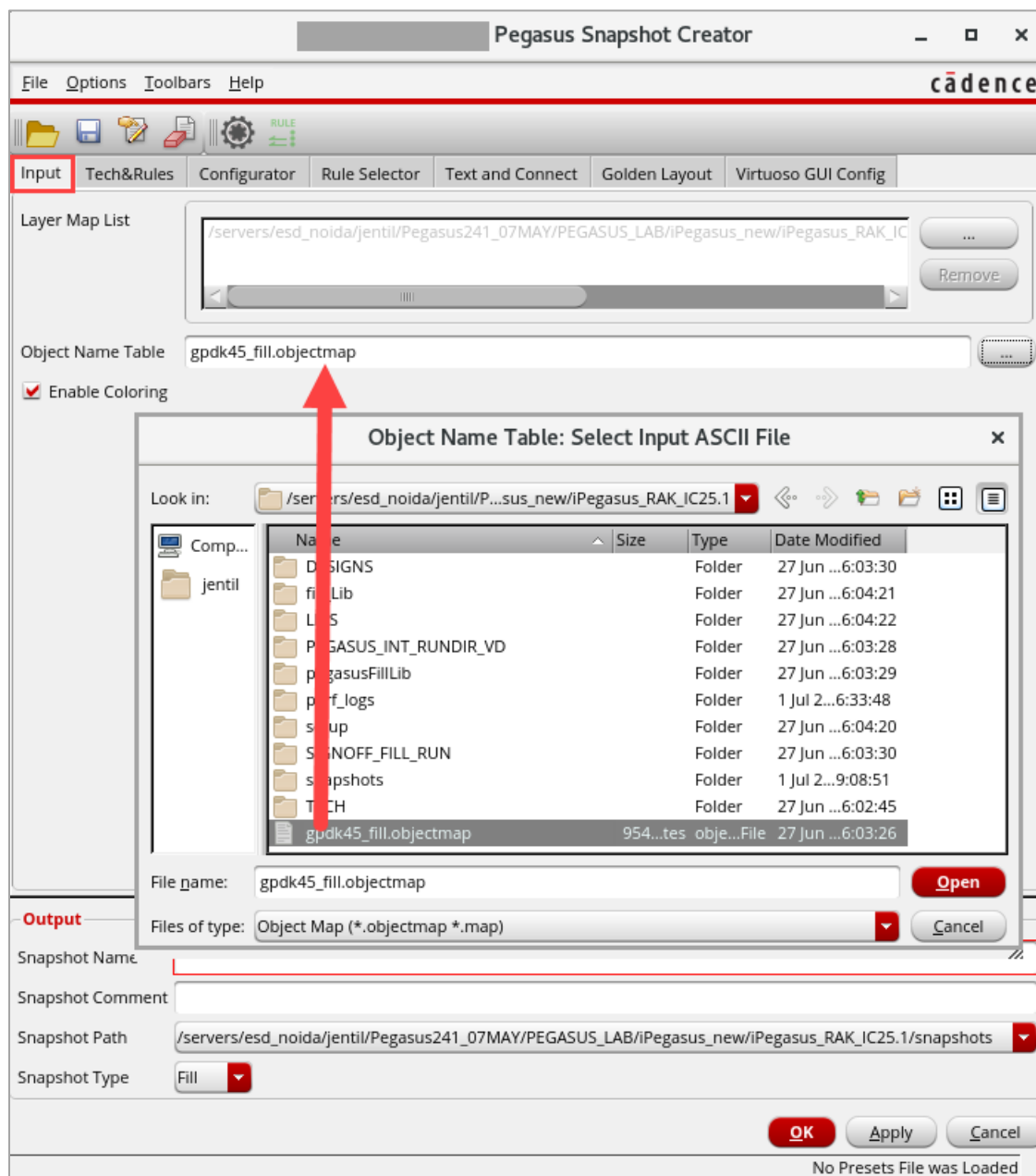
That opens the *iPegasus Options - Fill* form.

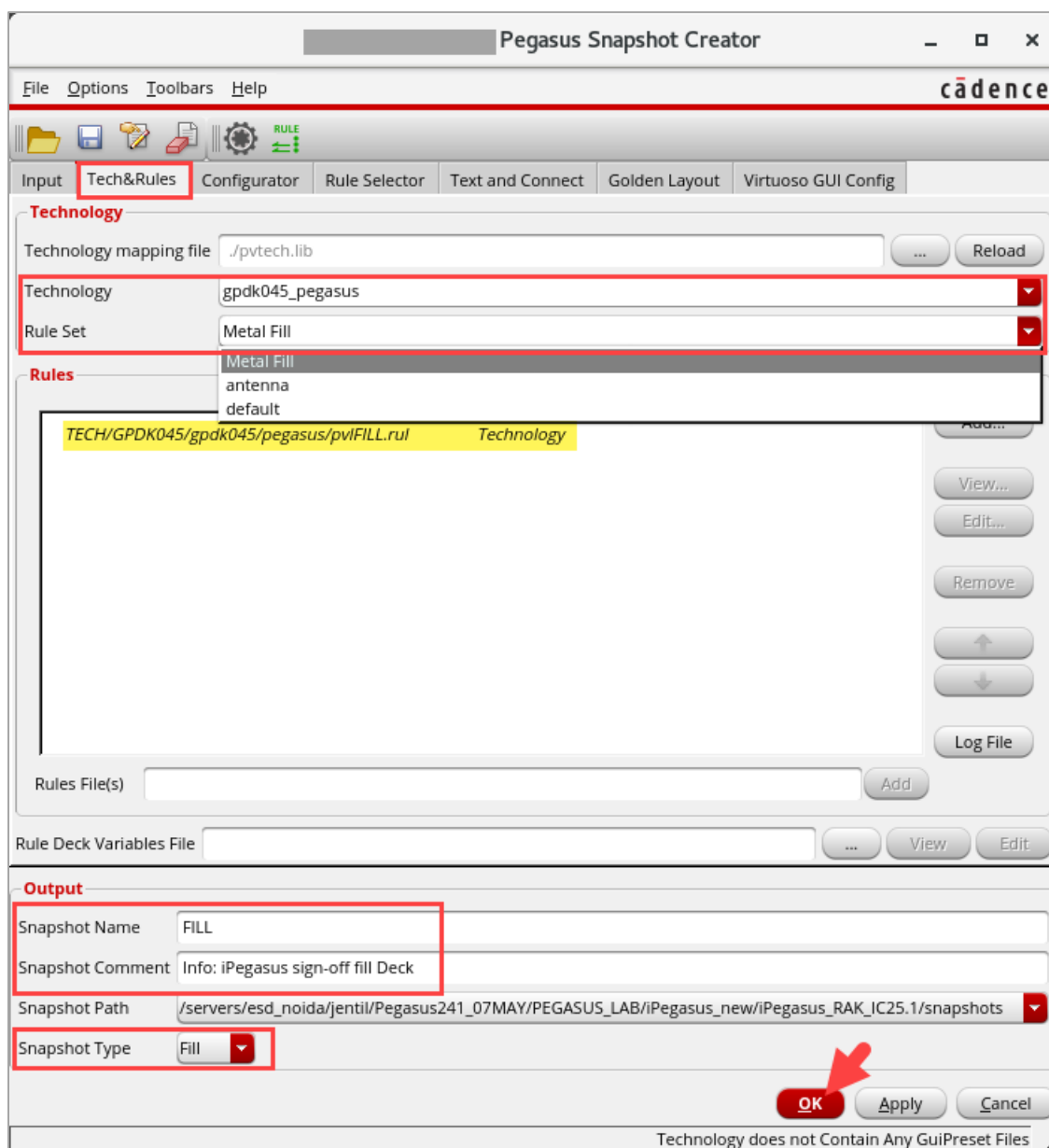
The screenshot shows the **iPegasus Options** dialog box with the **Fill** tab selected. The **Setting** section is expanded, showing the **Input** section with the **Layer Map** field set to `S_preRel_15DEC11/gpdk045/gpdk045.layermap`. The **Output** section includes fields for **Fill Instance Name**, **Library**, **Cell**, and **View**, along with checkboxes for **Create Hierarchical Fill** (checked) and **Clear Existing Fill**. The **Scope** section has radio buttons for **Area** (selected) and **Specified Area**, with sub-options **Current Cellview** and **Specified Area**. The **Advanced Options** section includes a **Symmetry** dropdown (set to **None**), checkboxes for **Mirrored** and **Specify Offset**, **X-Offset** and **Y-Offset** fields (both set to 0), a **Consider** dropdown (set to **All Layers**), **Exclude Cell File** and **Exclude Nets** fields, and checkboxes for **Override Sign-Off Area Limit** (unchecked) and **Load current snapshot settings** (unchecked). The **Snapshots** section has buttons for **Create**, **Manage**, and **Upgrade Snapshots**, with a red arrow pointing to the **Create** button. The **Configuration file** field is at the bottom, with **Save To** and **Load From** buttons. The **OK**, **Cancel**, **Defaults**, **Apply**, and **Help** buttons are at the bottom right.

4. In the **iPegasus Options - Fill** form:
 - a. Note that the Layermap file is already loaded from the Technology directory.
 - b. Click **Create** under the Snapshots section.

It will open the *Pegasus Snapshot Creator* form.

5. In the **Pegasus Snapshot Creator → Input** tab, under the Object Name Table field, use the Browse button to load `../gpd45_fill.objectmap` file.



6. In the **Pegasus Snapshot Creator** → **Tech & Rules** tab

a. Select *gpdK045_pegasus* as Technology.

b. Select *Metal Fill* as Rule Set.

Corresponding rule deck (*pvlFILL.rul*) will be loaded under the Rules section below.

c. Fill in Snapshot Name and Snapshot Comment as shown below:

Snapshot Name: *FILL*

Snapshot Comment: *Info: iPegasus sign-off fill Deck*

d. Note that Snapshot Type is set to *Fill*.

7. Click **OK** on the Pegasus Snapshot Creator.

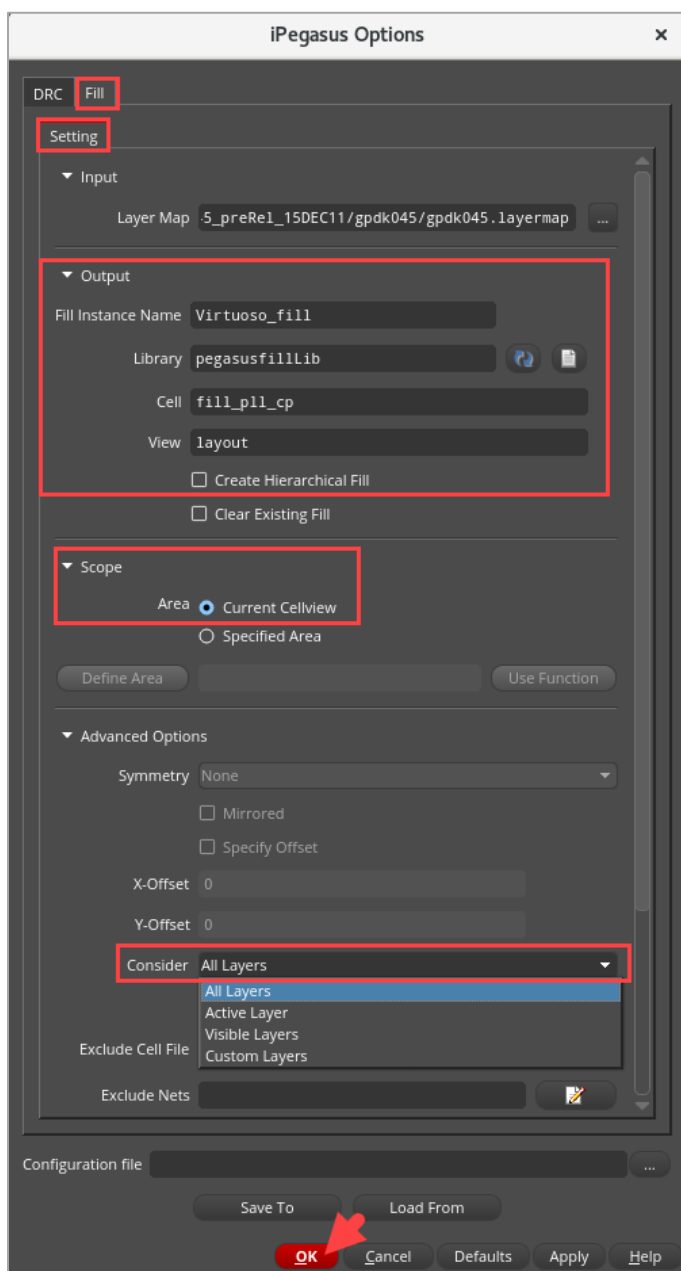
A progress bar will be displayed until snapshot creation is complete.

The new snapshot will be listed under the snapshot dropdown in the *iPegasus Fill Toolbar*.

- a. Select the 'FILL' snapshot in the iPegasus Fill Toolbar.



8. Make the following entries in the *iPegasus Options – Fill* form.

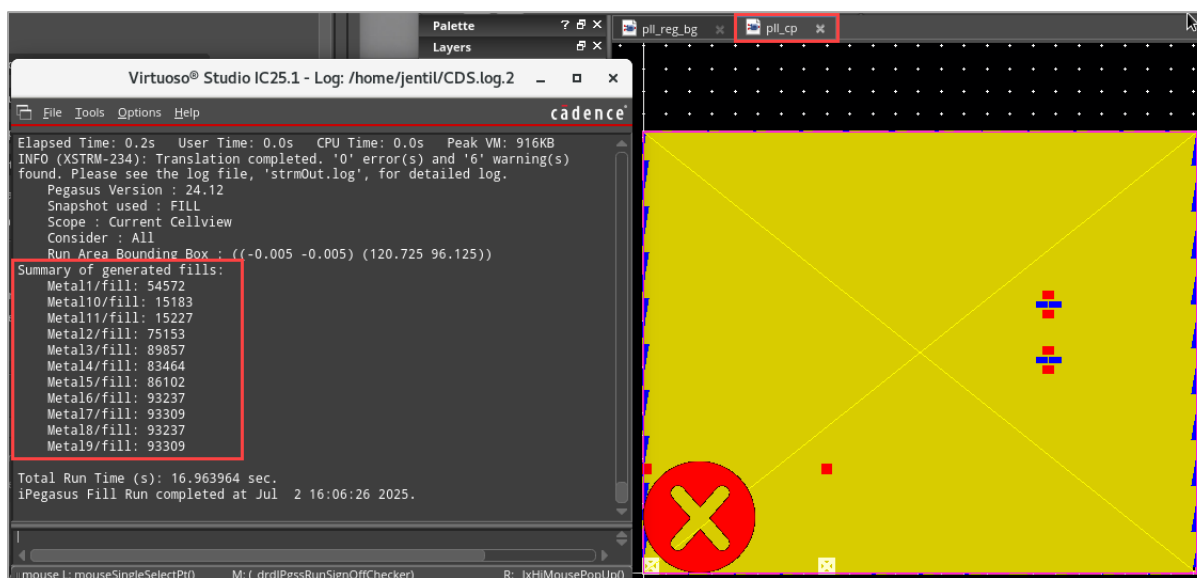


- a. Set the Fill Instance Name to *Virtuoso_fill*.
 - b. Set the output library to *pegasusfillLib*.
 - c. Set the fill Cell name to *fill_pll_cp*.
 - d. Set the View to *layout*.
 - e. Unselect the *Create Hierarchical Fill* field.
 - f. Ensure that the Checker Scope is set to *Current Cellview*.
 - g. Under Advanced Options, set to consider *All Layers* for fill.
 - h. Click **Apply**.
9. Ensure the following setup in the iPegasus Toolbar.



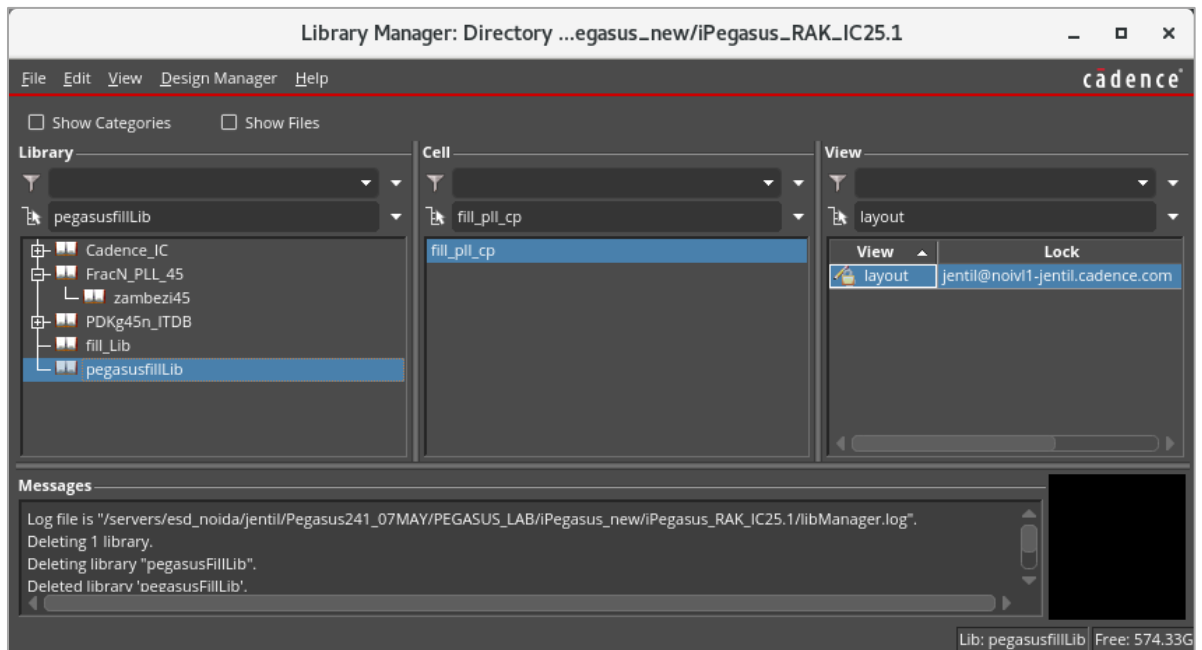
10. In the iPegasus Toolbar, click on the **Run iPegasus Fill** icon to start the run.

Once fill generation is completed,

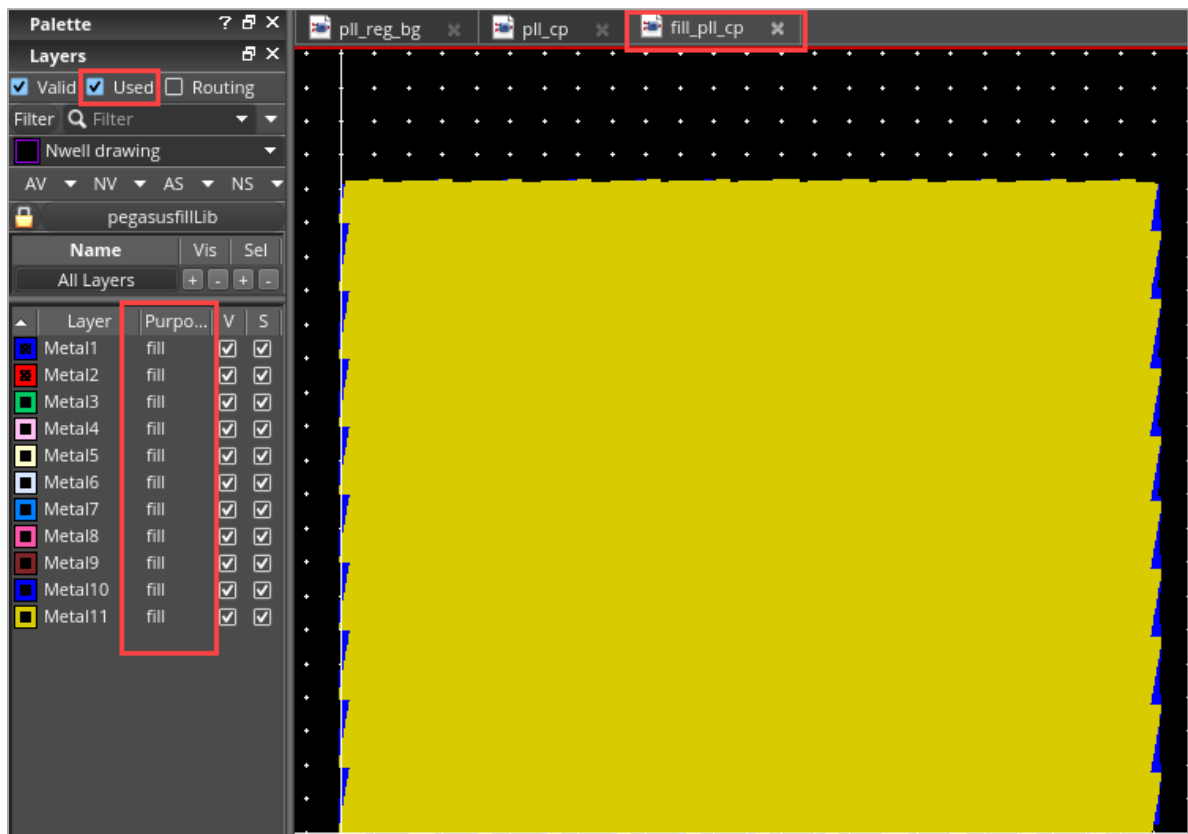


- a. Refer to the CIW window to check the generated fill summary. It will display the run summary and the number of fill shapes generated for each metal layer.
- b. Note that the newly generated fill cell is automatically instantiated over the core design.

11. Open the newly created fill cell in the Library Manager: `pegasusfillLib | fill_pll_cp| layout`.



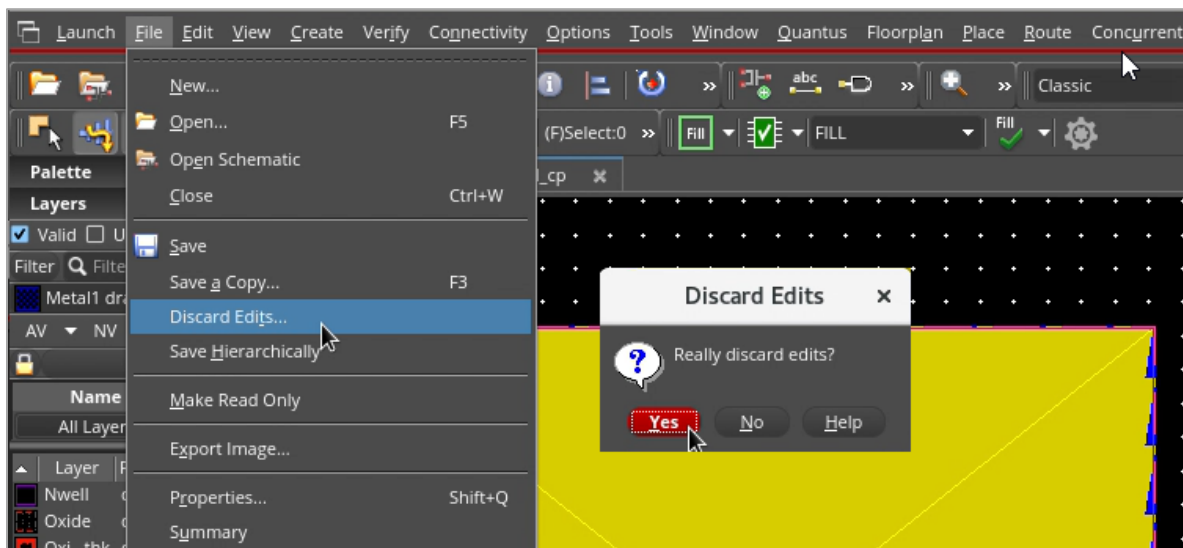
12. In the layout suite, show only the used layers by selecting **Used** in the Palette window.



Notice that this cell has only shapes of LPP ("Metal*" "fill").

13. Close cell: *pegasusfillLib* | *fill_pll_cp* | *layout*.

14. Switch to *zambezi45* | *pll_cp* | *layout* window, and remove the fill shapes by selecting **File → Discard Edits...**



a. Click **Yes** in the pop-up.

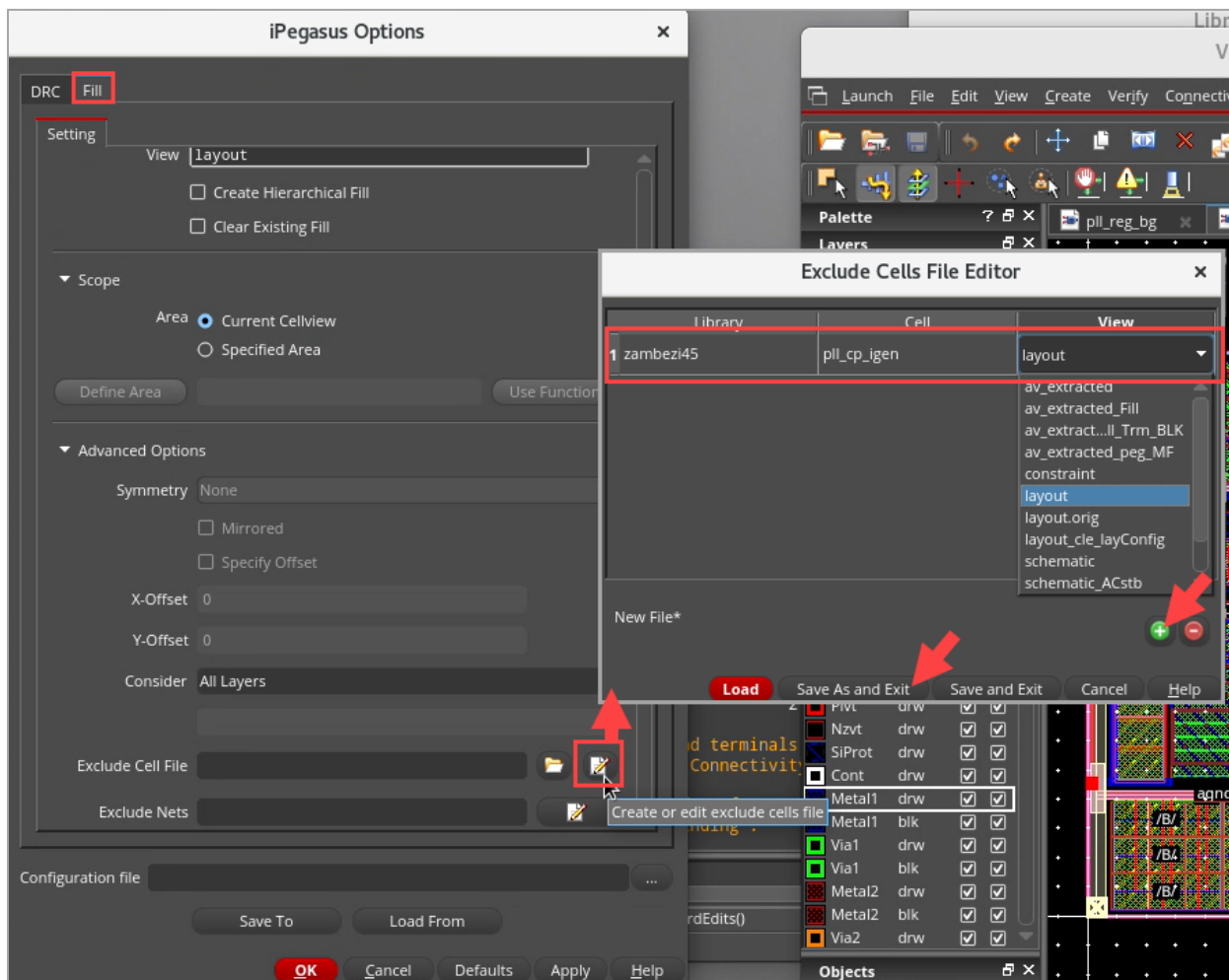
Do not close the iPegasus Options - Fill form.

Running iPegasus Fill – With Exclude Cells

In this lab, you will learn how to perform iPegasus Metal-Fill while excluding selected cells.

1. Ensure that *zambezi45* | *pll_cp* | *layout* is open in the layout suite. Press **F** to fit the design. Turn ON the visibility of all layers.

2. In the iPegasus Options - Fill form, click **Advance Options** → **Exclude Cell File** → **Create or edit exclude cells file**.



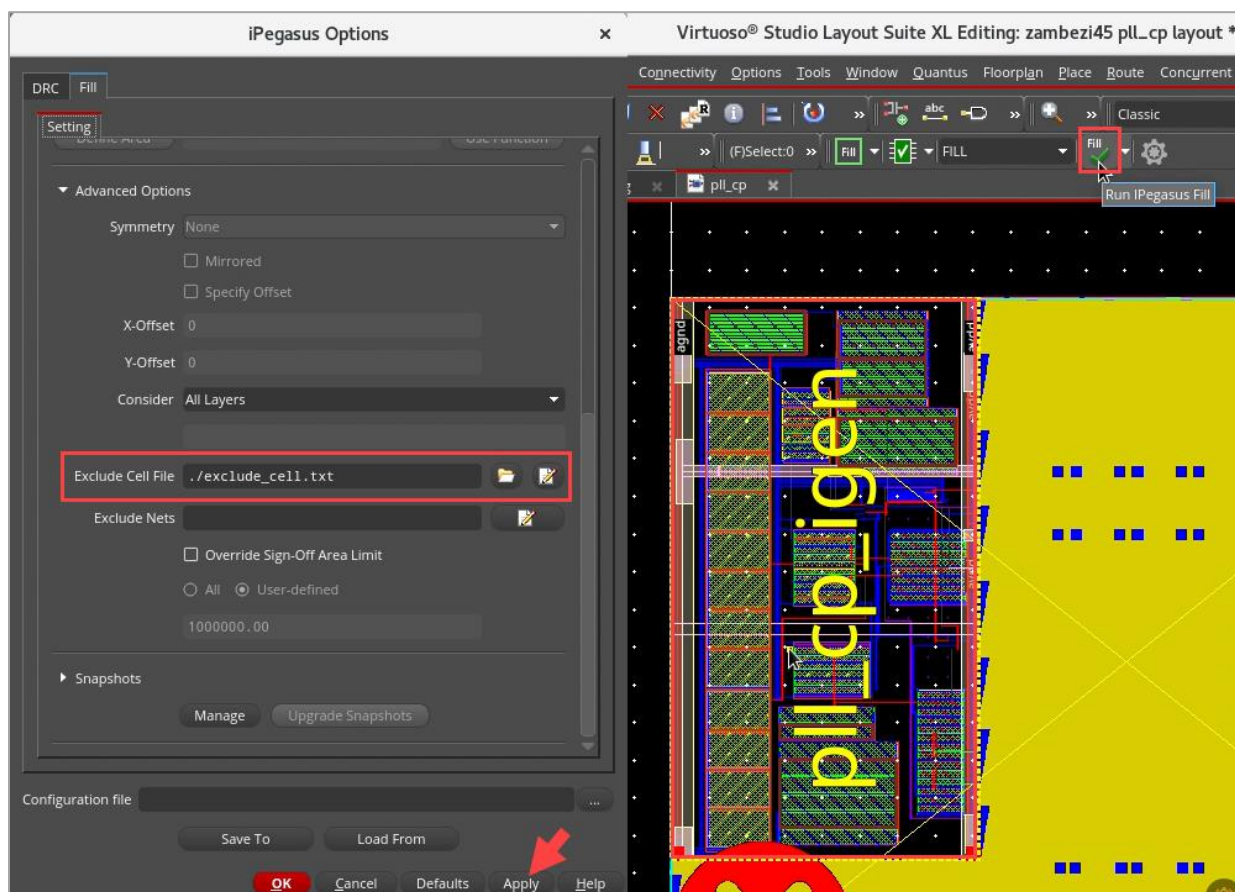
That will open the *Exclude Cells File Editor* form.

3. In the *Exclude Cells File Editor* form:
 - a. Click on the green plus field to add a new row to select a cell to exclude.
Double-click on the line to edit the value by selecting it from the menu.
 - b. Select the following cell in the library manager: *zambezi45 | pll_cp_igen | layout*.
 - c. Click **Save As and Exit**.
 - d. In the resultant form, **save** the exclude file as *exclude_cell.txt*.

That will close the *Exclude Cells File Editor* form.

We just created an exclude file with cell *pll_cp_igen* in it. This file will be populated in the *Exclude Cell file* field of the iPegasus Options - Fill form.

4. Back in the iPegasus Options - Fill form, click **Apply**.
5. In the iPegasus Toolbar, click on the **Run iPegasus Fill** icon to start the run.
That will generate the iPegasus fill.
6. In the Virtuoso layout suite:



- a. Zoom in to the border of the *pll_cp_igen* instance and note that no fill shapes are created in this cell region.
- b. Remove the fill shapes by selecting **File → Discard Edits**. Click **Yes** in the pop-up.

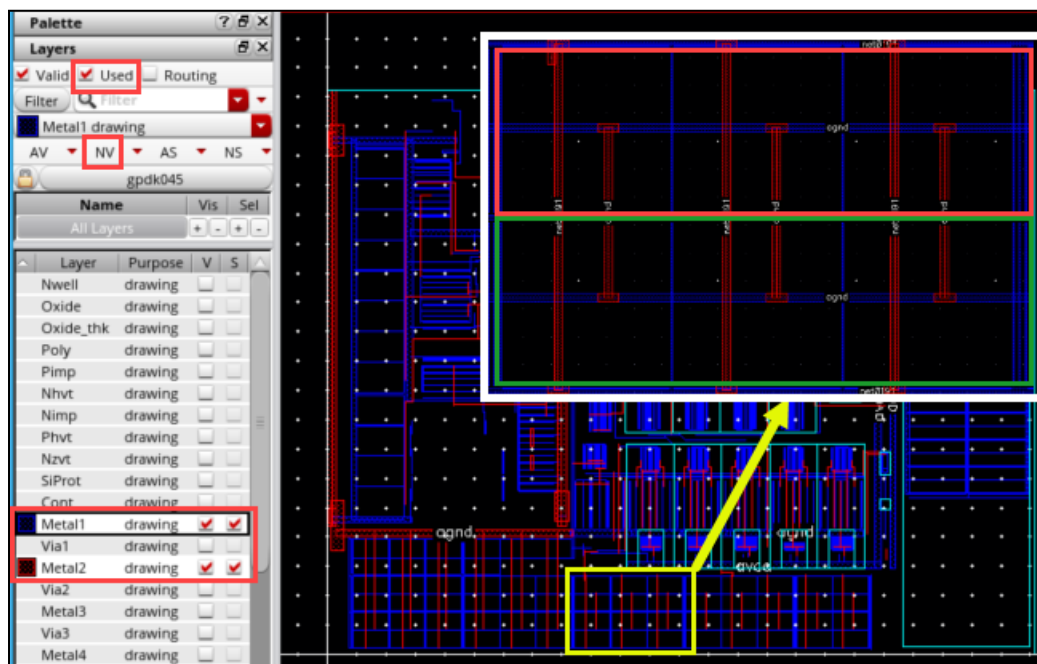
Do not close the iPegasus Options - Fill form.

Running iPegasus Fill – Create Symmetrical Fill

With this feature, iPegasus will create a symmetrical fill over a specific area. There are add-on options to perform the fill operation in a mirrored fashion, with XY offset, for selected layers, etc. This is important for sensitive designs, where metal fill mismatch can impact circuit timing performance.

1. Ensure that *zambezi45 | pll_cp | layout* is open in the layout suite. Press **F** to fit the design. Turn ON the visibility of all layers.
2. In the Virtuoso layout suite:
 - a. Under the Layers section, make only *Metal1* and *Metal2* drawing visible.

Note: Use the layer palette options such as *Used* and *NV*.
 - b. Zoom in on the highlighted area in the layout.



Notice that the *red* highlighted part looks just like the mirror image of the green highlighted part.

3. In the iPegasus Options - Fill form:

The screenshot shows the 'iPegasus Options' dialog box with the 'Fill' tab selected. The 'Output' section is highlighted with a red box, containing the following settings:

- Fill Instance Name: `Virtuoso_fill`
- Library: `pegasusfillLib`
- Cell: `fill_pll_cp`
- View: `layout`
- ☐ Create Hierarchical Fill
- ☒ Clear Existing Fill

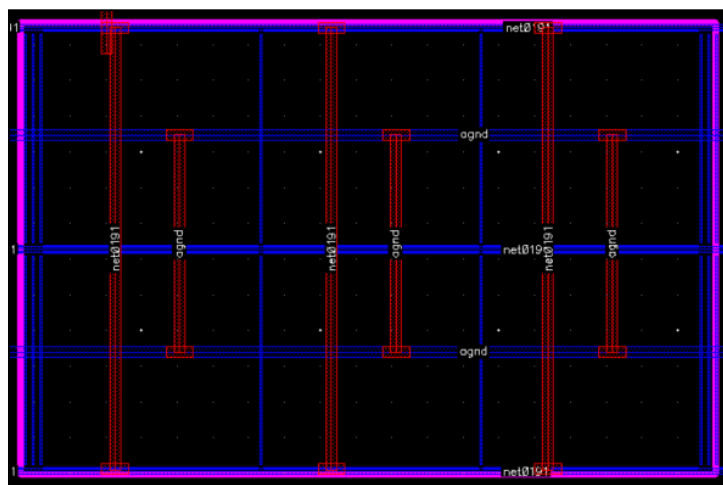
The 'Scope' section shows 'Specified Area' selected, with a 'Define Area' button and coordinates `41.585 0.745 61.330 13.775`.

The 'Advanced Options' section shows 'Symmetry' set to 'X-Axis' with 'Mirrored' checked. The 'Consider' dropdown is open, showing 'Visible Layers' selected.

At the bottom, there are buttons for 'Save To', 'Load From', 'OK', 'Cancel', 'Defaults', 'Apply', and 'Help'. Red arrows point to the 'Defaults' and 'Apply' buttons.

- Click on the **Defaults** icon to load the default state of the form.
- Set the *Fill Instance Name* to **Virtuoso_fill**.
- Set the output *Library* to **pegasusfillLib**.
- Set the fill *Cell* name to **fill_pll_cp**.

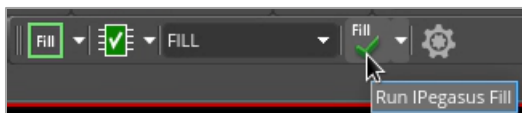
- e. Set the *View* to **layout**.
- f. Unselect the *Create Hierarchical Fill* field.
- g. Select the *Clear Existing Fill* checkbox.
- h. Switch the checker *Scope* to **Specified Area**.
- i. Click on **Define Area** and select the area as shown below.



The selected area is highlighted in the layout.

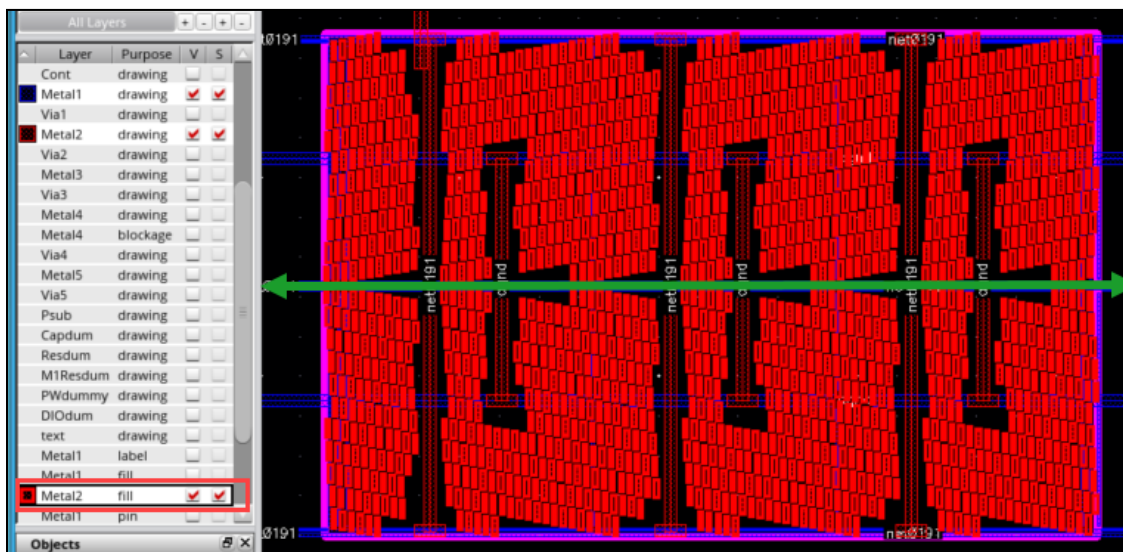
Respective coordinates are added to the *Define Area* field.

- j. Under Advance Options, select **X-Axis** for *Symmetry*.
 - k. Under Advance Options, select the *Mirrored* option.
 - l. Under Advance Options, set to *Consider Visible Layers* only.
With these settings, iPegasus should create a symmetrically mirrored fill whose reference axis for mirroring is the X-axis for only Metal1 and Metal2 within the defined area.
 - m. Click **Apply**.
4. In the iPegasus Toolbar, click on the **Run iPegasus Fill** icon to start the run.



That will generate the iPegasus fill.

5. In the Layer Palette, enable the visibility of only Metal2 fill and review the fill structure over the specified area.



Notice that all the Fills are generated symmetrically by keeping the highlighted green line as an X-axis reference.

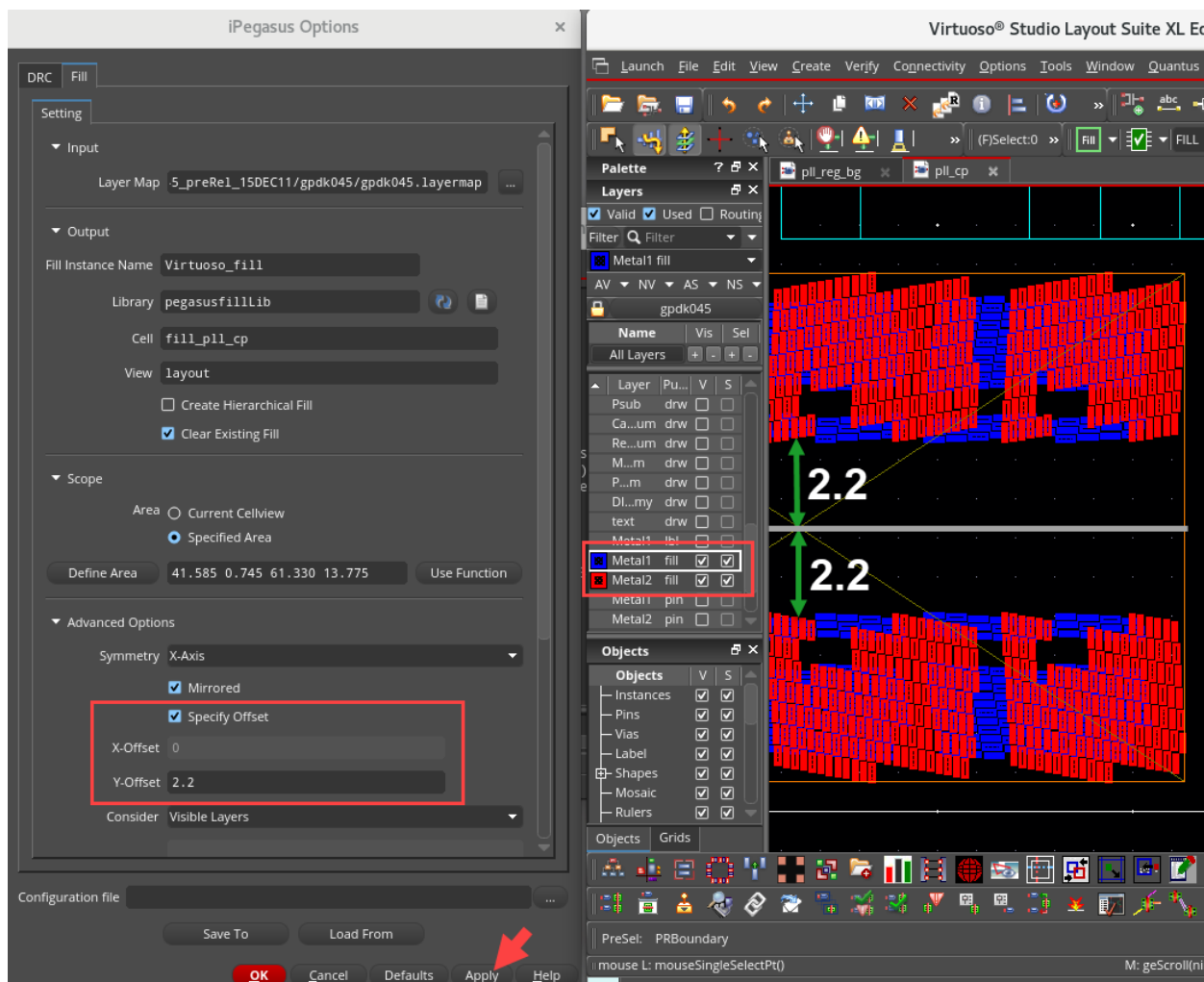
Symmetrical Fill with Offset

1. Back in the iPegasus Options - Fill form,
 - a. Under the Output section, select *Clear Existing Fill*.
 - b. Under Advance Options, select *Specify Offset*, and set *Y-Offset* to **2.2**.
 - c. Click **Apply**.
2. In the Virtuoso layout suite - Layers section, make only *Metal1* and *Metal2* drawing visible.
3. In the iPegasus Toolbar, click on the **Run iPegasus Fill** icon to start the run.

That will generate the fill.

In this case, iPegasus will clear the existing fill and regenerate the Metal Fill at an offset of 2.2 microns in the Y-direction.

4. In the Layer Palette, enable the visibility of Metal1 and Metal2 fills and review the fill structure.

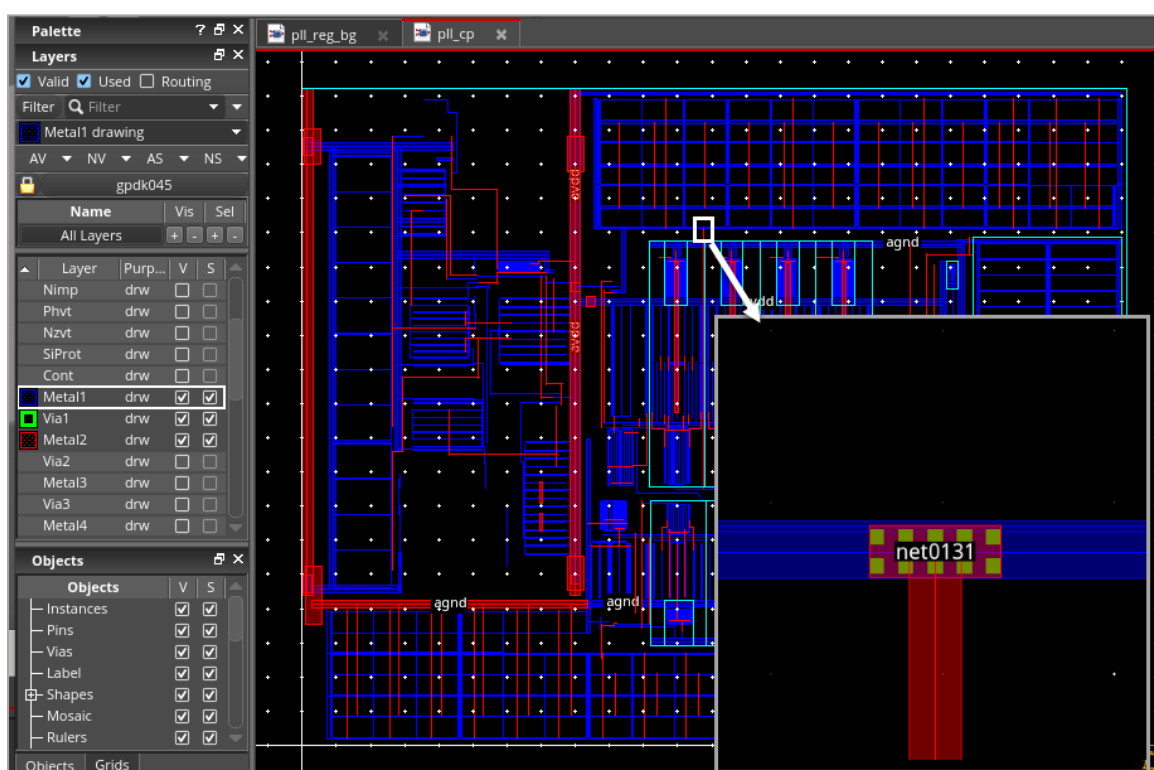


5. Remove the fill shapes by selecting **File** → **Discard Edits**. Click **Yes** in the pop-up.
6. Turn ON the visibility of all layers.
Do not close the iPegasus Options - Fill form.

Using Net Aware Feature

The Net Aware Fill feature of iPegasus can create a Fill with extra offset around critical nets, which can have a parasitic effect because of the Fill. For this lab, we will assume **net0131** is a critical net.

1. Ensure that *zambezi45 | pll_cp | layout* is open in the layout suite. Press **F** to fit the design. Turn ON the visibility of all layers.
2. In the layout suite,
 - a. Turn on the visibility of Metal1, Via1, and Metal2 only.
 - b. Zoom in to the area (58.9, 75.6).

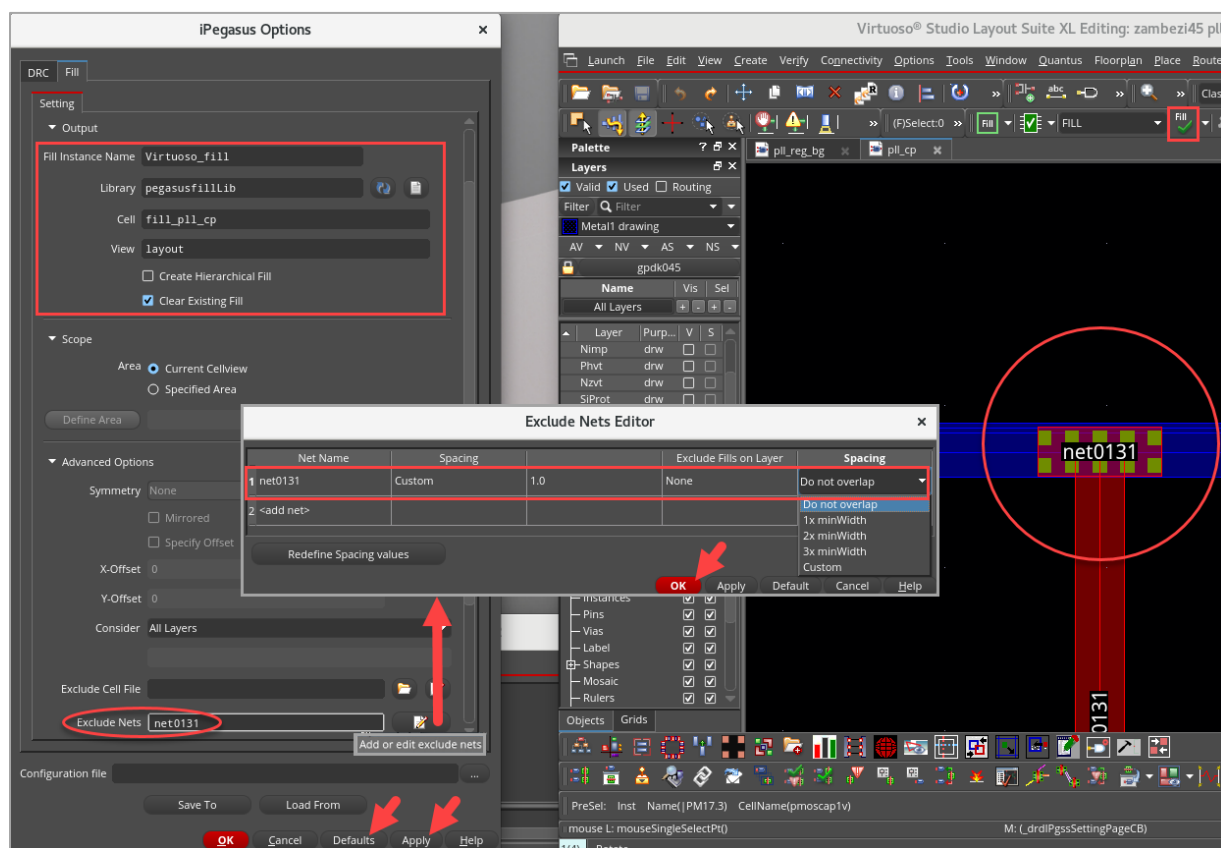


It is an intersecting area where net0131 goes from Metal1 to Metal2.

3. In the iPegasus Options - Fill form:
 - a. Click on the **Defaults** icon to load the default state of the form.
 - b. Set the *Fill Instance Name* to **Virtuoso_fill**.
 - c. Set the output *Library* to **pegasusfillLib**.
 - d. Set the fill *Cell* name to **fill_pll_cp**.
 - e. Set the *View* to **layout**.

f. Unselect the Create Hierarchical Fill field.

g. Select the *Clear Existing Fill* checkbox.



4. Click the *Exclude Nets* button in the iPegasus Options - Fill form.

That opens the Exclude Nets Editor.

5. In the **Exclude Nets Editor**,

a. Double click in the Net Name section to fill in the net name, **net0131**.

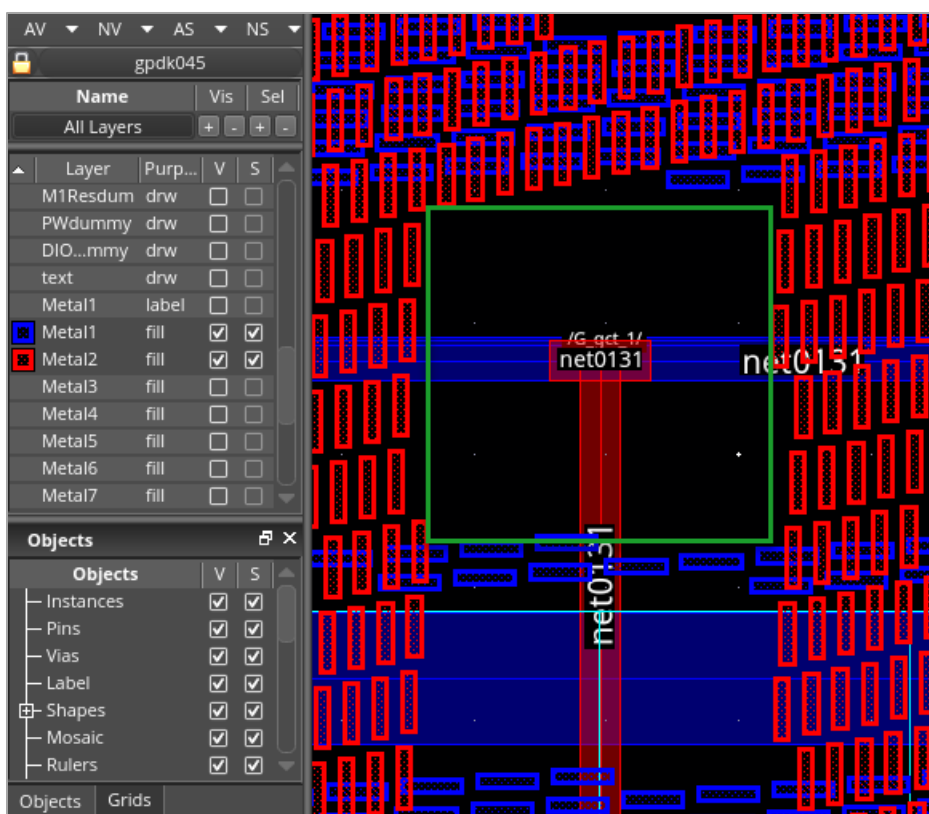
b. Similarly, double-click in every section to fill out the form as shown (Spacing—custom: 1.0, Exclude Fills on Layer—None, Spacing—Do not overlap).

c. Click **OK**.

That closes the Exclude Nets Editor.

Net name **net0131** is added to the Exclude Nets list in the form.

6. In the iPegasus Options - Fill form, click **Apply**.
7. In the iPegasus Toolbar, click on the **Run iPegasus Fill** icon to start the run.
That will generate the fill.
8. In the Layer Palette, enable the visibility of only Metal1 drawing, Metal2 drawing, Metal1 fill, and Metal2 fill. Review the fill structure around the **net0131**.



Notice the fill clearance around the selected net.

9. Remove the fill shapes by selecting **File → Discard Edits**. Click **Yes** in the pop-up.
10. In the CIW, choose **File → Close Data**.
The Close and Purge Data form opens:
 - a. Select **All** and click **OK**.
 - b. In the resultant *Save Cellview* form, select **Discard All**.
11. **Exit** the Virtuoso session.

